

Surreal Fields Across Set-Theoretic Universes

Omitted cuts, fresh-sign gaps and saturation spectra of old surreal fields; first new
birthdays; surcomplex real forms;
and a Boolean completion of reverse simplicity

22 September 2026

Abstract

Let $M \subseteq N$ be transitive universes of ZFC with the same ordinals, and view the old surreal field $\mathbf{No}^M$ as a proper class of N . All sizes, types and parameter sets are computed in N . For every uncountable cardinal κ of N , the ordered field $\mathbf{No}^M$ is κ -saturated exactly when N has no new ordinal-valued sequences of length less than κ . The same condition is equivalent to interpolation of every set-sized cut with just one side of size less than κ . The least length δ of a new sequence is a regular cardinal of both universes. A fresh sequence of length δ yields an omitted gap of exact cofinality pair (δ, δ) , by three explicit constructions: a nested interval tree, an absolute interval code with all ordinals as labels, and ordinal block codes of signs.

Every gap of $\mathbf{No}^M$ with set-sized cofinal and coinital witnesses comes from a unique fresh sign, a new sign sequence all of whose proper prefixes are old. Its fillers are exactly the sign extensions of that fresh sign, and every such gap is symmetric. The first new birthday β satisfies $\delta \leq \text{cf}^N \beta \leq \beta$. At the countable boundary, ω -saturation detects exactly the new reals. The pure field $\mathbf{No}^M[i]$ stays saturated over every N -set, while naming conjugation restores the real spectrum. The old field is closed and nowhere dense in $\mathbf{No}^N$, and each N -set of old numbers is uniformly discrete. We also treat bounded birthday fragments and the Cohen, $\text{Add}(\kappa, 1)$ and Prikry extensions. Under Global Choice, we give nonconjugate real forms of $\mathbf{No}^N[i]$ none of which has a set-sized cofinal subset. Finally we give a set-complete class Boolean algebra in which reverse simplicity is dense; class-completeness is impossible under Global Choice.

Status and provenance. An AI-assisted research report merged from four manuscripts of 22 September 2026 (sources 01, 03, 04 and 08 of the batch that placed them), each pinned to repository commit 4f26455; see section 1.6. The principal results are proposed contributions with written proofs, not certified priority claims. The report has not been refereed, and no statement in it is Lean-verified. Classical surreal facts, quantifier elimination, standard forcing facts, and the class back-and-forth and conjugacy criteria already in the collection are credited, not claimed. The finite Python checks test finite sign and interval formulas only.

Addition of 28 September 2026. Part VIII merges six further manuscripts (sources 09–14, pinned to ProveIt commit e21766d): full gap spectra of old surreal fields after forcing, the spectra of intermediate grounds, and old fields with equal full gap spectra that are not isomorphic (section 21.1). It is equally unrefereed and not Lean-verified.

Contents

1	The question, the answers, and how this report was assembled	1
1.1	The question	1
1.2	Standing hypotheses and the two levels of assumptions	1
1.3	Conventions and notation	2
1.4	The main results	2
1.5	What is offered as new, and what is not	4
1.6	Four manuscripts, one report	4
1.7	Relation to the collection	6
I	Foundations and the common spine	7
2	Foundations, notation, and imported facts	7
2.1	The two universes	7
2.2	Signs, birthdays and cones	7
2.3	Field theory and types	8
3	Absoluteness and ground-set bounds	9
4	The first new sequence length	11
5	Coding a fresh sequence by an omitted cut	12
5.1	Main route: an arbitrarily branching interval tree (source 01)	12
5.2	Second route A: an absolute all-ordinal interval code (source 04)	13
5.3	Second route B: ordinal block codes of signs (source 08)	15
6	The exact spectrum and its first gap	16
6.1	Proof of the main equivalence	16
6.2	Exact two-sided cofinalities	16
7	The countable boundary: no new reals is weaker	17
II	Fresh signs classify the set-presented gaps (source 03)	18
8	The gap–fresh-sign correspondence	18
9	The least gap	20
III	The first new birthday (source 08)	20
10	First new birthdays and their relation to saturation	21
IV	Surcomplex fields, fragments, forcing and topology	22
11	Surcomplex numbers: a language-dependent dichotomy	22
11.1	The pure field forgets the sequence obstruction	22
11.2	Naming conjugation restores the old ordered field	23
11.3	A nonisomorphism consequence	24
12	A sharp theorem for bounded birthday fields (source 01)	24

13 Forcing examples and distributivity	25
13.1 The forcing-theoretic formulation	25
13.2 A Cohen real destroys even finite-parameter saturation	26
13.3 Adding a subset of a regular cardinal	26
13.4 Prikry forcing separates the two countable notions	27
13.5 Comparison table	28
14 Why order gaps do not become Cauchy limits	28
V Class isomorphisms and real forms (Global Choice)	30
15 Pure-field class isomorphisms	30
16 Gap invariants of surcomplex real forms	31
16.1 Several ground real forms in one field	33
VI A Boolean completion of reverse simplicity (source 01)	34
17 A Boolean completion of reverse simplicity	34
17.1 A direct limit with set-coded elements	34
17.2 Why joins of all subclasses are impossible	35
VII Scope, questions and limits	36
18 Scope and significance	36
18.1 What the new information consists of	36
18.2 Limits of the assertions	36
18.3 A formalization route without class-sized type objects	37
19 Questions, and which of them the merge answered	37
20 What this report does not claim	39
VIII Full gap spectra, relative grounds and classification (sources 09–14)	43
21 How Part VIII was assembled	43
21.1 Six manuscripts, one addition	44
21.2 What the addition answers	46
21.3 Conventions for Part VIII	46
22 The cofinality transfer	48
22.1 Fresh-function spectra of a pair	48
22.2 Compression and three block codes	49
22.3 The transfer theorem	50
23 External density and single-coordinate spectra	52
23.1 A ceiling measured in the extension	52
23.2 Generalized Cohen forcing	53
23.3 The square chain condition	55
24 Easton products	57

25	Relative spectra of intermediate grounds	58
25.1	Forcing tools	58
25.2	Isolated coordinates (source 10)	60
25.3	Sparse products and finite complements (source 11)	60
25.4	Finite Boolean cubes (source 14)	62
25.5	All coordinate grounds of a countable product	62
25.6	Families of any size with one threshold	66
26	Set-sized cutoffs and local ubiquity	68
26.1	The exact birthday boundary (source 14)	68
26.2	Local ubiquity	69
27	The real trace and a Boolean family with one spectrum	70
27.1	The real trace	70
27.2	The Boolean family	71
28	Common small forcing and Cohen amplification	74
28.1	Two name facts and the base-change theorem	74
28.2	Cohen amplification	75
28.3	Two exact spectra: Prikry and Namba	77
29	A Cohen–Prikry tower	78
29.1	Three constructions of equal spectra without isomorphism (merge comparison)	79
30	Surcomplex real forms from families of old fields	80
30.1	Same spectrum, agreement on old constants (source 09)	81
30.2	Agreement on any set-sized core (source 10)	81
30.3	Agreement on $\mathbb{C}^N$ with one saturation spectrum (source 11)	82
30.4	Conjugacy and equivariant embeddings in the Boolean family (source 12)	83
30.5	A continuum of equal-threshold real forms (source 13)	83
30.6	Real forms of one set field, and the class version (source 14)	84
30.7	Relation to Part V and to the collection	85
31	Questions of sources 09–14, and which of them the cluster answered	85
32	What sources 09–14 do not claim	90
33	Provenance, repository comparison and finite checks of sources 09–14	93
33.1	Review scope of the sources	93
33.2	Repository statements checked against the current collection	94
33.3	Finite checks	94
33.4	Dependency ledger	95
A	Provenance and repository comparison	96
A.1	Pinned review scope of the sources	96
A.2	Statements corrected against the current collection	96
A.3	Literature boundary	97
B	Dependency and proof-obligation ledger	97
C	Finite checks	98
	References	100

1 The question, the answers, and how this report was assembled

1.1 The question

Forcing preserves ordinals, but it can add new functions on old ordinals. Consequently, it can add new surreal sign sequences without changing the ordinal labels used for birthdays. This raises a more delicate question than whether new surreal numbers exist:

How much of the old surreal field’s ability to realize cuts and first-order types survives when the surrounding universe acquires new sets?

An old field can remain an elementary subfield of a larger surreal field while losing saturation. Elementarity tests individual first-order formulas with finitely many parameters. Saturation also tests whole sets of formulas, and those sets can be new even when every formula and every parameter is old.

The familiar cut principle says that every pair of *sets* $A < B$ of surreals has a separator. The word “set” hides a parameter: which universe decides what counts as a set? When M is enlarged to N without changing the ordinals, every old surreal number remains available, but a new set of old numbers need not have belonged to M . The old cut principle promises nothing for such a collection. This report measures exactly what is lost.

1.2 Standing hypotheses and the two levels of assumptions

- (H) $N \models \text{ZFC}$, and M is a parameter-definable transitive inner class of N that contains every ordinal of N and satisfies ZFC (as a scheme). Equivalently for our purposes, M may be an available class predicate in a class theory that has the Separation and Replacement instances for that predicate, or a two-universe metatheory with these closure properties stipulated. Forcing extensions $N = M[G]$ are the intended examples, but no proof uses forcing except in section 13 and corollary 16.7.
- (GC) In addition, the ambient theory is GBC with Global Choice, and M is a class of that class universe; M need not be definable.

Everything in this report is proved under (H) except the following, which also use (GC): the class isomorphisms and real forms of part V, and the impossibility of a class-complete Boolean algebra (theorem 17.3). Sources 03 and 08 were written with (GC) as their ambient theory. Their own dependency statements confine Global Choice to the class back-and-forth and real-form arguments (source 03, its audit table; source 08, its foundations section). Every proof taken from them for a statement outside part V was re-read for this merge. It uses only set-sized choices, Choice inside M , and Replacement for the predicate M . One re-reading was needed: source 03 quantifies over gaps of No^M as class partitions. Under (H) a gap is instead given by its set presentation (definition 6.1), and the statements of part II become schemes over such presentations.

External means computed in N , never in a larger metatheory. If M and N are represented by countable transitive *set* models inside a larger universe, all claims about sets, cardinalities and saturation must still be read inside N . They are not claims of saturation with respect to every subset in that larger universe. We use no unrestricted truth predicate for class structures. In the languages used here, quantifier elimination reduces formulas to finite algebraic conditions and so supplies the satisfaction relation directly. Types are ordinary N -sets of finite syntactic expressions over set-sized parameter sets. This avoids the general class-satisfaction obstacle discussed by van den Dries–Ehrlich and by Ehrlich [3, 6].

The Prikry examples assume a measurable cardinal in M . The $\text{Add}(\kappa, 1)$ examples need no cardinal arithmetic; source 04’s extra hypothesis $\kappa^{<\kappa} = \kappa$ is used only for its additional preservation clause (example 13.3).

1.3 Conventions and notation

The four sources used different letters for the same objects, and several letters for two objects. This report fixes one symbol per notion.

Notion	Symbol here	In the sources
Inner and outer universe	$M \subseteq N$	$M \subseteq N$ (01, 03); $M \subseteq W$ (04); $M \subseteq V$ (08)
Old and new surreal fields	$\mathbf{No}^M, \mathbf{No}^N$	K, K^+ (01); F, E (03); S, T (04); F_M, F_V (08)
Old and new surcomplex fields	$\mathbf{SC}^M = \mathbf{No}^M[i],$ $\mathbf{SC}^N = \mathbf{No}^N[i]$	C, C^+ (01); K_M, K_N (03); K_M, K_W (04); K_M, K_V (08)
Canonical conjugations	c_M, c_N	c (01); c_M, c_N (03); c_M, c_W (04); c_M, c_V (08)
Transported conjugation	$j = \Phi c_M \Phi^{-1}$	j_M (03); d (04); c' (08)
First new length of an ordinal-valued sequence	$\delta = \delta(M, N)$	δ (01, 08); ν (03); $d(M, W)$ (04)
First new birthday	$\beta = \beta(M, N)$	β (08)
No new short sequences	$\mathbf{NS}_{<\kappa}(M, N)$	same (01, 08); $\mathbf{D}_{<\kappa}(M, W)$ (04)
Birthday (length) of a sign	$\mathbf{b}(s)$	$\mathbf{b}$ (01, 04); $\ell(s)$ (03); $\mathbf{lh}(s)$ (08)
Sign cone of s	$\mathbf{Cone}(s)$	(L_s, R_s) separators (01); $\mathbf{Cone}(s)$ (03, 08); C_s (04)
Ordinal block codes	$\text{code}(f), \text{code}^+(f)$	$C(f)$ (08); s_f (03)
Measurable cardinal (Prikry)	ρ	ρ (01); λ (03); ν (04); θ (08)
Regular cardinal of $\text{Add}(\kappa, 1)$	κ	ρ in $\text{Fn}_{<\rho}(\rho, 2)$ (01); κ (03, 04, 08)
Gap with set-sized witnesses	set-presented gap	set-character gap (01); set-presented gap (03); Dedekind gap with character (08)
Cut property	η_κ	η_κ (01, 03, 04); $\text{Cut}_{<\kappa}$ (08)

The letter K is not used for the proper-class fields $\mathbf{No}^M$ and $\mathbf{SC}^M$: most reports of the collection reserve K for a *set-sized* field, and the collection’s notation guide recommends $\mathbf{SC} = \mathbf{No}[i]$ for the surcomplex class (docs/NOTATION.md). The letter δ always denotes the first new length of an ordinal-valued sequence, never a birthday; the first new birthday is β , as in source 08. The sibling report docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets uses the same letter β for the first new birthday. Several of that report’s source manuscripts wrote δ for the first new birthday; that δ is the β of this report, not its δ . The letter θ denotes an ordinal bound, on the range of a sequence or on a set of birthdays, and ρ is reserved for the measurable cardinal of the Prikry examples. The letter ν is not used.

Tempting false readings. “Gap” here always means a gap of the class order $\mathbf{No}^M$ seen from N , with set-sized witnesses; it is not the proper-class “finite–infinite gap” of $\mathbf{No}$ itself studied in docs/surreal/genetic-gaps-and-primitives. “External” means computed in the outer universe N ; in the foundations report the same word refers to a stronger metatheory (remark 14.4). η_κ is indexed by a size bound, not by an aleph subscript. $\beta(M, N)$ is an ordinal attached to a pair of universes, not the birthday of a particular number.

1.4 The main results

For an infinite cardinal κ of N , write

$$\mathbf{NS}_{<\kappa}(M, N) \iff (\lambda\theta)^N \subseteq M \quad \text{for all ordinals } \lambda < \kappa \text{ and } \theta. \quad (1.1)$$

Every ordinal-valued sequence has bounded range, so this is exactly the absence of new ordinal-valued sequences of length less than κ . The range bound θ is *not* required to be below κ .

Theorem 1.1 (Main spectrum theorem; sources 01, 03, 04, 08). *For every infinite cardinal κ of N , the following are equivalent.*

- (i) $\text{NS}_{<\kappa}(M, N)$ holds.
- (ii) Every pair of N -sets $L, R \subseteq \mathbf{No}^M$ with $L < R$ and $\min(|L|, |R|) < \kappa$ has a separator in $\mathbf{No}^M$.
- (iii) Every such pair with $|L \cup R| < \kappa$ has a separator in $\mathbf{No}^M$.

For uncountable κ , these are also equivalent to κ -saturation of $\mathbf{No}^M$ in the ordered-field language, and to κ -saturation of the order $(\mathbf{No}^M, <)$ as a dense linear order, both evaluated in N . When they fail, an omitted pair as in (iii) can be chosen inside $(0, 1)$.

Condition (ii), from source 01, is stronger in appearance than the usual small-cut condition: the other side can be an arbitrarily large set. Neither side may be a proper class. In particular a cut is always filled if one side is an actual ground-model set (theorem 3.5). The two-sided form (iii) and the saturation clause were proved independently by all four sources; the dense-linear-order clause is source 08's and the $(0, 1)$ clause source 04's.

Theorem 1.2 (First loss and language sensitivity; sources 01, 03, 04, 08). *Suppose $M \neq N$, and let δ be the least ordinal length of a new ordinal-valued sequence. Then:*

- (i) δ is an infinite regular cardinal in both M and N .
- (ii) $\mathbf{No}^M$ has a set-presented gap of exact cofinality pair (δ, δ) , and no set-presented gap has either cofinality smaller than δ .
- (iii) For uncountable N -cardinals κ , $\mathbf{No}^M$ is κ -saturated exactly when $\kappa \leq \delta$.
- (iv) $\mathbf{No}^M$ is ω -saturated exactly when $\mathbb{R}^M = \mathbb{R}^N$.
- (v) SC^M in the pure field language is saturated over every N -set of parameters. The expansion (SC^M, c_M) by $c_M(a + bi) = a - bi$ has exactly the saturation spectrum of $\mathbf{No}^M$, including at ω .

Beyond this common spine, each source contributed results that the others do not have.

- **Fresh signs classify the gaps** (source 03, part II). The set-presented gaps of $\mathbf{No}^M$ correspond bijectively to the *fresh signs*, new sign sequences all of whose proper prefixes are old. The gap of a fresh sign s has both cofinalities equal to $\text{cf}^N \text{b}(s)$, and its fillers in $\mathbf{No}^N$ are exactly the extensions of s . Hence every set-presented gap is symmetric, and $\mathbf{No}^N \setminus \mathbf{No}^M$ is the disjoint union of the fresh cones.
- **The first new birthday** (source 08, part III). The least birthday β of a new surreal is a cardinal of both universes, and $\delta \leq \text{cf}^N \beta \leq \beta$.
- **Bounded birthday fragments** (source 01, section 12). For an uncountable regular cardinal κ of M that remains a cardinal in N ,

$$(\mathbf{No}_{<\kappa})^M \text{ is } \kappa\text{-saturated in } N \iff \begin{aligned} &\text{cf}^N(\kappa) = \kappa \text{ and} \\ &({}^\alpha 2)^N = ({}^\alpha 2)^M \quad (\alpha < \kappa). \end{aligned} \quad (1.2)$$

Thus the full-class theorem and the bounded-fragment theorem test different preservation properties.

- **Topology** (sources 01, 03, 04, section 14). Every N -set of old numbers is uniformly discrete and closed, and set-indexed Cauchy nets of old numbers are eventually constant. If $M \neq N$, the old field is closed and nowhere dense in $\mathbf{No}^N$, and its subspace topology is its own order topology.
- **Real forms under Global Choice** (sources 03, 04, 08, part V). $\mathbf{SC}^M \cong \mathbf{SC}^N$ as pure fields, over any N -set of common parameters. Transporting c_M gives involutions of $\mathbf{SC}^N$ that are not conjugate to c_N , although neither fixed field has a set-sized cofinal subset. Several such involutions can be separated by the gap invariant, and they can agree with c_N on every ordinary complex number.
- **A Boolean completion** (source 01, section 17). The reverse-simplicity order on signs embeds densely in an explicit set-complete class Boolean algebra. Under Global Choice no class-complete such algebra exists.

1.5 What is offered as new, and what is not

The proposed contributions are the exact external spectrum of theorem 1.1 with its one-sided form; the three explicit codings of a fresh sequence by an omitted cut (the arbitrary-alphabet interval tree of source 01, the absolute all-ordinal interval code of source 04, and the ordinal block codes of sources 03 and 08); the exact gap threshold; the fresh-sign classification of all set-presented gaps; the separation of the first new birthday from the first saturation obstruction; the distinction between ω - and ω_1 -saturation under no-real extensions; the resulting surcomplex language comparison; the bounded-fragment criterion; the external topological statements; the real-form invariants; and the Boolean completion. Each source stated its own contributions as proposed, with priority not certified.

Their ingredients are not claimed as new: Conway's cut theorem and sign expansions, real closedness of $\mathbf{No}$ and algebraic closedness of $\mathbf{No}[i]$ [1, 2]; quantifier elimination for real closed and algebraically closed fields [9]; saturation and homogeneous universality of surreal class models [6, 7]; standard forcing facts [10, 11, 12, 13]; the forcing notion of a fresh function [14]; and, from the collection, class back-and-forth for proper-class algebraically closed fields, the conjugacy criteria for real forms, and a real form with a countable cofinal subset [19]. Several statements are deliberately labelled consequences rather than independent discoveries. Regularity of the first new length is an elementary set-theoretic fact, not a new invariant.

The Boolean section answers a concrete question in Berenbeim's 2020 *notes*, not a conjecture of a peer-reviewed paper. Its construction is a natural direct limit and may be folklore in class forcing; its value here is an explicit, size-correct answer, not a claim that Boolean completions were previously unavailable. General class-forcing completion theory predates those notes [16]. Berenbeim also proposes studying surreals under forcing in the final section of the same notes [15, pp. 32, 34]; this report develops one concrete direction of that program and does not settle the notes' broader suggestions about embeddings of universes.

1.6 Four manuscripts, one report

This report is merged from four manuscripts, all dated 22 September 2026 and all written against commit 4f2645599121fa872c7104995e47f86f0382351f of VladimirReshetnikov/Surreal. The source numbers are those of the batch in which they arrived. Delivered code and data files keep those numbers as prefixes. No source manuscript is shipped.

Source	Manuscript	Contribution here
01 (base)	<i>Forcing, Omitted Cuts, and Old Surreal Fields: ordinal-sequence spectra, surcomplex conjugation, and a Boolean completion of reverse simplicity</i> (25 pages)	The foundations and spine (sections 2 to 7), the one-old-side theorem, the interval tree, the surcomplex dichotomy (section 11), bounded fragments (section 12), the forcing framework (section 13), external Cauchy rigidity, the Boolean completion (section 17), and most of section 18. LaTeX Makefile, as <code>code/01-forcing-omitted-cuts-Makefile</code> .
03	<i>Surreal Fields Across Set-Theoretic Universes: fresh gaps, exact saturation thresholds, and surcomplex real forms</i> (21 pages)	All of part II (fresh-sign gap classification, filling cones, the terminal-plus block code, the least-gap theorem); old ordinal bounds (lemma 3.3); the ordinal-bound form of the cut criterion; the $\text{Add}(\kappa, 1)$ first-gap calculation without cardinal arithmetic; the Prikry fresh sign; the proper-class ACF saturation theorem; the real-form invariant, agreement with c_N on $\mathbb{C}^N$, several ground real forms and finite forcing towers; the closed, nowhere-dense and subspace-topology statements. Files prefixed <code>03-fresh-sign-gaps-</code> .
04	<i>Surreal Fields Across Set-Theoretic Universes: exact saturation, forcing, and surcomplex real forms</i> (25 pages)	The absolute all-ordinal interval code with closed finite-branch formulas (second route A, section 5.2); the $(0, 1)$ clause; the forcing characterization for preserved uncountable cardinals; $\text{Add}(\kappa, 1)$ under $\kappa^{<\kappa} = \kappa$; the elementarily equivalent nonconjugate involutions with different spectra and the several-spectra corollary; external discreteness in the product topology, open prefix cones, and nowhere-denseness of SC^M in SC^N . Files prefixed <code>04-universe-saturation-</code> .
08	<i>Forcing and Saturation of Surreal and Surcomplex Fields: exact preservation criteria and a real-form dichotomy</i> (24 pages)	The ordinal block code with its two-bound cone identity and exact separator class (second route B, section 5.3); the dense-linear-order clause; the first new birthday and $\delta \leq \text{cf}^N \beta \leq \beta$ (part III); the Cohen, $\text{Add}(\kappa, 1)$ and Prikry values of β ; the lottery-sum caution; the non-definability of conjugation in the pure field; the saturated back-and-forth route and the isomorphism over an arbitrary old set. Files prefixed <code>08-forcing-saturation-</code> .

Why one report. All four manuscripts prove the same spine in the same setting: the cut criterion, the regular first new length, an omitted (δ, δ) gap, the ω boundary, the pure/conjugation contrast, the distributivity corollary and the Cohen and Prikry examples. Their extras are complementary. **Source 01 is the base:** its hypotheses are the weakest (plain ZFC with a definable inner model, Global Choice only for one negative Boolean result), and its main criterion is the strongest in form, since one small side suffices.

Printed once. The following results were proved by two or more sources and appear once, with all proving sources named at the statement: the absoluteness of old arithmetic and elementarity (proposition 3.1); ground-set bounds and envelopes (lemma 3.2); coding of short families (lemma 3.4); regularity of δ (proposition 4.2); cuts versus real-closed saturation (lemma 2.3); the main spectrum (theorem 1.1); the first gap (theorem 6.2); the ω criterion (theorem 7.1); pure surcomplex saturation (theorem 11.2); the conjugation spectrum (theorem 11.4); nonisomorphism (corollary 11.6); the distributivity corollary (corollary 13.1); the Cohen, $\text{Add}(\kappa, 1)$ and Prikry examples; external discreteness (proposition 14.1); closedness and nowhere-denseness (theorem 14.3); the class isomorphism (theorem 15.2); the transported real form (theorem 16.2); and the several-grounds statements (theorem 16.6).

Marked second routes. The converse direction of the spine has three genuinely different constructions. The interval tree of source 01 is printed as the main route (section 5). The absolute interval code of source 04 (section 5.2) and the ordinal block codes of sources 08 and 03 (section 5.3 and proposition 9.1) are kept as marked second routes. Two existence proofs for δ (Mostowski collapse, sources 01 and 03; least new rank, sources 04 and 08) and two class back-and-forth routes (algebraic/transcendental extension, sources 03 and 04 as in the collection; saturated elementary maps, source 08) are kept as well.

Where the merge had to choose. (a) Symbols: every renaming is listed in section 1.3, and none changes a normalization. (b) Hypotheses: statements are printed at the weakest hypotheses that any source proved them under. Where sources differ, the difference is recorded at the statement: regularity of δ in M (sources 01, 03, 08; source 04 states N only); cardinal arithmetic for $\text{Add}(\kappa, 1)$ (only source 04 assumes it); the range of κ in the cut criterion (every infinite cardinal in source 01, every infinite ordinal in source 03, uncountable in sources 04 and 08). (c) Open questions: those answered inside the cluster are recorded as answered, with the answering source (section 19). (d) Stale repository statements in the sources are corrected in section A. Text written for the merge is confined to this section, the framing remarks that relate sources and reports, remark 10.4, the question statuses, the non-claims ledger and the appendices. It adds no mathematical claim beyond the two-line observation of remark 10.4.

Addition of 28 September 2026. Six further manuscripts of that date, numbered sources 09–14 in this report’s local sequence and pinned to ProveIt commit `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9` (placed in `0e53d1034`), answer parts of questions 19.2, 19.4 and 19.7. They are merged into one addition, Part VIII (part VIII), placed after section 20 so that all earlier section, theorem and question numbers stay fixed; its provenance, bases, notation and non-claims are in sections 21, 32 and 33. Its labels carry the sub-prefix `univ:gs:`.

1.7 Relation to the collection

Surcomplex field automorphisms (`docs/surcomplex/surcomplex-field-automorphisms`, prefix `saut:`). Its Theorem 9.1 (`saut:thm:acfhomogeneity`) is the class back-and-forth used in part V; its Section 10 (`saut:sec:realforms`) proves that involutions are conjugate exactly when their fixed real fields are isomorphic, and constructs a real form with a countable cofinal subset (Theorem 10.2, `saut:thm:inequivalentrealform`). Its Proposition B.1 (`saut:prop:conjugacycriterion`) says that an involution is conjugate to c if and only if its fixed real field has the set-cut interpolation property. None of the four sources cites that proposition. Read in N , its easy direction already gives the non-conjugacy statements of theorems 16.2 and 16.3, because the transported fixed field is isomorphic to $\mathbf{No}^M$, which omits an N -set cut. What is new is recorded in remark 16.4. These results *continue*, but do not settle, that report’s open “real-form and continuity classifications” (its Section 14). Its batch-32 finite-symmetry sections classify the involutions preserving the valuation, Hahn summation and $\mathbf{Oz}[i]$ (note after remark 16.4).

First- κ coefficients (`docs/surcomplex/first-kappa-coefficients`, prefix `fk:`; batch 32). Its Part II builds, in one universe and without forcing, a proper class of pairwise nonconjugate pure-field involutions of $\mathbf{No}[i]$ from fields of surreals with fewer than κ terms; this answers one clause of question 19.7 (status note there).

Foundations (`docs/foundations-and-computation/foundations`, prefix `found:`). Proposition 14.1 is an external version of its Theorem 12.1 (`found:thm:discrete`); see remark 14.4 for the exact relation, including its hypothesis on bounds and the passage of its Section 7.7 on external countability.

Computable surreals (`docs/foundations-and-computation/computable-surreals`). Its `prop:topology` is the universe-relative version of set-indexed convergence, with an index type small in the

lower universe. In the present same-ordinal setting no such smallness is needed, because $\mathbf{No}^M$ is still a proper class of N ; the two statements are consistent.

Birthday cutoffs and hereditary sets (docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets), the sibling report placed in the same batch. It studies bounded birthday fields *with a birthday symbol*, their interpretation of hereditary sets, and their behaviour under outer models. It also proves, for every same-ordinal outer model, that the first new birthday is the least ordinal acquiring a new subset. This report keeps source 08’s statement and proof of that fact, because source 08 adds cardinality in N and the inequality $\delta \leq \text{cf}^N \beta$ (remark 10.5). The two reports study different structures: here the pure ordered field and its external saturation, there the birthday-expanded fields and their interpretations.

Genetic gaps and primitives (docs/surreal/genetic-gaps-and-primitives) uses “gap” for a proper-class gap of $\mathbf{No}$; the gaps here are set-presented gaps of $\mathbf{No}^M$ seen from N . The same report cites Berenbeim’s 2022 dissertation; the 2020 notes answered in section 17 are not cited elsewhere in the collection.

Part I

Foundations and the common spine

2 Foundations, notation, and imported facts

2.1 The two universes

In N , $\mathbf{No}^M$ is a proper class: it contains an old all-plus sign of every ordinal length. The superscript in $\mathbf{No}^M$ means that the sign functions, their construction and the operations are computed in M . It is not an additional set-sized model assumed to have a cardinality. We never form a set of all surreals, a proper-class support, or a class-object whose members are all class maps.

2.2 Signs, birthdays and cones

A surreal number is a sign function $s : \alpha \rightarrow \{-, +\}$ for an ordinal α ; its birthday is $\text{b}(s) = \alpha$. The empty function represents zero. At the first place where two signs differ, numerical comparison uses

$$- < \text{termination} < +.$$

Write $s \preceq t$ when s is an initial segment (prefix) of t , allowing equality. Simplicity is the proper-prefix relation, not numerical order and not merely a smaller birthday. The ordinal α is represented by the all-plus function of length α , written $\hat{\alpha}$ when confusion with field arithmetic is possible. Its surreal successor agrees with $\alpha + 1$ as an ordinal; arbitrary ordinal addition and surreal field addition must not be conflated. Concatenation $\frown$ is ordinal concatenation.

For sets L, R of surreals, $L < R$ means $\ell < r$ for all $\ell \in L$ and $r \in R$. Empty sides are allowed. A separator means a *strict* separator: $L < x < R$. The classical cut theorem supplies the unique simplest separator $\{L \mid R\}$, a prefix of every separator [1, 2, 3]. We use the birthday bound

$$\text{b}(\{L \mid R\}) \leq \sup\{\text{b}(y) + 1 : y \in L \cup R\}, \quad (2.1)$$

with the supremum of the empty set equal to zero. Put

$$\text{Cone}(s) = \{t : s \preceq t\}.$$

Lemma 2.1 (Cones are convex; the prefix characterization; sources 03, 04). (i) *The cone $\text{Cone}(s)$ is numerically convex. That is, if $s \preceq t$ and u lies numerically between s and t , then $s \preceq u$.*

(ii) *Let I be a convex class of surreals, let $s \in I$, and suppose no proper prefix of s belongs to I . Then every $x \in I$ extends s , and s is the unique simplest element of I .*

Proof. (i) Two extensions of s have the same signs before $b(s)$; anything between them cannot first disagree with s at an earlier position.

(ii) If x is a proper prefix of s , the hypothesis excludes it. If neither sequence extends the other, their maximal common prefix u is a proper prefix of s . At the first disagreement one sequence has $-$ and the other $+$, so u lies strictly between them, and convexity gives $u \in I$, a contradiction. The remaining possibility is that x extends s ; such an extension has birthday at least $b(s)$, with equality only for $x = s$. $\square$

Lemma 2.2 (The cone of a sign; source 01). *For $s : \alpha \rightarrow \{-, +\}$ put*

$$L_s = \{s \upharpoonright \gamma : \gamma < \alpha, s(\gamma) = +\}, \quad R_s = \{s \upharpoonright \gamma : \gamma < \alpha, s(\gamma) = -\}.$$

A surreal x satisfies $L_s < x < R_s$ if and only if $s \preceq x$.

Proof. If x extends s , its comparison with each prefix in the displayed sets is decided by the next sign of s , so the strict inequalities hold. Conversely, if x terminates at $\gamma < \alpha$ while agreeing up to that point, then $x = s \upharpoonright \gamma$ violates the corresponding strict inequality. If it first disagrees at $\gamma < \alpha$, compare x with the prefix $s \upharpoonright \gamma$: the opposite signs place x on the wrong side of it. These exhaust the ways in which x could fail to extend s . $\square$

The numerical bounds

$$-\widehat{b(x)} \leq x \leq \widehat{b(x)} \tag{2.2}$$

follow by comparing a sign with the all-minus and all-plus signs of the same length.

2.3 Field theory and types

The full surreal field is real closed, and its extension by a square root of -1 is algebraically closed. Also $\mathbf{No}_{<\lambda}$ is a real closed subfield whenever λ is an epsilon number; in particular for every uncountable cardinal λ , since such cardinals satisfy $\omega^\lambda = \lambda$ in ordinal arithmetic. We use the corrected birthday-fragment results of van den Dries–Ehrlich and their restatement by Bournez–Guilmant [3, 4, 5]. Nothing about exponential functions is needed.

We use quantifier elimination for real closed fields in $\mathcal{L}_{\text{or}} = \{0, 1, +, -, \cdot, <\}$, for algebraically closed fields of characteristic zero in $\mathcal{L}_{\text{fld}} = \{0, 1, +, -, \cdot\}$, and for dense linear orders without endpoints [9]. Embeddings between real closed ordered fields, and between algebraically closed fields of the same characteristic, are elementary. Over a real closed parameter subfield a nonalgebraic one-variable type is determined by an order cut; over an algebraically closed field one-variable types are algebraic or transcendental.

A class structure is κ -saturated (in N) if every complete one-variable type over an N -set of parameters of cardinality less than κ that is finitely satisfiable in the structure is realized there. As usual this gives the finite-variable version: realize the projection of a type in $n + 1$ variables to the first n , then realize the specialized one-type over the enlarged parameter set. All syntactic data and parameter sets at every step are sets, and satisfaction is supplied by quantifier elimination. ω -saturation concerns finite parameter sets; countable parameter sets are ω_1 -saturation. Even a type over the empty set can contain countably many formulas.

An ordered class H has the η_κ property if every pair of N -sets $A, B \subseteq H$ with $A < B$ and $|A \cup B|^N < \kappa$ has a strict separator in H . Empty sides are allowed. Following source 03 we allow an infinite ordinal bound κ when needed.

Lemma 2.3 (Cuts versus saturation; sources 01, 03, 04, 08). (i) *For an uncountable cardinal κ , a real closed ordered field — including the definable class fields used here — is κ -saturated if and only if its order is η_κ .*

(ii) (source 08) *For a dense linear order without endpoints, the same equivalence holds for every infinite κ .*

(iii) (source 04) *In each universe of set theory, its own full surreal field is set-saturated as an ordered field.*

Proof. (i) Saturation realizes the finitely satisfiable partial type of the inequalities of a small cut; a complete extension exists by compactness applied to the set of formulas with those parameters, inside a set-sized real closed subfield containing them.

Conversely, let A have size less than κ . The real closure F of $\mathbb{Q}(A)$ inside the field is a set of cardinality at most $\max(|A|, \aleph_0) < \kappa$; there are set many polynomials over $\mathbb{Q}(A)$ and finitely many roots of each. A complete type over A extends to one over F that is finitely satisfiable in the ambient field. If the extended type is algebraic, it is realized in F . Otherwise it declares, for every $a \in F$, exactly one of $a < X$ and $X < a$; these inequalities form a separated cut of size less than κ . A separator y realizes the entire type: every nonzero polynomial over the real closed F factors into linear factors and irreducible quadratics; the quadratics have constant sign, and the signs of the linear factors are fixed by the cut. Quantifier elimination completes the argument. Restrict back to A .

(ii) A complete one-type of a dense linear order over A either says $X = a$ for some $a \in A$ or specifies a cut of A . There is no countable field closure to add, so the argument works even at $\kappa = \omega$.

(iii) The cut theorem realizes every set-sized separated cut; apply the proof of (i) with no cardinal bound. $\square$

The uncountability hypothesis in (i) cannot be dropped: the real closure of even the empty parameter set is countable, not finite. Every dense order without endpoints has η_ω , whereas a real closed field need not be ω -saturated (section 7).

3 Absoluteness and ground-set bounds

Proposition 3.1 (Old arithmetic and elementary inclusions; sources 01, 03, 04, 08). (i) $\mathbf{No}^M = \mathbf{No}^N \cap M$ is an initial subclass of the sign tree: every prefix of an old sign is old. Birthdays and numerical comparison of old signs are absolute.

(ii) If $L, R \in M$ are sets of old surreals with $L < R$, then $\{L \mid R\}^M = \{L \mid R\}^N$.

(iii) The inclusion $\mathbf{No}^M \subseteq \mathbf{No}^N$ preserves zero, one, negation, addition, multiplication and inversion. Every ordinal is old, so $\mathbf{No}^M$ is cofinal and cointial in $\mathbf{No}^N$.

(iv) In N , $\mathbf{No}^M$ is real closed and $\mathbf{SC}^M$ is algebraically closed, and

$$\mathbf{No}^M \preccurlyeq \mathbf{No}^N \quad \text{in } \mathcal{L}_{\text{or}}, \quad \mathbf{SC}^M \preccurlyeq \mathbf{SC}^N \quad \text{in } \mathcal{L}_{\text{fld}}.$$

Proof. (i) Being an ordinal-length function into $\{-, +\}$, its domain, and first disagreement comparison are absolute for transitive models. If $s \in M$ and $\alpha \leq b(s)$, then $\alpha \in M$ and $s \upharpoonright \alpha \in M$.

(ii) Let $u = \{L \mid R\}^M$. It still separates L, R in N . The simplest separator v computed in N is a prefix of u by the simplicity theorem in N , hence old by (i), hence an available separator in M , so $u \preceq v$ by the

simplicity theorem in M . Thus $u = v$ (source 08). Equivalently, every proper prefix of u fails in M to separate, witnessed by an old element of L or R , and that witness still works in N ; lemma 2.1(ii) applied to the convex class of N -separators gives the same conclusion (source 04). The argument does not minimize birthdays over old candidates, which would not be absolute since N can contain new short signs; the prefix witnesses exclude new candidates.

(iii) Conway's addition and multiplication are defined from canonical options by well-founded, set-like recursion. For old inputs the options are old, the earlier recursive values agree by induction, the option families belong to M , and (ii) identifies the resulting cuts. Every recursion needed for particular inputs has a set-sized dependency closure; no class-recursion principle is assumed. The inverse of an old nonzero number remains its inverse by uniqueness. A large all-plus or all-minus ordinal lies above or below any given new surreal.

(iv) A polynomial over $\mathbf{No}^M$ has a finite old coefficient tuple, which belongs to M . In M positive elements have square roots and odd-degree polynomials have roots, and those witnesses work in N . Thus $\mathbf{No}^M$ is externally real closed, and the same finite-polynomial argument gives external algebraic closedness of $\mathbf{SC}^M$. Quantifier elimination gives both elementary inclusions. $\square$

This proposition asserts elementarity of the *fields*. It does not assert $M \preccurlyeq N$ as models of set theory, which is generally false for a forcing extension. “Initial” is a simplicity-tree assertion: $\mathbf{No}^M$ is not a numerical interval of $\mathbf{No}^N$, and it need not contain a new surreal whose birthday is below some old birthday.

Lemma 3.2 (Every external set of old numbers is ground-bounded; sources 01, 04, 08). (i) *If A is an N -set and $A \subseteq \mathbf{No}^M$, there are an ordinal α and a ground set $B \in M$ with $A \subseteq B = (\mathbf{No}_{<\alpha})^M$. Every N -set of elements of $\mathbf{SC}^M$ is likewise contained in a ground set of old surcomplex numbers, and these ground sets have well-orderings in M .*

(ii) *Such an A is contained in a set-sized real closed subfield $E \in M$ of $\mathbf{No}^M$, and an N -set of elements of $\mathbf{SC}^M$ in a set-sized algebraically closed subfield $E[i] \in M$; these are elementary substructures. There is an old element transcendental over any such $E[i]$.*

Proof. (i) In N form $\alpha = \sup\{b(a) + 1 : a \in A\}$. It belongs to M because the universes have the same ordinals. The set of old signs of length below α exists in M by Power Set and Replacement. For $\mathbf{SC}^M$ bound both coordinates. Choice in M gives the well-orderings.

(ii) Inside M , take the field generated by the cover from (i) and its relative real algebraic closure in $\mathbf{No}^M$; there are set many polynomials and finitely many roots of each. Quantifier elimination gives elementarity. An all-plus ordinal above every element of E is not in E , hence transcendental over E (a real closed subfield is relatively algebraically closed in an ordered extension), hence over $E[i]$. $\square$

The lemma is not a covering property of a forcing notion, and it does not assert $A \in M$. It holds even when a forcing badly fails familiar small-set covering properties, and the containing ground set can be vastly larger than A . It works because a set of birthdays is bounded and the two universes have the same ordinals.

Lemma 3.3 (Old bounds for new sets; source 03). *Every N -set of elements of $\mathbf{No}^N$ has an upper bound that is an old ordinal. Consequently every N -set of old numbers is bounded above and below in $\mathbf{No}^M$, and $\mathbf{No}^M$ has no set-sized cofinal or coinital subset in N . Old positive elements are coinital in $\mathbf{No}_{>0}^N$.*

Proof. For a set $X \subseteq \mathbf{No}^N$, Replacement bounds all birthdays by an ordinal θ . The all-plus ordinal $\theta + 1$ exceeds every member of X : at any earlier minus sign it wins the first comparison, and it exceeds an all-plus sequence of smaller length. It is old. Apply the same argument to $-X$. For $r \in \mathbf{No}_{>0}^N$ choose an old positive ordinal $\gamma > 1/r$; then $0 < 1/\gamma < r$ and $1/\gamma \in \mathbf{No}^M$. $\square$

Lemma 3.4 (Coding short families; sources 01, 03, 04). *Assume $\text{NS}_{<\kappa}(M, N)$. Every N -set of old surreal or surcomplex numbers of cardinality less than κ is a member of M . More generally, every sequence of length less than κ with values in a fixed ground set is a ground sequence, and every N -subset of cardinality less than κ of a ground set belongs to M .*

Proof. For a ground set B fix in M an injection $e : B \rightarrow \theta$ into an ordinal. Composing a short sequence in B with e gives an ordinal-valued sequence of the same length, hence an old one, and decoding with e shows that the original sequence is old. For an external set A of size less than κ , use lemma 3.2 and an enumeration of A in N of length less than κ ; the enumeration is old, and so is its range. $\square$

The following asymmetric fact is the efficient direction of the main theorem.

Theorem 3.5 (One old side suffices; source 01). *Let L, R be N -sets contained in $\mathbf{No}^M$ with $L < R$. If either $L \in M$ or $R \in M$, then some $x \in \mathbf{No}^M$ satisfies $L < x < R$.*

Proof. Suppose $L \in M$. Choose a ground set $B \supseteq R$ by lemma 3.2 and define, in M , $U = \{b \in B : (\forall \ell \in L) \ell < b\}$. Order is absolute, so $R \subseteq U$ and $L < U$. The ground cut $x = \{L \mid U\}$ satisfies $L < x < U$, and therefore $L < x < R$. This also covers empty sides. If $R \in M$, apply the proved case to $-R < -L$ and negate the separator. $\square$

Corollary 3.6 (Source 01). *If $\text{NS}_{<\kappa}(M, N)$ holds, every external set-sized cut with at least one side of size less than κ is filled in $\mathbf{No}^M$.*

Proof. The small side belongs to M by lemma 3.4; apply theorem 3.5. $\square$

In particular, every N -set of positive elements of $\mathbf{No}^M$ has a positive lower bound in $\mathbf{No}^M$: take $L = \{0\}$. Every N -set of old numbers has an old strict upper bound. These statements survive every extension under consideration, including ones that destroy countable saturation.

4 The first new sequence length

Definition 4.1. When $M \neq N$, the *first new length* is

$$\delta(M, N) = \min\{\lambda : \exists \theta \ (^{\lambda}\theta)^N \not\subseteq M\}. \quad (4.1)$$

No value is assigned when $M = N$. A sequence $f : \lambda \rightarrow \text{Ord}$ in N is *fresh over M* (source 04) if $f \notin M$ but $f \upharpoonright \xi \in M$ for every $\xi < \lambda$. This is the forcing-theoretic notion of a fresh function [14]; the name is not claimed.

The minimum exists. *First route* (sources 01, 03). Every set of N can be coded in N by a well-founded extensional relation on an ordinal together with a marked point, and such a relation is coded by a set of ordinals. If all sets of ordinals were old, the code and its Mostowski collapse would be old, so $M = N$. *Second route* (sources 04, 08). Choose a new set x of least rank α . Every set of rank below α is old, so $x \subseteq V_\alpha^M$, and an M -well-ordering of V_α^M identifies x with a new subset of an ordinal. Either way, a new set of ordinals gives a new characteristic sequence.

No finite sequence of old ordinals is new, and a sequence of successor length is determined by its shorter prefix and one old ordinal. Hence $\delta(M, N)$ is an infinite limit ordinal, and every new sequence of length δ is fresh. More generally, a fresh sequence has infinite limit length.

Proposition 4.2 (Regularity of the first new length; sources 01, 03, 04, 08). *The ordinal $\delta(M, N)$ is an infinite regular cardinal in both M and N .*

Proof. Write $\delta = \delta(M, N)$ and choose a new $f : \delta \rightarrow \theta$.

If δ were not an N -cardinal, choose in N a bijection $e : \mu \rightarrow \delta$ with $\mu < \delta$. By minimality $e \in M$, and $f \circ e$, also of length $\mu < \delta$, belongs to M . Decoding with e gives $f \in M$, a contradiction.

If $\text{cf}^N(\delta) = \mu < \delta$, choose a cofinal $e : \mu \rightarrow \delta$; it is old by minimality. Every $f \upharpoonright e(\xi)$ is old and lies in the ground set $T = ({}^{<\delta}\theta)^M$. Well-order T in M . The sequence of codes of $f \upharpoonright e(\xi)$, $\xi < \mu$, has length less than δ and is old; its ground decoding and union reconstruct f , a contradiction. Hence δ is regular in N . A ground bijection or a shorter ground cofinal sequence would survive to N , so δ is a regular cardinal of M as well. $\square$

Sources 01, 03 and 08 state regularity in both universes; source 04 states the N clause, and its proof gives the M clause by the last sentence above. This is an elementary minimal-length fact about extensions, not a new characterization of regular cardinals and not a new forcing invariant.

Remark 4.3 (Ordinal sequences, not binary ones; source 08). Ordinal-valued sequences in (4.1) must not be replaced by subsets of small ordinals. A new countable cofinal sequence in a large cardinal may exist without any new real; Prikry forcing is the decisive example (section 13.4). The codings below preserve the number of bounds rather than encode large ordinal entries into a short binary string.

5 Coding a fresh sequence by an omitted cut

The converse direction of the main theorem needs more than the observation that a new binary sign has a new cut. A short sequence can take values in a very large ordinal: a new countable cofinal sequence can appear while no real is added. Encoding it as one binary sign might require a long ordinal domain, which would destroy the cardinal bound on the cut. Each construction below keeps the number of inequalities equal to the length of the sequence, independently of the size of its values. Section 5.1 is the main route (source 01); sections 5.2 and 5.3 are marked second routes (sources 04 and 08, with source 03's variant in proposition 9.1).

5.1 Main route: an arbitrarily branching interval tree (source 01)

Lemma 5.1 (Arbitrarily branching interval tree). *Let $\theta \geq 2$ be an ordinal and λ a nonzero limit ordinal. In M there is a set-sized table of pairs*

$$(a_s, b_s) \in (\mathbf{No}^M)^2, \quad s \in T := ({}^{<\lambda}\theta)^M,$$

such that:

- (i) $a_\emptyset = 0 < b_\emptyset = 1$;
- (ii) if s is a proper initial segment of t , then $a_s < a_t < b_t < b_s$;
- (iii) the intervals of distinct immediate successors of one node are pairwise disjoint and ordered by their successor labels.

Proof. Recurse on the levels below λ . Given (a_s, b_s) put $w_s = b_s - a_s > 0$, regard ξ and θ as their all-plus surreal images, and set

$$a_{s \smallfrown \xi} = a_s + w_s \frac{3\xi + 1}{3\theta + 3}, \quad b_{s \smallfrown \xi} = a_s + w_s \frac{3\xi + 2}{3\theta + 3} \quad (\xi < \theta). \quad (5.1)$$

Every operation here is surreal field arithmetic. If $\xi < \eta < \theta$, the equality of ordinal and surreal successor gives $\xi + 1 \leq \eta$, hence $0 < 3\xi + 1 < 3\xi + 2 < 3\eta + 1 < 3\eta + 2 < 3\theta + 3$. This proves the interior and sibling-separation properties.

At a nonzero limit level $\gamma < \lambda$ and $s \in (\gamma\theta)^M$, the induction hypothesis gives separated ground sets $A_s = \{a_{s \upharpoonright \epsilon} : \epsilon < \gamma\}$ and $B_s = \{b_{s \upharpoonright \epsilon} : \epsilon < \gamma\}$. Define inside M

$$a_s = \{A_s \mid B_s\}, \quad b_s = \{A_s \cup \{a_s\} \mid B_s\}. \quad (5.2)$$

Then $A_s < a_s < b_s < B_s$. Each level and its prefix data are sets in M ; Replacement and set-length recursion produce the table in M . No recursion along all ordinals is used. $\square$

Theorem 5.2 (Branch decoder and omitted cut). *Suppose $f : \lambda \rightarrow \theta$ belongs to $N \setminus M$, where λ is a nonzero limit ordinal and every proper initial segment $f \upharpoonright \alpha$ belongs to M . Then $\mathbf{No}^M$ omits the finitely satisfiable partial type*

$$p_f(X) = \{a_{f \upharpoonright \alpha} < X < b_{f \upharpoonright \alpha} : \alpha < \lambda\}. \quad (5.3)$$

It uses at most $|\lambda|$ parameters on each side; the left endpoints strictly increase and the right endpoints strictly decrease.

Proof. All endpoints are old. A finite subset of the inequalities is satisfied by the midpoint of its deepest interval, which is old. Strict nesting proves monotonicity.

Suppose an old x realizes all inequalities. Using only the ground table and x , define $\text{dec}_x : \lambda \rightarrow \theta$ by recursion in M : at a successor step, with prefix s constructed, choose the unique $\xi < \theta$ with $a_{s \frown \xi} < x < b_{s \frown \xi}$ if it exists, and 0 otherwise; at limits take unions. Sibling intervals are disjoint, so the choice is unique, and the default value makes this a *total ground recursion* that does not presuppose a branch in M .

In N , induction gives $\text{dec}_x \upharpoonright \alpha = f \upharpoonright \alpha$ for every $\alpha < \lambda$: at the next step x lies in the interval of $f \upharpoonright (\alpha + 1)$ by the type, and $\alpha + 1 < \lambda$ because λ is limit. Thus $\text{dec}_x = f$, contradicting $\text{dec}_x \in M$ and $f \notin M$. $\square$

The proof does not infer that an external branch is old because all its nodes are old; that inference would be false. An assumed *old realizer* supplies a total, deterministic ground decoder for the whole branch. There is also no assertion that the interval widths tend to zero in the fine topology; section 14 shows they have a common positive old lower bound.

5.2 Second route A: an absolute all-ordinal interval code (source 04)

The tree of lemma 5.1 fixes an alphabet bound θ and is built level by level inside M . Source 04's code instead lets the children of a node be indexed by *all* ordinals and defines the interval of each individual sequence s by set recursion along $\text{dom}(s) + 1$, uniformly in whichever universe contains s . At the root $a_\emptyset = 0$, $b_\emptyset = 1$. With $w_s = b_s - a_s$ and α an ordinal,

$$a_{s \frown \langle \alpha \rangle} = a_s + \frac{w_s}{2\hat{\alpha} + 3}, \quad b_{s \frown \langle \alpha \rangle} = a_s + \frac{w_s}{2\hat{\alpha} + 2}, \quad (5.4)$$

all operations being surreal field operations. At a nonzero limit length $\gamma = \text{dom}(s)$,

$$u_s = \{a_{s \upharpoonright \xi} : \xi < \gamma \mid b_{s \upharpoonright \xi} : \xi < \gamma\}, \quad e_s = \{0 \mid u_s - a_{s \upharpoonright \xi}, b_{s \upharpoonright \xi} - u_s : \xi < \gamma\}, \quad (5.5)$$

and $a_s = u_s - e_s/2$, $b_s = u_s + e_s/2$; the right option set of e_s is the union of the two displayed families. Write $I_s = (a_s, b_s)$.

Theorem 5.3 (Absolute ordinal-branching interval code; source 04). (i) *For every proper extension t of s , $[a_t, b_t] \subset (a_s, b_s)$.*

(ii) *If $\alpha < \beta$ then $b_{s \smallfrown \langle \beta \rangle} < a_{s \smallfrown \langle \alpha \rangle}$; distinct children have disjoint closed intervals.*

(iii) *Every nonroot closed interval lies inside $(0, 1)$.*

(iv) *If $s \in M$, both endpoints belong to M and have the same values when computed in M or in N .*

For a fixed input s the construction needs only set recursion along $\text{dom}(s) + 1$; it forms no set of all nodes at a level and no class-valued transfinite recursion.

Proof. At a successor node both denominators in (5.4) exceed one, with $2\widehat{\alpha} + 3 > 2\widehat{\alpha} + 2 > 0$, so $a_s < a_{s \smallfrown \langle \alpha \rangle} < b_{s \smallfrown \langle \alpha \rangle} < b_s$. If $\alpha < \beta$ then $\widehat{\beta} \geq \widehat{\alpha} + 1$, hence $\frac{1}{2\widehat{\beta}+2} \leq \frac{1}{2\widehat{\alpha}+4} < \frac{1}{2\widehat{\alpha}+3}$, which is (ii). At a limit stage the earlier closed intervals are strictly nested, so the cut u_s is legitimate and lies strictly inside every earlier interval; all right options of e_s are positive, so $0 < e_s$ is smaller than each of those distances, and $a_{s \upharpoonright \xi} < u_s - e_s/2 < u_s + e_s/2 < b_{s \upharpoonright \xi}$. This proves (i)–(iii) by induction. For (iv), run the recursion on the prefixes of the given s : successor steps use absolute arithmetic (proposition 3.1(iii)) and limit steps use absoluteness of old cuts (proposition 3.1(ii)) twice; all option sets belong to M by Replacement in M . The values on the prefixes of one input form a set-length sequence of sets, constructed by the ordinary recursion theorem. $\square$

Theorem 5.4 (Fresh-branch omitted cut; source 04). *Let $f : \mu \rightarrow \text{Ord}$ be fresh over M , and put $A_f = \{a_{f \upharpoonright \xi} : \xi < \mu\}$, $B_f = \{b_{f \upharpoonright \xi} : \xi < \mu\}$. Then A_f, B_f are N -sets of old surreals of cardinality at most $|\mu|^N$, $A_f < B_f$, and their strict cut is finitely satisfiable in $\mathbf{No}^M$ but has no separator in $\mathbf{No}^M$. Removing the root endpoints leaves an omitted cut inside $(0, 1)$.*

Proof. All endpoints are old by theorem 5.3(iv), and nesting gives $A_f < B_f$; the midpoint of a deep enough $I_{f \upharpoonright \eta}$ satisfies finitely many constraints. Suppose $x \in \mathbf{No}^M$ separates the whole cut, so $x \in I_{f \upharpoonright \xi}$ for all $\xi < \mu$. Define $g : \mu \rightarrow \text{Ord}$ by set recursion in M : given $g \upharpoonright \xi$, let $g(\xi)$ be the unique ordinal α with $x \in I_{(g \upharpoonright \xi) \smallfrown \langle \alpha \rangle}$ if one exists, and 0 otherwise. The existential quantifier ranges over ordinals, not over a set of all children, and uniqueness comes from (ii). By induction in N , $g(\xi) = f(\xi)$: since μ is a limit, $x \in I_{f \upharpoonright (\xi+1)} = I_{(g \upharpoonright \xi) \smallfrown \langle f(\xi) \rangle}$, an interval computed absolutely from old data. Hence $g = f \in M$, a contradiction. For the last clause omit the index 0 on both sides. $\square$

Proposition 5.5 (Closed formulas along a finite branch; source 04). *For an ordinal α put $r_\alpha = \frac{1}{2\widehat{\alpha}+3}$ and $\ell_\alpha = \frac{1}{(2\widehat{\alpha}+2)(2\widehat{\alpha}+3)}$. If $s = \langle \alpha_0, \dots, \alpha_{n-1} \rangle$, then*

$$a_s = \sum_{j < n} r_{\alpha_j} \prod_{k < j} \ell_{\alpha_k}, \quad b_s = a_s + \prod_{k < n} \ell_{\alpha_k},$$

with the empty product one and the empty sum zero.

Proof. The child interval is the image of $(0, 1)$ under $t \mapsto a_s + w_s(r_\alpha + \ell_\alpha t)$, since $\frac{1}{2\widehat{\alpha}+2} - \frac{1}{2\widehat{\alpha}+3} = \ell_\alpha$. Induct on n . $\square$

For a fresh sequence of length ω , such as a Prikry sequence (section 13.4), all endpoints of the omitted cut are given by these finite rational expressions in the embedded ordinals; no limit-stage cut is needed to write them down. The absence of an old separator is still an infinite assertion, proved by the decoder, not by finite experimentation.

5.3 Second route B: ordinal block codes of signs (source 08)

Here the fresh sequence is written into a single sign sequence whose prefixes provide the bounds.

Lemma 5.6 (Two bounds force a prefix; source 08). *For every sign sequence u , the numerical interval $(u \frown -, u)$ equals $\text{Cone}(u \frown - \frown +)$.*

Proof. An extension of $u \frown - \frown +$ is below u , since its first sign after u is minus, and above $u \frown -$, since its first sign after that is plus. Conversely, if $u \frown - < y < u$ and y first differed from or terminated before the end of u , its comparisons with u and with $u \frown -$ would agree, contradicting the two inequalities. So $u \preceq y$; the upper inequality forces the next sign to be minus, and the lower inequality, excluding $y = u \frown -$, forces the following sign to be plus. $\square$

Let λ be an infinite limit ordinal and $f : \lambda \rightarrow \text{Ord}$. By set recursion put $p_0 = \emptyset$, $p_{\xi+1} = p_\xi \frown +^{f(\xi)} \frown - \frown +$, and $p_\eta = \bigcup_{\xi < \eta} p_\xi$ at limits, where $+^\alpha$ is a run of plus signs of ordinal length α . Set

$$\text{code}(f) = p_\lambda, \quad u_\xi = p_\xi \frown +^{f(\xi)}, \quad l_\xi = u_\xi \frown -. \quad (5.6)$$

Lemma 5.7 (Decoding and nesting; source 08). *$\text{code}(f)$ determines f and its block boundaries by a set-recursive decoder. For $\xi < \eta < \lambda$, $l_\xi < l_\eta < \text{code}(f) < u_\eta < u_\xi$, and $\bigcap_{\xi < \lambda} (l_\xi, u_\xi) = \text{Cone}(\text{code}(f))$. These assertions are absolute for transitive universes containing f .*

Proof. From a block boundary, read the run of plus signs up to the next minus; its order type is $f(\xi)$. Consume that minus and the following plus; the next block begins there, and at limits take suprema of boundaries. By lemma 5.6, $(l_\xi, u_\xi) = \text{Cone}(p_{\xi+1})$. Every later endpoint and $\text{code}(f)$ extend $p_{\xi+1}$; comparing at the final plus marker of block η gives $l_\eta < \text{code}(f)$, and at its minus marker $\text{code}(f) < u_\eta$. A sequence extends every $p_{\xi+1}$ iff it extends their union. $\square$

Theorem 5.8 (Fresh-sequence cut and its separators; source 08). *Let $f : \lambda \rightarrow \text{Ord}$ be fresh over M , λ an infinite limit ordinal. The endpoints l_ξ, u_ξ are old, but the cut $L_f = \{l_\xi : \xi < \lambda\}$, $R_f = \{u_\xi : \xi < \lambda\}$ has no separator in $\mathbf{No}^M$. In $\mathbf{No}^N$ its class of separators is exactly $\text{Cone}(\text{code}(f))$. Every proper prefix of $\text{code}(f)$ is old, while $\text{code}(f)$ is new. If $\lambda = \delta(M, N)$, the cut determines a set-presented gap of $\mathbf{No}^M$ of exact cofinality pair (λ, λ) , and $\text{cf}^N \text{b}(\text{code}(f)) = \lambda$. (For an arbitrary fresh sign, the cofinalities of its gap are computed in lemma 8.3.)*

Proof. Since λ is a limit, each $f \upharpoonright (\xi + 1)$ is old, and computing its code inside M gives $p_{\xi+1}, u_\xi, l_\xi$. The two endpoint sets need not belong to M . By lemma 5.7 the separators in $\mathbf{No}^N$ are the extensions of $\text{code}(f)$. An old separator y would give $\text{code}(f) \preceq y$, hence $\text{code}(f) \in M$ by initiality, and decoding inside M would recover f . The same decoding shows $\text{code}(f) \notin M$. Any proper prefix of $\text{code}(f)$ is a restriction of some old p_α .

For the gap, recall that $\lambda = \delta$ is a regular cardinal of N (proposition 4.2). The classes $D^- = \{y \in \mathbf{No}^M : \exists \xi y \leq l_\xi\}$ and $D^+ = \{y \in \mathbf{No}^M : \exists \xi y \geq u_\xi\}$ partition $\mathbf{No}^M$, since a point in neither would separate the cut; strict nesting shows that neither has an endpoint toward the other. A cofinal $X \subseteq D^-$ of size below λ is impossible: its indices ξ_x with $x \leq l_{\xi_x}$ are bounded below λ by regularity, and a later l_η exceeds X ; symmetrically on the right. Finally $\text{b}(\text{code}(f)) = \sum_{\xi < \lambda} (f(\xi) + 2)$; its increasing partial sums are cofinal, and fewer than λ cofinal ordinals would, by regularity, bound the partial-sum indices below λ . $\square$

The distinction between a family of old endpoints and an old *set* of endpoints is the whole mechanism: internal saturation fills cuts whose bounding sets are in M , and the cut constructed here is an external

set of old bounds. Each possibly enormous value $f(\xi)$ is stored in a single old sign prefix; what is compressed is the *number of parameters*, not the birthdays of their values. Source 03 uses the variant $\text{code}^+(f) = \cap_{\xi} (+^{f(\xi)+1} \cap -)$, with endpoints read off before the last plus and before the minus of each block; it is printed with its proof in proposition 9.1, where it serves the least-gap theorem. Both codes use ordinal sums, not the natural sums that describe addition of ordinal surreals.

6 The exact spectrum and its first gap

6.1 Proof of the main equivalence

Proof of theorem 1.1. (i) $\Rightarrow$ (ii) is corollary 3.6, and (ii) $\Rightarrow$ (iii) is immediate. Suppose (i) fails. Choose the least ordinal $\lambda < \kappa$ for which there is a new ordinal-valued sequence of length λ , choose one such sequence, and bound its range by an ordinal $\theta \geq 2$. As observed in section 4, λ is an infinite limit ordinal and the sequence is fresh. Theorem 5.2 produces an unfilled cut whose two sides together have cardinality at most $|\lambda| < \kappa$ in N , so (iii) fails; theorem 5.4 gives one inside $(0, 1)$. For uncountable κ , proposition 3.1 and lemma 2.3(i) identify (iii) with ordered-field κ -saturation, and lemma 2.3(ii) with κ -saturation of the dense order. $\square$

κ need not be regular. Only the conversion to real closed saturation requires $\kappa > \omega$; at $\kappa = \omega$ both sides of (i) $\Leftrightarrow$ (iii) hold trivially (source 08), since finite sequences of ordinals and finite sets of old numbers are old. Source 03 states the cut equivalence for every infinite *ordinal* bound:

$$\mathbf{No}^M \text{ has } \eta_{\kappa} \iff \forall \alpha < \kappa \ (\alpha \text{Ord})^N \subseteq M, \quad (6.1)$$

where the right side is shorthand for the class statement that every ordinal-valued sequence of length α in N belongs to M . The proof is the same: a new sequence of length below κ gives $\delta < \kappa$ and an omitted pair of size δ (theorem 6.2).

6.2 Exact two-sided cofinalities

Definition 6.1 (Set-presented gaps). A *set-presented gap* of $\mathbf{No}^M$ is a partition $\mathbf{No}^M = D \sqcup U$ with $D < U$, neither D having a greatest nor U a least element, together with N -sets $A \subseteq D$ cofinal in D and $B \subseteq U$ coinital in U . Under (H) a gap is given by such a presentation (A, B) , and $D = \{x \in \mathbf{No}^M : \exists a \in A \ x \leq a\}$, $U = \mathbf{No}^M \setminus D$ are defined from it; statements about all gaps are schemes over presentations. Its *cofinality pair* ($\text{cf}^N D, \text{ci}^N U$) is the pair of least sizes, computed in N , of cofinal and coinital N -subsets; these are infinite regular cardinals. D and U themselves are never required to be sets. This is source 01's "set-character gap", source 03's "set-presented gap" and source 08's "Dedekind gap" with its character.

Theorem 6.2 (First gap theorem; sources 01, 03, 08). *Let $M \neq N$ and $\delta = \delta(M, N)$. Every set-presented gap of $\mathbf{No}^M$ has both cofinalities at least δ , and $\mathbf{No}^M$ has one of exact cofinality pair (δ, δ) .*

Proof. Apply theorem 5.2 to a new sequence of minimal length δ and write its endpoints as a_{α}, b_{α} . Define in N $D = \{x \in \mathbf{No}^M : \exists \alpha < \delta \ (x \leq a_{\alpha})\}$ and $U = \{x \in \mathbf{No}^M : \exists \alpha < \delta \ (b_{\alpha} \leq x)\}$. A point in neither class would realize the omitted type, so $D \cup U = \mathbf{No}^M$ and $D < U$; strict monotonicity and the limit length show that neither class has an endpoint toward the other. The endpoint sequences are witnesses of size δ .

For the lower bound, take any set-presented gap with witnesses A and B . If either could have size less than δ , theorem 1.1(ii) at $\kappa = \delta$ would give an old strict separator between A and B . A point of the lower class cannot lie strictly above a cofinal subset of it, and a point of the upper class cannot lie strictly below a coinital subset of it; since the classes partition $\mathbf{No}^M$, this is impossible. $\square$

The block code gives a second proof of the upper bound, with the exact separator class $\text{Cone}(\text{code}(f))$ (theorem 5.8, source 08).

Corollary 6.3 (Recovering the threshold from the order; sources 01, 03, 08). *For $M \neq N$, $\delta(M, N)$ is the least cardinal μ for which $\mathbf{No}^M$ has an unfilled set-sized cut with both sides of size at most μ ; equivalently, the least cofinality on either side of a set-presented gap of $\mathbf{No}^M$.*

Proof. The upper bounds follow from theorem 6.2. A smaller unfilled cut contradicts the main theorem at δ , and a smaller gap cofinality contradicts the first gap theorem. $\square$

Corollary 6.4 (The uncountable spectrum; sources 01, 03, 04, 08). *For uncountable N -cardinals κ , $\mathbf{No}^M$ is κ -saturated exactly when $\kappa \leq \delta(M, N)$. Thus if $\delta > \omega$ it is δ -saturated but not $(\delta^+)^N$ -saturated, and if $\delta = \omega$ it is not ω_1 -saturated. $\mathbf{No}^M$ is externally set-saturated if and only if $M = N$ (source 04).*

Proof. There are no new sequences of length less than δ , and there is one of length δ ; apply the main theorem. If $M = N$, use lemma 2.3(iii); if $M \neq N$, any uncountable cardinal above δ fails. $\square$

That $\mathbf{No}^M$ has no smaller set-presented gap does not exclude cuts with proper-class cofinality on one side. Those are outside the invariant and outside the main interpolation theorem. The definition of the invariant uses only the order of $\mathbf{No}^M$, external sets and their cardinalities, yet its value is the first length at which N has new ordinal data. Since a real closed field has a unique field order, it is preserved even by isomorphisms stated only in the field language (source 03).

7 The countable boundary: no new reals is weaker

The absence of new reals and the absence of new countable ordinal-valued sequences are different preservation properties, and the old surreal field records the difference exactly.

Theorem 7.1 (Finite-parameter saturation; sources 01, 03, 04, 08).

$$\mathbf{No}^M \text{ is } \omega\text{-saturated in } N \iff \mathcal{P}(\omega)^M = \mathcal{P}(\omega)^N \iff \mathbb{R}^M = \mathbb{R}^N.$$

In contrast, the pure order $(\mathbf{No}^M, <)$ is always ω -saturated (source 08).

Proof. Equality of the power sets is equivalent to equality of the reals by ordinary countable coding in both transitive universes.

Suppose no new subsets of ω are added. Let A be a finite set of old parameters and p a complete one-type over A finitely satisfiable in $\mathbf{No}^M$ according to N . The finite set A is in M , the formulas over A have a countable enumeration in M , and the code of p is a subset of ω , so $p \in M$. For each finite $p_0 \subseteq p$, solvability of its conjunction is an existential ordered-field formula over old parameters, with the same truth value in M and N by proposition 3.1 and quantifier elimination. Hence M recognizes p as finitely satisfiable, and in M the full surreal field realizes every set-sized type (lemma 2.3(iii)).

Conversely let $r \in \mathbb{R}^N \setminus \mathbb{R}^M$ and consider the rational-cut partial type

$$\{q < X : q \in \mathbb{Q}, q < r\} \cup \{X < q : q \in \mathbb{Q}, r < q\}. \quad (7.1)$$

It is finitely satisfiable by rationals, and it has no parameters, since $m/n < X$ is equivalent to $m < nX$ for $n > 0$. If an old x realized it, the ground set $\{q \in \mathbb{Q} : q < x\}$ would be the lower rational cut of r , and ground Dedekind completeness would put r in M . For a complete omitted type, take the type of the image of r in $\mathbf{No}^N$

over the empty set; every finite part is realized in $\mathbf{No}^M$ because $\mathbf{No}^M \preceq \mathbf{No}^N$. No class compactness principle is required. Source 04 gives the same argument with the dyadic prefixes q_n of a new sign $s : \omega \rightarrow \{-, +\}$, using the parameter-free formulas $q_n < X$ for $s(n) = +$ and $X < q_n$ for $s(n) = -$.

Finally, all finite cuts of the old order are filled, so lemma 2.3(ii) gives ω -saturation of the pure order. $\square$

Corollary 7.2 (Source 01). *An extension that adds no reals but adds a countable ordinal-valued sequence leaves $\mathbf{No}^M$ ω -saturated and destroys ω_1^N -saturation. It creates an (ω, ω) set-presented gap in $\mathbf{No}^M$, but none of its omitted types can be witnessed over a fixed finite parameter set.*

Proof. The first new length is ω . Apply theorems 6.2 and 7.1 and corollary 6.4. The last assertion is the surviving ω -saturation. $\square$

A new countable ordinal-valued sequence need not code a new real. The new countable parameter family in the interval tree carries information that no old finite parameter tuple can code; a bare new-real argument would miss this example. One must not set $\kappa = \omega$ in the field-saturation clause of theorem 1.1: the real closure of a finite parameter set is usually countable, not finite. This is not a conflict between two definitions of saturation but an exact distinction between finite and countable parameter sets in the same language.

Part II

Fresh signs classify the set-presented gaps (source 03)

8 The gap–fresh-sign correspondence

This part is source 03's. It classifies *all* set-presented gaps of $\mathbf{No}^M$, not only the first. It answers source 01's question about asymmetric cofinality pairs (section 19). Its proofs use only the sign order, prefix-initiality, the old constant-tail closure and the set-cut principle in N ; they need no Global Choice.

Definition 8.1 (Fresh signs). *A fresh sign over M in N is $s \in \mathbf{No}^N \setminus \mathbf{No}^M$ such that $s \upharpoonright \alpha \in \mathbf{No}^M$ for every $\alpha < b(s)$. Write $\text{Fresh}(M, N)$ for the class of fresh signs. This is the binary special case of a fresh function (definition 4.1); the forcing notion of freshness and its spectra are due to Fischer–Koelbing–Wohofsky [14], and the name is not claimed as a new invariant. For $s \in \mathbf{No}^N \setminus \mathbf{No}^M$ put*

$$D_s = \{x \in \mathbf{No}^M : x < s\}, \quad U_s = \{x \in \mathbf{No}^M : s < x\}.$$

Write $\text{GapSpec}_N(\mathbf{No}^M)$ for the class of cofinality pairs of set-presented gaps of $\mathbf{No}^M$ (definition 6.1), computed in N .

Lemma 8.2 (Both signs are cofinal). *If $s \in \text{Fresh}(M, N)$ and $\lambda = b(s)$, then λ is a nonzero limit ordinal, and both $P_s = \{\alpha < \lambda : s(\alpha) = +\}$ and $Q_s = \{\alpha < \lambda : s(\alpha) = -\}$ are cofinal in λ .*

Proof. A successor-length sequence is determined by its immediately preceding prefix and one sign; the empty sequence is old. If, say, P_s were bounded by $\alpha < \lambda$, then s would be the old prefix $s \upharpoonright \alpha$ followed by minus signs up to the old ordinal λ , an operation in M . Interchange the signs for Q_s . $\square$

Lemma 8.3 (Canonical cofinal presentations). *For a fresh sign s of length λ put $A_s = \{s \upharpoonright \alpha : \alpha \in P_s\}$ and $B_s = \{s \upharpoonright \alpha : \alpha \in Q_s\}$. Then A_s is cofinal in D_s , B_s is coinital in U_s , both are N -sets of old numbers, $A_s < s < B_s$, and*

$$\text{cf}^N D_s = \text{ci}^N U_s = \text{cf}^N \lambda. \quad (8.1)$$

Proof. At a plus position the prefix ends before a plus, so lies below s ; at a minus position above. These prefixes are old by freshness, and no old element extends s , since restriction to λ would place s in M .

Take $x \in D_s$. Either x is a proper prefix of s or it first disagrees with s at some $\gamma < \lambda$. Choose a plus position beyond γ (or beyond $b(x)$); comparison at the earlier position gives $x < s \upharpoonright \alpha < s$. A late minus position handles U_s . This also shows that neither side has an endpoint.

Write $\mu = \text{cf}^N \lambda$. A cofinal sequence of positions of length μ can be replaced by later plus and later minus positions, giving witnesses of size at most μ . Conversely, if $X \subseteq D_s$ has size below μ , record for each $x \in X$ a position where x ends as a prefix or first disagrees with s ; these positions are bounded below λ , and a plus prefix beyond the bound exceeds all of X and lies in D_s . The right side is symmetric. $\square$

Theorem 8.4 (Gap–fresh–sign correspondence). *The assignment $s \mapsto (D_s, U_s)$ is a bijection between fresh signs over M in N and the set-presented gaps of $\mathbf{No}^M$. The inverse sends a gap with any presentation $A \subseteq D$, $B \subseteq U$ to the simplest separator $\{A \mid B\}$ computed in N , and the result does not depend on the chosen presentation. Every set-presented gap is symmetric, with cofinalities given by (8.1).*

Proof. Lemma 8.3 shows the assignment is well defined. For surjectivity, let (D, U) be presented by A, B . No member of $\mathbf{No}^M$ separates A, B : a member of D is at most some member of A , and a member of U is at least some member of B . Let $s = \{A \mid B\}^N$; then $s \notin \mathbf{No}^M$. Every proper prefix $y = s \upharpoonright \gamma$ is old: by simplicity y does not separate A, B ; if $y < s$, all inequalities $y < b$ already hold, so there is $a \in A$ with $y \leq a < s$, and convexity of $\text{Cone}(y) \ni s$ gives $y \preceq a$, so y is a prefix of an old number; if $s < y$, argue with some $b \in B$. Thus s is fresh. Cofinality of A gives $D \subseteq D_s$ and coinitality of B gives $U \subseteq U_s$; both are partitions of $\mathbf{No}^M$, so the gap is (D_s, U_s) .

For injectivity, take distinct fresh s, t . Neither is a proper prefix of the other, since then a new sign would be a proper prefix of a fresh sign. Their common prefix u is old and lies strictly between them, on opposite sides of the two cuts. The same argument shows independence of the presentation. Symmetry is lemma 8.3. $\square$

Theorem 8.5 (The whole filling class). *If s is fresh, then $\{y \in \mathbf{No}^N : D_s < y < U_s\} = \text{Cone}(s)$. Every new $x \in \mathbf{No}^N \setminus \mathbf{No}^M$ lies in exactly one such cone, that of its shortest new prefix. Thus $\mathbf{No}^N \setminus \mathbf{No}^M$ is the disjoint union of the cones $\text{Cone}(s)$, $s \in \text{Fresh}(M, N)$, as class-defined convex pieces.*

Proof. An old element cannot extend s , so its comparison with any extension of s is decided before $b(s)$ and equals its comparison with s ; this gives $\supseteq$. If y does not extend s , either it is a proper prefix of s , hence old and not a filler of its own ground cut, or it first disagrees before $b(s)$; a later plus prefix in D_s or a later minus prefix in U_s then lies on the wrong side of y . For a new x , the least length of a new prefix gives a fresh prefix, and distinct fresh signs are incomparable, so their cones are disjoint. The union is shorthand for membership in exactly one piece; it posits no set of all pieces. $\square$

Corollary 8.6 (Exact gap spectrum). *With all cofinalities taken in N ,*

$$\text{GapSpec}_N(\mathbf{No}^M) = \{(\text{cf}^N b(s), \text{cf}^N b(s)) : s \in \text{Fresh}(M, N)\}. \quad (8.2)$$

In particular, an asymmetrically set-presented gap cannot arise in a same-ordinal ground surreal field.

This is stronger than the existence of a missing cut after adding a real. It classifies all set-presented gaps, including gaps with long sign codes and small outer cofinality, and it explains why two new surreals can have exactly the same cut over the whole old field: they lie in one fresh cone, which can contain a proper class of extensions. The fresh signs themselves, not only their lengths, parametrize the gaps. The symmetry concerns the two sides of a bounded gap, not the global cofinality of the field, which is a proper class (lemma 3.3). Cofinalities must be computed in N : the old universe’s cofinality of $b(s)$ can be the wrong one (section 13.4).

9 The least gap

Proposition 9.1 (Terminal-plus block code; source 03). *Let μ be an infinite regular cardinal of N , and let $f : \mu \rightarrow \text{Ord}$ be new with $f \upharpoonright \xi \in M$ for every $\xi < \mu$. Define, by ordinal concatenation,*

$$\text{code}^+(f) = \bigcap_{\xi < \mu} (+^{f(\xi)+1} -), \quad \text{b}(\text{code}^+(f)) = \sum_{\xi < \mu} (f(\xi) + 2).$$

Then $\text{code}^+(f)$ is fresh over M and $\text{cf}^N \text{b}(\text{code}^+(f)) = \mu$. With $\sigma_\xi = \sum_{\eta < \xi} (f(\eta) + 2)$, the old prefixes

$$a_\xi = \text{code}^+(f) \upharpoonright (\sigma_\xi + f(\xi)), \quad b_\xi = \text{code}^+(f) \upharpoonright (\sigma_\xi + f(\xi) + 1)$$

satisfy $a_\xi < a_\eta < \text{code}^+(f) < b_\eta < b_\xi$ for $\xi < \eta < \mu$, and present an omitted (μ, μ) gap of $\mathbf{No}^M$.

Proof. Each block is a nonempty plus run of successor length followed by one minus. A proper prefix of the concatenation lies inside some block and is determined by the old restriction $f \upharpoonright (\xi + 1)$ and ordinal positions; since μ is a limit, $\xi + 1 < \mu$, so the prefix is old. If $\text{code}^+(f)$ were in M , its minus positions would delimit the blocks, and the unique predecessors of the successor run lengths would recover f in M . So it is fresh.

The partial sums σ_ξ increase and are cofinal, so $\text{cf}^N \text{b}(\text{code}^+(f)) \leq \mu$; fewer than μ cofinal positions would, by regularity, bound the block indices below μ . At length $\sigma_\xi + f(\xi)$ the next sign is the last plus of block ξ , and at $\sigma_\xi + f(\xi) + 1$ it is the minus; first-disagreement comparison gives the nesting. The endpoint lengths are cofinal, so, as in lemma 8.3, the endpoints are cofinal and coinital in the two sides, and theorem 8.4 gives an omitted gap with the asserted cofinalities. $\square$

The extra plus in each run is not cosmetic: for a limit value $f(\xi)$, a run of length $f(\xi)$ has no last plus, and the uniformly specified left endpoint would not exist. Source 08's code in section 5.3 achieves the same with the two terminal signs $- \frown +$.

Theorem 9.2 (Exact least-gap theorem; source 03). *For $M \neq N$,*

$$\delta(M, N) = \min\{|A \cup B|^N : A < B \text{ is an omitted set pair in } \mathbf{No}^M\} \quad (9.1)$$

$$= \min\{\text{cf}^N \text{b}(s) : s \in \text{Fresh}(M, N)\}. \quad (9.2)$$

There is an omitted gap of exact cofinality pair (δ, δ) .

Proof. The lower bound in (9.1) is the main theorem at $\kappa = \delta$ (equivalently lemma 3.4: a pair of size below δ is old, and its old cut fills it). Apply proposition 9.1 to a new $f : \delta \rightarrow \text{Ord}$, using proposition 4.2; its endpoint families have size δ . Every fresh sign gives an omitted presentation of size $\text{cf}^N \text{b}(s)$ by lemma 8.3 and theorem 8.4, so no such cofinality is below δ , and the block code attains δ . $\square$

The recovery is deliberately limited. The invariant recovers a threshold, and corollary 8.6 recovers cofinality data. It does not recover the inner model, the forcing notion, or the actual sign lengths from a bare ordered-field presentation. Different grounds can have the same threshold, and equality of gap spectra is not proved to classify the fields. *Status (batch 37, 28 September 2026).* It does not classify them: old fields in one universe can have equal full gap spectra, even equal saturation at every cardinal, and fail to be isomorphic as ordered or pure fields (theorems 27.4, 28.5 and 29.2, Part VIII); the full spectrum of a pair is the outer-cofinality image of its fresh-function spectrum (theorem 22.7).

Part III

The first new birthday (source 08)

10 First new birthdays and their relation to saturation

The first new surreal number, measured by birthday, need not appear at the same ordinal as the first omitted-cut size.

Definition 10.1 (First new birthday). For $M \neq N$ put

$$\beta(M, N) = \min\{\alpha : \mathcal{P}(\alpha)^M \neq \mathcal{P}(\alpha)^N\}, \quad (10.1)$$

the power sets being compared as actual sets of subsets in N .

Proposition 10.2 (Birthday threshold; source 08). $\beta(M, N) = \min\{b(s) : s \in \mathbf{No}^N \setminus \mathbf{No}^M\}$. It is an infinite cardinal in both M and N . With $\delta = \delta(M, N)$ and $\beta = \beta(M, N)$,

$$\delta \leq \text{cf}^N(\beta) \leq \beta. \quad (10.2)$$

In particular, the first new birthday can be much larger than the least size of an omitted cut.

Proof. Subsets of α and signs of length α are uniformly interdefinable by characteristic signs. This gives the birthday characterization; existence follows from section 4. No finite subset is new.

If β were not an M -cardinal, an M -bijection between β and a smaller ordinal would pull a new subset of β back to a new subset of that smaller ordinal. Suppose there were an N -bijection $e : \mu \rightarrow \beta$ with $\mu < \beta$. Transporting the order of β along e gives a well-order $\triangleleft$ of μ of type β . Since β is an infinite M -cardinal, M codes $\mu \times \mu$ by an ordinal $\gamma < \beta$, so by minimality $\triangleleft \in M$. Its order type is absolute, so M would see a well-order of μ of type β , contradicting that β is an M -cardinal.

For the first inequality, choose a new $X \subseteq \beta$ and let $\mu = \text{cf}^N(\beta)$. If $\mu < \delta$, a cofinal $c : \mu \rightarrow \beta$ belongs to M , and every $X \cap c(\xi)$ is old by minimality of β . In M well-order $S = \bigcup_{\alpha < \beta} \mathcal{P}(\alpha)^M$ and code these restrictions by ordinals; the code sequence has length $\mu < \delta$, so it is old, and decoding and taking the union in M recovers X . Hence $\delta \leq \text{cf}^N(\beta)$. $\square$

Remark 10.3 (Two resources; source 08). β measures the least *length of a new sign sequence*; δ measures the least *number of old parameters needed for an omitted cut*. Long old parameters can carry large ordinal entries, as the block codes show. Equality of all low birthday fragments therefore does not by itself guarantee countable saturation of the whole old field (corollary 13.4).

Remark 10.4 (β is the least length of a fresh sign; merge observation). A new sign of least birthday β has all its proper prefixes old, since they are signs of smaller birthday; so it is fresh. Every fresh sign is new, so $\beta = \min\{b(s) : s \in \text{Fresh}(M, N)\}$. By theorem 9.2, δ is the least *cofinality* $\text{cf}^N b(s)$ of a fresh sign. This two-line observation, added in the merge, places source 08's β in source 03's fresh-sign picture. It does not answer source 03's question whether an intrinsic ordered-field invariant recovers such lengths (question 19.5): β is defined through birthdays, which are not part of the field language.

Remark 10.5 (Relation to the sibling report). The sibling report docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets proves, for every same-ordinal outer model, that the first new birthday is the least ordinal acquiring a new subset; it also writes that ordinal β . Proposition 10.2 keeps source 08's statement and proof because source 08 adds that β is a cardinal of N and the inequality $\delta \leq \text{cf}^N \beta$

with the first new length δ of this report. The sibling report also studies bounded birthday fields in a language with a birthday symbol; section 12 here concerns the ordered-field saturation of such fragments, without that symbol.

Part IV

Surcomplex fields, fragments, forcing and topology

11 Surcomplex numbers: a language-dependent dichotomy

Let $c_M(a + bi) = a - bi$ on SC^M and c_N likewise on SC^N . The collection's automorphism report already emphasizes the difference between a pure algebraically closed field and a field with a distinguished surreal real axis [19]. That distinction is standard algebra and is not claimed as new. The point here is its exact effect on *external saturation under a change of universe*.

11.1 The pure field forgets the sequence obstruction

Lemma 11.1 (Algebraic closure stays a set; source 03). *Let H be a class field and $A \subseteq H$ an N -set. The field generated by A is a set, and the elements of H algebraic over it form a set. If H is algebraically closed, this relative algebraic closure is an algebraically closed set subfield of H .*

Proof. Finite rational expressions in A form a set, there are set many polynomials over that field, and a nonzero polynomial of degree d has at most d roots. Collect the root sets by Replacement. A root of a polynomial with algebraic coefficients is algebraic over the original field. $\square$

Theorem 11.2 (Set-saturation of the old pure surcomplex field; sources 01, 03, 04, 08). *For every N -set $A \subseteq SC^M$, every complete one-variable pure-field type over A that is finitely satisfiable in SC^M is realized in SC^M . Consequently SC^M is κ -saturated in N for every N -cardinal κ , with no assumption on new sequences or new reals. More generally (source 03), every definable proper-class algebraically closed field is saturated in the pure field language over every N -set of parameters.*

Proof. Choose a ground set $B \supseteq A$ by lemma 3.2. In M let E be the set of elements of SC^M algebraic over $\mathbb{Q}(B)$; it is an algebraically closed ground subfield by lemma 11.1. Choose an old $t \notin E$ without a class-sized transcendence basis: bound the birthdays of the coordinates of all elements of E and take a larger all-plus ordinal. Then t is transcendental over $\mathbb{Q}(B)$, and this persists in N , because a polynomial witnessing algebraicity has a finite old coefficient tuple. In particular t is transcendental over $\mathbb{Q}(A)$ as formed in N .

If the given type p contains no equation $P(X) = 0$ with $P \neq 0$ over $\mathbb{Q}(A)$, completeness and quantifier elimination make p the unique transcendental type over $\mathbb{Q}(A)$, realized by t . Otherwise choose such an equation. Its finitely many roots lie in SC^M by proposition 3.1. If none realized p , choose for each root a formula of p that it fails; together with $P(X) = 0$ these form an unsatisfiable finite subset of p . The argument covers algebraic parameters and any new organization of A .

For a general proper-class algebraically closed field H , let D be the algebraic closure in H of the field generated by A (lemma 11.1); extend the type to D by compactness in that set model, realize an algebraic extension by a root in D , and a transcendental one by any $y \in H \setminus D$, which exists because H is a proper class. $\square$

There is no cardinal paradox: SC^M is a proper class of N , and the statement is a scheme over set-sized parameter sets, not a claim that a set of cardinality μ realizes types over itself of size μ . The theorem is an established model-theoretic mechanism rather than a new principle about algebraically closed fields; what matters is its contrast with the real field. Nonalgebraic one-types over an algebraically closed field have no order cut to remember. Nor does the theorem by itself give a class isomorphism $SC^M \cong SC^N$; that needs a class back-and-forth and Global Choice (part V).

11.2 Naming conjugation restores the old ordered field

Lemma 11.3 (Real coordinates are interpretable; sources 01, 03, 04, 08). *The fixed field of c_M is definable in (SC^M, c_M) by $c_M(x) = x$, and on it the order is definable by*

$$x < y \iff \exists u (c_M(u) = u \wedge u \neq 0 \wedge u^2 = y - x). \quad (11.1)$$

After naming i , the map $z \mapsto (\operatorname{re} z, \operatorname{im} z)$ with

$$\operatorname{re} z = \frac{z + c_M(z)}{2}, \quad \operatorname{im} z = \frac{z - c_M(z)}{2i} \quad (11.2)$$

is a definable bijection $SC^M \rightarrow (\mathbf{No}^M)^2$, and field operations and conjugation become polynomial operations on coordinates: $(a, b)(u, v) = (au - bv, av + bu)$ and $c_M(a, b) = (a, -b)$.

Proof. The fixed-point equation gives zero imaginary coordinate; in a real closed field the nonzero squares are exactly the positive elements. The coordinate formulas follow from $i^2 = -1$, with inverse $(a, b) \mapsto a + bi$. Quantifiers over SC^M become pairs of quantifiers over $\mathbf{No}^M$, and quantifiers over $\mathbf{No}^M$ become quantifiers over the definable fixed field. $\square$

Theorem 11.4 (Conjugation spectrum; sources 01, 03, 04, 08). *For every infinite N -cardinal κ ,*

$$(SC^M, c_M) \text{ is } \kappa\text{-saturated in } N \iff \mathbf{No}^M \text{ is } \kappa\text{-saturated in } N.$$

For uncountable κ this is equivalent to $NS_{<\kappa}(M, N)$, and for $\kappa = \omega$ to the absence of new reals. The statements remain equivalent after naming i . Moreover the inclusion $(SC^M, c_M) \preccurlyeq (SC^N, c_N)$ is elementary (sources 03, 04, 08).

Proof. Naming a fixed finite tuple of constants preserves and reflects κ -saturation for infinite κ : absorb the tuple into the parameter set, or extend a reduct type by compactness and realize it. So we may name i . If $\mathbf{No}^M$ is κ -saturated, translate a type in (SC^M, c_M, i) over fewer than κ parameters by lemma 11.3: the real and imaginary coordinates of the parameters are still fewer than κ , the translated type has two field variables, and finite-variable saturation of $\mathbf{No}^M$ realizes it. Conversely translate a type over a small subset of $\mathbf{No}^M$ into (SC^M, c_M) by (11.1), restricting quantifiers to the fixed field and adding $c_M(X) = X$; saturation gives a fixed-field realization. The last assertions follow from theorems 1.1 and 7.1. Elementarity: the same coordinate translation reduces each formula with old parameters to an ordered-field formula, and $\mathbf{No}^M \preccurlyeq \mathbf{No}^N$. $\square$

In particular the interval type (5.3) remains omitted in the conjugation expansion when written with the definable fixed field and definable positive squares. In the pure field those order inequalities are unavailable, and the one-variable transcendental type is realized by theorem 11.2.

Remark 11.5 (Conjugation is not definable in the pure field; source 08). In a pure algebraically closed field every definable subset of one variable, with any finite tuple of parameters, is finite or cofinite. The fixed field of c_M is infinite with infinite complement, so c_M is not definable in the pure field, even with parameters, and naming a set of constants does not change this. The contrast in table 3 therefore does not violate preservation of saturation under definitional expansions.

Table 3: Saturation in N by structure and language (source 08). “No new reals” means $\mathbb{R}^M = \mathbb{R}^N$.

Structure and language	ω -saturation	Uncountable κ -saturation
$(\mathbf{No}^M, <)$, pure order	Always	$\text{NS}_{<\kappa}(M, N)$
$\mathbf{No}^M$, ordered field	No new reals	$\text{NS}_{<\kappa}(M, N)$
SC^M , pure field	Always	Always
(SC^M, c_M)	No new reals	$\text{NS}_{<\kappa}(M, N)$

11.3 A nonisomorphism consequence

Corollary 11.6 (Sources 01, 04, 08). *Assume $M \neq N$. There is no N -class order isomorphism $\mathbf{No}^M \rightarrow \mathbf{No}^N$ whose restrictions to sets have set-sized images. Consequently there is no such field isomorphism, and no such isomorphism $(\text{SC}^M, c_M) \cong (\text{SC}^N, c_N)$.*

Proof. The first gap theorem supplies two N -sets of old endpoints with no separator in $\mathbf{No}^M$. An isomorphism as stated sends them to separated N -sets in $\mathbf{No}^N$, whose cut has a separator; its inverse image would separate the original cut. A field isomorphism of real closed fields preserves order, since positivity is being a nonzero square, and an isomorphism preserving conjugation restricts to an isomorphism of the definable fixed fields. $\square$

The set-image condition is automatic for definable class maps and for class maps of GBC with class Replacement; sources 04 and 08, working in a class theory, state the corollary without it. It is stated here to avoid quantifying over unrestricted external graphs without an ambient class theory. The corollary compares an inner universe with its ambient universe only; it says nothing about whether $\mathbf{No}^{M_1} \cong \mathbf{No}^{M_2}$ forces $M_1 = M_2$ for two inner models of a common universe, whose spectra could agree without settling their isomorphism problem (source 08).

12 A sharp theorem for bounded birthday fields (source 01)

The previous theorems concern all ground birthdays at once. A set-sized fragment has a new obstruction: its birthdays can become cofinally approachable by a short sequence, even if every individual sign of length below the bound remains old.

Let κ be an uncountable regular cardinal in M that remains a cardinal in N , and consider the set $(\mathbf{No}_{<\kappa})^M$. It is a real closed set field in M by the classical birthday-fragment theorem, and remains real closed in N by the finite-coefficient argument. Regularity in N is *not* assumed in advance.

Theorem 12.1 (Bounded-birthday preservation). *The following are equivalent.*

- (i) $(\mathbf{No}_{<\kappa})^M$ is κ -saturated in N as an ordered field.
- (ii) The order $(\mathbf{No}_{<\kappa})^M$ has the η_κ property in N .

- (iii) $\text{cf}^N(\kappa) = \kappa$ and $({}^\alpha 2)^N = ({}^\alpha 2)^M$ for every $\alpha < \kappa$.
- (iv) κ is regular in N and $(\mathbf{No}_{<\kappa})^M = (\mathbf{No}_{<\kappa})^N$ as sets of sign sequences.

Proof. (i) $\Leftrightarrow$ (ii) is lemma 2.3(i), and (iii) $\Leftrightarrow$ (iv) is the representation of surreals by binary sign sequences.

Assume (iv), and let $L, R \subseteq (\mathbf{No}_{<\kappa})^M$ be N -sets with $L < R$ and $|L \cup R| < \kappa$. By regularity in N , $\gamma = \sup\{b(y) + 1 : y \in L \cup R\} < \kappa$. The cut constructed in N has birthday at most γ by (2.1), so it lies in $(\mathbf{No}_{<\kappa})^N = (\mathbf{No}_{<\kappa})^M$. This proves (ii).

For necessity, suppose first $\text{cf}^N(\kappa) = \mu < \kappa$. A cofinal sequence of ordinals below κ , viewed as all-plus surreals, is cofinal in the fragment by (2.2), has size below κ , and has no strict upper bound there; with an empty right side it violates (ii). Next suppose a new binary function of length below κ exists, and take one $s : \alpha \rightarrow \{-, +\}$ of least length. Its proper prefixes are old and have birthday below κ , so L_s, R_s of lemma 2.2 lie in the fragment and have size at most $|\alpha| < \kappa$. Any separator, even in $\mathbf{No}^N$, extends s ; an old separator cannot, since restricting it to α would put s in M . This violates (ii). $\square$

Remark 12.2 (Why the two criteria differ). In the full-class theorem, $\mathbf{NS}_{<\kappa}$ bans short sequences with *arbitrarily large range*. The fragment theorem bans new binary signs below a fixed birthday bound and separately preserves the cofinality of that bound. A short sequence with unbounded range at a much larger cardinal may be invisible to one small fragment but visible to the full old field. Conversely, singularizing the bound can destroy fragment saturation without adding any new sign below it (corollary 13.4). In the notation of part III, condition (iii) says $\text{cf}^N \kappa = \kappa$ and $\kappa \leq \beta(M, N)$ when $M \neq N$.

Proposition 12.3 (Pure complexification of a set fragment; source 01). *Let $\mu = |(\mathbf{No}_{<\kappa})^M|^N$. If μ is uncountable, the algebraically closed field $(\mathbf{No}_{<\kappa})^M[i]$ is μ -saturated in the pure field language in N , whether or not the conditions of theorem 12.1 hold.*

Proof. An algebraically closed field of characteristic zero and uncountable cardinality μ has transcendence degree μ over $\mathbb{Q}$: a basis of size τ would give an algebraic closure of size $\max(\tau, \aleph_0)$. Over fewer than μ parameters there is still a transcendental element, and algebraic types are realized by algebraic closedness; quantifier elimination gives μ -saturation. This is the usual dimension argument, included as a consequence rather than a new field theorem. $\square$

13 Forcing examples and distributivity

The theorems above apply to every pair of universes in their hypotheses. For set forcing they translate the usual preservation of short ordinal sequences into field-theoretic statements. The forcing facts used are standard [10, 11, 12, 13] and are not reproved.

13.1 The forcing-theoretic formulation

A forcing $\mathbb{P}$ is $< \kappa$ -distributive if the intersection of fewer than κ dense open subsets is dense, equivalently if its complete Boolean algebra is $< \kappa$ -distributive. With the usual separative interpretation this is equivalent to adding no new ordinal-valued sequences of length less than κ in any generic extension [10]; conventions about whether κ itself is included vary, and the sequence formulation is the one used here. For the forward direction, take for each coordinate of a name for a short sequence the dense set of conditions deciding it; a condition in the intersection decides the whole sequence as a ground sequence. Conversely, refine the dense open sets by maximal antichains in the complete Boolean algebra; a generic choice through them is a short

ordinal-coded sequence, and if all such sequences are old, conditions deciding a whole choice are dense below all the antichains. This is the standard Boolean form of the equivalence, not a new forcing theorem.

Corollary 13.1 (Forcing characterization; sources 01, 03, 04, 08). *Let κ be an uncountable cardinal of M and $\mathbb{P} \in M$.*

- (a) (source 03) *If κ is regular in M , then $\mathbb{P}$ adds no new ordinal sequences of length below κ in any generic extension if and only if, in every generic extension, $\mathbf{No}^M$ has the η_κ property. The preservation direction preserves κ and its regularity.*
- (b) (sources 04, 08) *If $\mathbb{P}$ preserves κ as an uncountable cardinal, the following are equivalent: $\mathbb{P}$ is $< \kappa$ -distributive in M ; in every generic extension $\mathbf{No}^M$ is externally κ -saturated; in every generic extension $(\mathbf{SC}^M, c_M)$ is externally κ -saturated.*

Without conjugation, $\mathbf{SC}^M$ is externally set-saturated in every generic extension, regardless of distributivity.

Proof. Apply (6.1), theorem 1.1, theorem 11.4 and theorem 11.2 in each generic extension. A new short cofinal map or a collapse to a smaller ordinal would be a forbidden short sequence, which gives the preservation clause in (a). $\square$

The phrase “in every generic extension” is indispensable. A property of one extension does not identify the behaviour of every condition of a nonhomogeneous forcing. For example, the lottery sum of trivial forcing and Cohen forcing has a generic branch that adds nothing and another that adds a real; the trivial branch preserves all saturation, but the whole forcing is not countably distributive (source 08). The invariants $\delta(M, N)$ and $\beta(M, N)$ belong to a particular pair of universes, not to the name of a poset, and the theorems are formulated pair by pair. One can also apply corollary 13.1 below a specified condition.

13.2 A Cohen real destroys even finite-parameter saturation

Let $N = M[g]$ add a Cohen real, for instance with $\mathbb{P} = \text{Fn}(\omega, 2, <\omega)$. The generic union is a new subset of ω : for every ground binary sequence the conditions disagreeing with it somewhere are dense. No finite ordinal sequence is new, so

$$\delta(M, N) = \beta(M, N) = \omega.$$

$\mathbf{No}^M$ has an (ω, ω) gap and is not even ω -saturated; the rational-cut type (7.1) witnesses this directly. The pure old order is ω -saturated but not ω_1 -saturated. Nevertheless $\mathbf{No}^M \preccurlyeq \mathbf{No}^N$ and $(\mathbf{SC}^M, c_M) \preccurlyeq (\mathbf{SC}^N, c_N)$ remain true, and the pure field $\mathbf{SC}^M$ remains saturated over every N -set; naming conjugation makes it fail ω -saturation again. Neither loss of real closedness nor failure of elementarity explains the omitted type: the change lies in which countable sets of formulas are available. Nothing here depends on Cohen combinatorics beyond adding a real.

13.3 Adding a subset of a regular cardinal

Proposition 13.2 (Any prescribed regular first gap; sources 01, 03, 04, 08). *Let κ be an infinite regular cardinal in M , and let $N = M[G]$ be obtained by $\text{Add}(\kappa, 1) = \text{Fn}_{<\kappa}(\kappa, 2)$, the partial functions $\kappa \rightarrow 2$ with domain of size below κ , ordered by reverse inclusion. Then*

$$\delta(M, N) = \beta(M, N) = \kappa.$$

$\mathbf{No}^M$ has an omitted (κ, κ) gap and no omitted pair of size below κ . If $\kappa > \omega$, no real is added, $\mathbf{No}^M$ and $(\mathbf{SC}^M, c_M)$ are ω -saturated and κ -saturated, and they are not $(\kappa^+)^N$ -saturated. No cardinal arithmetic assumption is needed.

Proof. For uncountable κ a descending chain of fewer than κ conditions has its union as a lower bound, by regularity. To decide a name for an ordinal-valued sequence of length $\lambda < \kappa$, strengthen a condition coordinate by coordinate, taking unions at limits and at the end; the result decides the sequence as a ground sequence. So no such sequence is added, and κ stays a regular cardinal. The generic union $g : \kappa \rightarrow 2$ is total and new: given an old h , it is dense to put the opposite value at an unused coordinate. Hence $\delta = \kappa$. No subset of an ordinal below κ is added, and g codes a new subset of κ , so $\beta = \kappa$. Apply theorems 6.2, 7.1, 9.2 and 11.4 and corollary 6.4. For $\kappa = \omega$ this is the Cohen case. The generic sign of length κ is itself fresh, so its gap can also be read from theorem 8.4. $\square$

No assumption such as $2^{<\kappa} = \kappa$ is needed for this first-gap calculation; the saturation failure is indexed by the successor computed in the actual extension. The proposition does not compute the full gap spectrum, does not claim that no larger gaps occur, and asserts nothing about the preservation of cardinals above κ . For $\kappa = \omega_1^M$, no reals are added, ω - and ω_1 -saturation survive, and ω_2^N -saturation is lost: failures at different parameter sizes. *Status (batch 37, 28 September 2026)*. The full spectrum is now computed: every set-presented gap of $\mathbf{No}^M$ in this extension has character (κ, κ) , with no cardinal arithmetic and for every ground alphabet (theorem 23.6, Part VIII).

Example 13.3 (Under $\kappa^{<\kappa} = \kappa$; source 04). If moreover $M \models \kappa^{<\kappa} = \kappa$ with κ uncountable, then $|\text{Add}(\kappa, 1)| = \kappa$, so the forcing has the κ^+ -chain condition. Together with closure this preserves κ , κ^+ and the regularity of κ [10], so the failure of saturation occurs at the ground model's κ^+ . This hypothesis is stronger than needed for proposition 13.2; it is recorded because source 04 uses it for this extra preservation clause.

13.4 Prikry forcing separates the two countable notions

Assume M has a measurable cardinal ρ , and force with classical Prikry forcing from a normal measure on ρ . Its standard properties are that it preserves cardinals, adds an increasing cofinal sequence $g : \omega \rightarrow \rho$, changes $\text{cf}(\rho)$ to ω , and adds no bounded subsets of ρ ; in particular it adds no reals [11, 12, 13]. These are imported facts, not reproved here. The sequence g is new: M regards ρ as regular and uncountable. The range of g is a new subset of ρ , since in M its order type ω and cofinality in ρ would contradict regularity. Consequently

$$\delta(M, N) = \omega, \quad \beta(M, N) = \rho, \quad \mathbb{R}^M = \mathbb{R}^N. \quad (13.1)$$

Corollary 13.4 (Prikry separation; sources 01, 03, 04, 08). *In this Prikry extension:*

- (i) $\mathbf{No}^M$ and $(\mathbf{SC}^M, c_M)$ are ω -saturated but not ω_1 -saturated, while $\mathbf{SC}^M$ is set-saturated.
- (ii) $(\mathbf{No}_{<\rho})^N = (\mathbf{No}_{<\rho})^M$, yet this set field is not ρ -saturated in N : its all-plus ordinals acquire a countable cofinal sequence (source 01).
- (iii) Applying the block code to g gives an omitted (ω, ω) gap whose simplest new separator $\text{code}(g)$ has birthday exactly ρ (source 08).
- (iv) The characteristic sign s of $\text{ran}(g)$, of length ρ , is fresh, and its gap has cofinalities $(\text{cf}^N \rho, \text{cf}^N \rho) = (\omega, \omega)$, not (ρ, ρ) (source 03).
- (v) The endpoints of the omitted cut obtained from the absolute interval code are the finite rational expressions of proposition 5.5 with $\alpha_j = g(j)$ (source 04).

Proof. (i) Theorem 7.1 gives ω -saturation, the new countable sequence and theorem 1.1 give failure at ω_1 , and theorems 11.2 and 11.4 give the surcomplex clauses. (ii) Every binary sign of length below ρ codes a bounded subset of ρ ; the regularity clause of theorem 12.1 fails. (iii) Every finite initial segment of g is old, so theorem 5.8 applies with $\lambda = \omega = \delta$; $b(\text{code}(g)) = \sum_{n < \omega} (g(n) + 2) = \rho$, since the finite partial sums lie below the cardinal ρ and are cofinal in it. (iv) Every proper initial segment of $\text{ran}(g)$ is finite, so every proper prefix of s is old, and $\text{ran}(g)$ is new; apply lemma 8.3. (v) is proposition 5.5. $\square$

The parameters of the omitted cut all have birthdays below ρ and were all present in M ; what is new is their countable arrangement. The alphabet in the interval tree must be allowed to reach ρ : a test restricted to new reals, or to new binary signs of countable length, would miss this example. The old universe's cofinality of the birthday ρ is the wrong cofinality for the gap computation. Corollary 13.4(ii) shows directly why regularity is an independent clause in theorem 12.1, and (iii) shows how far $\beta = \rho$ can lie above $\delta = \omega$.

13.5 Comparison table

Extension	δ	β	New reals?	Old ordered field $\mathbf{No}^M$	Old pure $\mathbf{SC}^M$
Cohen real	ω	ω	Yes	Not ω -saturated	Set-saturated
Add($\kappa, 1$), $\kappa > \omega$ regular	κ	κ	No	κ -saturated, not $(\kappa^+)^N$ -saturated	Set-saturated
Prikry at a measurable ρ	ω	ρ	No	ω -saturated, not ω_1 -saturated	Set-saturated

Adding conjugation to the last column reproduces the ordered-field column. The table describes external saturation in the stated outer universe; it does not alter the internal theorem that each universe's own surreal field realizes all of that universe's set-sized cuts.

14 Why order gaps do not become Cauchy limits

The native fine topology of a surreal field uses all positive radii of that field, not only real ones; its additive uniformity has the entourages $|x - y| < \varepsilon$ for $\varepsilon \in \mathbf{No}_{>0}^M$. On the surcomplex field use either the surreal-valued modulus $|a + bi| = \sqrt{a^2 + b^2}$ (source 01) or $\|a + bi\|_\infty = \max(|a|, |b|)$ (source 04); they define the same product topology and uniformity. All claims are about neighborhoods and set-indexed families; no set of all open classes is formed.

Proposition 14.1 (External discreteness and Cauchy rigidity; sources 01, 03, 04). *For every pair $M \subseteq N$ under (H) :*

- (i) *every N -set $A \subseteq \mathbf{No}^M$ is closed in the fine topology of $\mathbf{No}^M$ and uniformly discrete: some old $\varepsilon > 0$ is below every nonzero distance in A ;*
- (ii) *every N -set-indexed Cauchy net in $\mathbf{No}^M$ is eventually constant, and every convergent such net is eventually equal to its limit;*
- (iii) *the same holds for $\mathbf{SC}^M$ with either surcomplex modulus.*

No short-sequence hypothesis is needed.

Proof. The nonzero pairwise distances in A form an N -set of positive old numbers. Theorem 3.5 with left side $\{0\}$ (source 01), or directly lemma 3.3 applied to the reciprocals (source 04), gives a positive old lower

bound ε . For $x \in \mathbf{No}^M \setminus A$ apply the same argument to $\{|x - a| : a \in A\}$ to get a neighborhood of x missing A . The range of a set-indexed net is an N -set; choosing ε below all nonzero pairwise distances in the range, the Cauchy condition for that one entourage makes late values equal. For a convergent net, apply the separation argument to its range together with its limit. For $\mathbf{SC}^M$, the modulus of an old nonzero number is old and positive; apply the same argument. $\square$

For the interval tree, the widths $b_{f \upharpoonright \alpha} - a_{f \upharpoonright \alpha}$, $\alpha < \lambda$, are positive old numbers with a common positive old lower bound; the endpoint sequences are not Cauchy in the fine uniformity, and the omitted cut is not evidence of a missing fine-Cauchy limit. Even a Prikry cut with countably many endpoints leaves the proposition intact. A topology that looks only at selected scales could behave differently, but it is a different structure and is not used. Elementarity, saturation and fine completeness must not be treated as interchangeable: the old field remains real closed and elementary, loses specified types, and keeps this set-indexed rigidity.

Lemma 14.2 (Prefix cones are open; source 04). *For every $s \in \mathbf{No}^N$, $\text{Cone}(s)$ is open in the fine order topology of $\mathbf{No}^N$.*

Proof. By lemma 2.2, $\text{Cone}(s)$ is the class of separators of $L_s < R_s$. For $x \in \text{Cone}(s)$, the set $\{x - l : l \in L_s\} \cup \{r - x : r \in R_s\}$ of positive distances has a positive lower bound in $\mathbf{No}^N$ by the cut theorem, and a smaller interval about x keeps all the canonical inequalities. If the set is empty, $\text{Cone}(s) = \mathbf{No}^N$. $\square$

Theorem 14.3 (Closed, nowhere dense, and still set-discrete; sources 03, 04). *$\mathbf{No}^M$ is closed in $\mathbf{No}^N$, and its subspace topology is its own order topology. If $M \neq N$, $\mathbf{No}^M$ is nowhere dense in $\mathbf{No}^N$. The same statements hold for $\mathbf{SC}^M \subseteq \mathbf{SC}^N$ in the product topologies.*

Proof. If $x \in \mathbf{No}^N \setminus \mathbf{No}^M$, every extension of x is new, because an old sign has only old prefixes; so the open cone $\text{Cone}(x)$ misses $\mathbf{No}^M$. Source 03's route uses the open interval $(x \frown -, x \frown +)$, all of whose points extend x .

For every positive $r \in \mathbf{No}^N$ there is an old $0 < e < r$ (lemma 3.3); this gives equality of the intrinsic and subspace topologies at every old point. Assume $M \neq N$ and choose a new z ; replacing it by $h = |z|/(1 + |z|)$ gives a new $h \in (0, 1)$, since otherwise $|z| = h/(1 - h)$ would be old. Given $a \in \mathbf{No}^M$ and $r > 0$, the point $a + eh$ lies in $(a - r, a + r)$ and is new, since otherwise $h = ((a + eh) - a)/e$ would be old. So no old point is interior; a closed class with empty interior is nowhere dense. (Source 03 instead uses $a + z/\gamma$ for an old ordinal $\gamma > |z|/r$.) For the complex fields use $\mathbf{No}^M \times \mathbf{No}^M \subseteq \mathbf{No}^N \times \mathbf{No}^N$: products of closed coordinate classes are closed, the radius comparison gives the subspace claim, and perturbing one real coordinate by eh shows empty interior. $\square$

The old field is therefore simultaneously cofinal, simplicity-initial, elementary, closed and nowhere dense in $\mathbf{No}^N$. Order-cofinality must not be mistaken for topological density, nor elementary inclusion for density. In the outer field the whole interval of fillers of a fresh gap, the cone of theorem 8.5, is separated from every old point. No convergence of an ordinary sequence to a missing point is asserted.

Remark 14.4 (Relation to the foundations report). The internal statement is Theorem 12.1 (`found:thm:discrete`) of `docs/foundations-and-computation/foundations`, which proves (i)–(iii) of proposition 14.1 for sets of the ambient universe, assuming its Lemma 3.3 (`found:lem:bounds`, `bounds` and `small positive lower bounds`) “for the permitted small families”. Proposition 14.1 extends it to *new* N -sets of old numbers. Its only input is the existence of old positive lower bounds, which holds in every same-ordinal extension even where cut filling fails. In the Lean ledger (`docs/FORMALIZATION.md`), the Cauchy-net clause of `found:thm:discrete` is stated under the hypothesis `HasSmallCutFillers`, that is, under small cut

filling. The external version shows that positive lower bounds suffice for that clause. This is recorded as information for formalization; nothing here is formalized.

The foundations report’s Section 7.7 (`found:sub:countable`) warns: “‘every set family has a bound’ means every family represented by a set *in the relevant model*; it is not an absoluteness claim about every family visible from a stronger metatheory.” That warning concerns countable models and different ordinals, and it stands. For same-ordinal outer universes this report sharpens it: bounds of outer sets of old numbers *are* absolute (lemmas 3.2 and 3.3), while cut filling is not (theorem 1.1). The different-height analogue, the embedding $\mathbf{No}_U \rightarrow \mathbf{No}_V$ for Grothendieck universes, is treated in that report and is not used here. The catalogue’s remark that universe-relative formulations “require smaller-universe subsets and index types” also concerns carriers of different heights; in the present setting the index sets need not be ground-small.

Part V

Class isomorphisms and real forms (Global Choice)

15 Pure-field class isomorphisms

Throughout this part assume (GC): the ambient theory is GBC with Global Choice, and M is a class of the class universe. Source 04 works with a definable M and definable fields, which is a special case. All assertions concern individual class maps; $\text{Aut}(\cdot)$ is shorthand for quantification over class field automorphisms, and no class-object of all class maps is formed. At every ordinal stage of the extension arguments, the partial fields and maps are sets; their histories below a given ordinal are sets by class Replacement, and deterministic choices from a fixed set-like global well-order permit set-valued transfinite recursion with class parameters and its class union. No general recursion with class-valued stages, no class Zorn lemma and no unrestricted class truth predicate is assumed.

Theorem 15.1 (Class back-and-forth; prior result). *Let H_1, H_2 be proper-class algebraically closed fields of the same characteristic. Every isomorphism between set-sized subfields of H_1 and H_2 extends to a class field isomorphism $H_1 \rightarrow H_2$.*

This is Theorem 9.1 (`saut:thm:acfhomogeneity`) of `docs/surcomplex/surcomplex-field-automorphisms`, with the same hypothesis (Global Choice). Sources 03 and 04 reprove it by the same route: enumerate both fields along Ord by set-like global well-orders, extend at a forth step by a root of the transported irreducible polynomial or by an element transcendental over the current target field — which exists by lemma 11.1 and proper-class size — do the symmetric back step, and take unions at limits. Source 03 included the proof because applying a theorem stated for set-sized fields to proper-class fields without such a check would leave a foundational gap; the collection’s statement is already for proper-class fields.

Second route (source 08). Let H_1, H_2 be proper-class algebraically closed fields of characteristic zero that are saturated over every set of parameters. Every set-sized partial elementary map $H_1 \rightarrow H_2$ extends to a class isomorphism. At a forth step, transport the complete type of the next uncovered element over the current domain, a set by quantifier elimination; it is finitely satisfiable in H_2 by partial elementarity, and set-saturation of H_2 realizes it, outside the current range because the type records inequality with every element of the domain. Back steps are symmetric, and the least realization in a fixed global well-order makes each step specified. The method is classical [6, 7].

Theorem 15.2 (Old and new pure surcomplex fields; sources 03, 04, 08). *For every N -set $A \subseteq \text{SC}^M$ there is a class field isomorphism $\Phi : \text{SC}^M \rightarrow \text{SC}^N$ fixing A pointwise. If $M \neq N$, no such isomorphism maps $\mathbf{No}^M$ onto $\mathbf{No}^N$, and none satisfies $\Phi \circ c_M = c_N \circ \Phi$.*

Proof. Both fields are proper classes, since each contains every surreal ordinal in its real coordinate, and are algebraically closed in N (proposition 3.1). The identity on the field generated by A is an isomorphism of set-sized subfields; apply theorem 15.1. (Along the second route: the identity on A is partial elementary by proposition 3.1, both fields are set-saturated by theorem 11.2.) If $\Phi[\mathbf{No}^M] = \mathbf{No}^N$, its restriction would be an isomorphism of real closed fields, contrary to corollary 11.6; class Replacement supplies the set-image condition. An isomorphism intertwining the conjugations would carry one fixed field onto the other. $\square$

The inclusion $\text{SC}^M \hookrightarrow \text{SC}^N$ and the isomorphism Φ are different maps: for $M \subsetneq N$ the inclusion is a proper embedding, and the existence of Φ does not make it surjective. Φ is noncanonical. It need not preserve old coordinates, the natural valuation, the real axis, simplicity, an exponential, or a derivation, and no such compatibility is claimed. Theorem 15.2 settles, under Global Choice, source 01's question on class isomorphisms $\text{SC}^M \cong \text{SC}^N$ (section 19).

16 Gap invariants of surcomplex real forms

A *real form* of an algebraically closed field H of characteristic zero is a real closed subfield $F \subseteq H$ with $H = F(i)$; its conjugation fixes F and sends i to $-i$. This is the usage of the automorphism report (`saut:sec:realforms`). For a real closed class field F , let $\text{GapSpec}_N(F)$ be the class of cofinality pairs of its set-presented gaps, and let $g_N(F)$ be the least size of an omitted separated N -set pair, with the bookkeeping value $g_N(F) = \infty$ (not a cardinal) when every set pair is filled.

Lemma 16.1 (Invariance; source 03). *A field isomorphism between real closed class fields preserves their gap spectra and least omitted-pair sizes. Two involutions of SC^N are conjugate by a class field automorphism if and only if their fixed real fields are isomorphic.*

Proof. A field isomorphism preserves nonzero squares and hence the unique order; its restriction to an N -set is a set bijection by class Replacement, so it preserves cardinalities, cofinalities and the existence of separators. The second statement is the conjugacy criterion of `saut:sec:realforms`: restrict a conjugating automorphism, or extend $f : F_1 \rightarrow F_2$ by $x + iy \mapsto f(x) + if(y)$. It is the transport mechanism, not a new invariant. $\square$

Theorem 16.2 (A forcing-sensitive real form without set-sized cofinal subsets; sources 03, 04, 08). *Let $M \subsetneq N$, choose $\Phi : \text{SC}^M \rightarrow \text{SC}^N$ by theorem 15.2, and put*

$$j = \Phi c_M \Phi^{-1}, \quad H = \Phi(\mathbf{No}^M).$$

Then j is a field involution of SC^N with fixed field H and $\text{SC}^N = H(i)$, and

$$\text{GapSpec}_N(H) = \{(\text{cf}^N \mathbf{b}(s), \text{cf}^N \mathbf{b}(s)) : s \in \text{Fresh}(M, N)\}, \quad (16.1)$$

$$g_N(H) = \delta(M, N). \quad (16.2)$$

H has no N -set cofinal or coinital subset, and every N -set of elements of H is bounded above and below in H . Nevertheless j is not conjugate to c_N by any class field automorphism of SC^N .

Proof. Transport of structure gives $\text{Fix}(j) = \Phi(\text{Fix}(c_M)) = H$ and $\text{SC}^N = H(i)$; even if $\Phi(i) = -i$ the quadratic extension is unchanged. The displayed equations follow from corollary 8.6, theorem 9.2, and lemma 16.1. A cofinal N -set in H would pull back to a cofinal set in $\mathbf{No}^M$, contrary to lemma 3.3; the same holds for coinitial sets and for bounds. Finally $\mathbf{No}^N = \text{Fix}(c_N)$ fills every N -set cut, so $g_N(\mathbf{No}^N) = \infty \neq \delta(M, N) = g_N(H)$; the fixed fields are not isomorphic, and lemma 16.1 gives non-conjugacy. $\square$

Theorem 16.3 (Elementarily equivalent, not conjugate; source 04). *In the situation of theorem 16.2:*

- (i) $\text{Fix}(c_N) = \mathbf{No}^N$ and $\text{Fix}(j)$ is isomorphic as an ordered field to $\mathbf{No}^M$; both fixed fields are real closed, and $\text{SC}^N = \text{Fix}(c_N)(i) = \text{Fix}(j)(i)$;
- (ii) c_N and j are not conjugate by any class field automorphism;
- (iii) (SC^N, c_N) and (SC^N, j) are elementarily equivalent, but their external saturation spectra differ: for uncountable κ the second is κ -saturated exactly when $\kappa \leq \delta(M, N)$, whereas the first is set-saturated; at ω the second is saturated exactly when $\mathbb{R}^M = \mathbb{R}^N$.

Proof. (i) and (ii) are theorem 16.2; source 04 derives (ii) from corollary 11.6. Both expansions are interpreted, with the same coordinate rules, in real closed fields, and RCF is complete, so they are elementarily equivalent. Their spectra are those of their fixed fields by theorem 11.4 transported along Φ , and corollary 6.4, theorem 7.1, and lemma 2.3(iii) compute them. $\square$

Three statements must not be confused: the embedded real axes are different; the real axes are not isomorphic as fields; and the involutions are not conjugate. Applying a phase automorphism to one real axis proves the first but produces a *conjugate* involution and an isomorphic fixed field. The nonisomorphism is detected by external types, not by a first-order sentence.

Remark 16.4 (What is new relative to the automorphism report). Proposition B.1 (saut:prop:conjugacycriterion) of docs/surcomplex/surcomplex-field-automorphisms states that an involution of $\mathbf{No}[i]$ is conjugate to c if and only if its fixed real field has the set-cut interpolation property. It was present at the sources' pin as prop:conjugacycriterion, and none of the four sources cites it. Read in N , its easy direction (the property holds in $\mathbf{No}^N$ by Conway cuts and is preserved by ordered isomorphisms) already shows that j is not conjugate to c_N : $\text{Fix}(j) \cong \mathbf{No}^M$ omits an N -set cut whenever $M \neq N$ (theorem 1.1). The non-conjugacy clauses of theorems 16.2 and 16.3 are therefore consequences of that proposition together with the spectrum theorem. What is genuinely new in this part is:

- (a) *no set-sized cofinal subset*: the transported real forms, like $\mathbf{No}^N$ itself, have no set-sized cofinal subset, so the cofinality obstruction of the automorphism report's Theorem 10.2 (saut:thm:inequivalentrealform, a real form with a countable cofinal subset) cannot separate them; the obstruction lies inside a bounded cut;
- (b) *several forms separated from one another* by the invariant g_N and the gap spectrum (theorem 16.6), which the interpolation criterion alone does not do, since it only separates a form from c ;
- (c) *agreement with c_N on ordinary complex numbers* (corollary 16.5);
- (d) *elementary equivalence with different external spectra* (theorem 16.3), and the exact values of the spectra.

These continue, but do not settle, the automorphism report's open "real-form and continuity classifications" (its Section 14): no classification of real forms or involutions is obtained.

Related (batch 32). The automorphism report's finite-symmetry sections classify a different class of involutions: those of $\mathbf{No}[i]$ that preserve the valuation, Hahn summation and $\mathbf{Oz}[i]$. Two of them are conjugate by such an automorphism if and only if their restrictions to $\mathbb{C}$ are conjugate in $\text{Aut}(\mathbb{C})$, equivalently if and only if the coefficient fixed fields are isomorphic, and the fixed pair of each is a Hahn field over a real closed coefficient field of transcendence degree $2^{\aleph_0}$ with its integer part (`saut:fs:thm:main-involutions`, `saut:fs:prop:conjugacy`). That classification assumes the valuation data; it says nothing about the pure-field real forms transported here, whose compatibility with a valuation structure is question 19.6.

Corollary 16.5 (Agreement on prescribed ordinary data; source 03). *Let $A \subseteq \text{SC}^M$ be a set-sized subfield stable under c_M (as a subfield of SC^N it is then stable under c_N). The involution j can be chosen to agree with c_N on every element of A . If $\mathbb{R}^M = \mathbb{R}^N$, it can be chosen to agree with c_N on all of $\mathbb{C}^N$.*

Proof. Choose Φ fixing A pointwise. For $a \in A$, $c_M(a) \in A$, so $j(a) = \Phi(c_M(\Phi^{-1}(a))) = \Phi(c_M(a)) = c_M(a) = c_N(a)$. When no reals are added, $\mathbb{C}^M = \mathbb{C}^N$ is a set-sized common subfield stable under conjugation. $\square$

Thus agreeing with ordinary conjugation on every ordinary complex constant does not identify the surreal real axis even up to conjugacy. The obstruction is not an unusual choice of sign for i or a moved ordinary coefficient.

16.1 Several ground real forms in one field

Theorem 16.6 (Different thresholds separate real forms; sources 03, 04). (i) (source 03) *Let $r \geq 1$, and suppose that the models M_a , $a < r$, are proper transitive inner models of N with the same ordinals, each available as a class and satisfying the standing hypotheses, and that the cardinals $\delta(M_a, N)$ are pairwise distinct. Then SC^N has $r + 1$ pairwise nonconjugate field involutions, including c_N , none of whose fixed fields has a set-sized cofinal or coinital subset. If each M_a has the same reals as N , all of them can be chosen to agree with c_N on $\mathbb{C}^N$.*

(ii) (source 04) *Suppose $M_1, \dots, M_n$ satisfy the standing hypotheses and their old surreal fields have pairwise different external saturation spectra in N . Then SC^N has n pairwise nonconjugate real-closed-field involutions realizing those spectra, and c_N supplies one more whenever none of the listed spectra is the full set-saturation spectrum.*

Proof. Apply theorem 16.2 to each M_a with its own isomorphism $\Phi_a : \mathbf{No}^{M_a}[i] \rightarrow \text{SC}^N$. The fixed fields have least omitted-pair sizes $\delta(M_a, N)$, and that of c_N is ∞ ; all values differ, so lemma 16.1 separates the conjugacy classes. Cofinality and agreement come from theorem 16.2 and corollary 16.5. For (ii), a conjugacy between two transported involutions would give an isomorphism of fixed fields, which preserves external saturation; $\mathbf{No}^N$ is set-saturated. $\square$

Part (ii) realizes a given finite family of inner models; it does not assert that arbitrary families of saturation spectra can be produced by forcing without further hypotheses.

Corollary 16.7 (Arbitrarily long finite lists; source 03). *For every positive integer r there are finite forcing-tower constructions whose final surcomplex field admits $r + 1$ involutions as in theorem 16.6(i), all agreeing on every ordinary complex number. No large-cardinal assumption is needed. This is a relative forcing construction over a suitable ambient model, not an assertion that every universe contains the requisite grounds.*

Proof. Starting from M_0 , build $M_0 \subset M_1 \subset \dots \subset M_r = N$: at stage a choose in M_a an uncountable regular cardinal κ_a above all previously chosen ones, and force with $\text{Add}(\kappa_a, 1)$. Each later stage is closed under decreasing sequences of length below its own larger regular cardinal, so it adds no ordinal sequences of length below κ_a and preserves κ_a as a regular cardinal: a sequence in the final model of length below κ_a was present before the last forcing by closure, and repeating through the finitely many later stages puts it in M_a . The generic subset of κ_a added at stage a remains new over M_a in N . Thus $\delta(M_a, N) = \kappa_a$, pairwise distinct in N . No stage adds reals, so all the models have the same ordinary complex field. Apply theorem 16.6(i). $\square$

The finite tower is intentional: an infinite iteration needs support and preservation arguments not supplied by the finite proof, and no claim is made that an arbitrary proper class of gap invariants can be realized simultaneously. *Status (batch 37, 28 September 2026).* Infinite set-indexed families are obtained in Part VIII from product forcing rather than iteration, with one common threshold and distinct higher spectra (section 30); infinite iterations remain unanalysed (question 31.21).

Part VI

A Boolean completion of reverse simplicity (source 01)

17 A Boolean completion of reverse simplicity

Berenbeim’s Question 2 asks for a complete class Boolean algebra in which the reverse-simplicity order of surreal numbers embeds densely [15, p. 32]. The answer depends on the meaning of “complete.” We give an explicit positive answer for joins of sets, and a negative answer for joins of all subclasses under Global Choice. This section works in a single universe and does not use the pair $M \subseteq N$.

This is a different use of the sign tree from numerical cuts: *stronger* means extending a sign, regardless of its numerical value. Identifying $-$ and $+$ with 0 and 1, the partial order is

$$P = {}^{<\text{Ord}}2, \quad p \leq_P q \iff q \subseteq p.$$

Each condition is a set, and P is a proper class; it is the reverse of the non-strict initial-segment simplicity order.

17.1 A direct limit with set-coded elements

For each ordinal α let $B_\alpha = \mathcal{P}({}^\alpha 2)$, and for $\alpha \leq \gamma$ define the cylinder embedding

$$j_{\alpha\gamma}(A) = \{s \in {}^\gamma 2 : s \restriction \alpha \in A\}. \quad (17.1)$$

These are injective Boolean homomorphisms preserving arbitrary set-indexed unions and intersections, with $j_{\gamma\epsilon}j_{\alpha\gamma} = j_{\alpha\epsilon}$. Identify pairs (α, A) with $A \subseteq {}^\alpha 2$ when their images agree at a common later level. A proper-class equivalence class must not be made an element; instead represent each class by its unique *least-level pair*. Among the levels at most a representative’s level there is a least one from which it is a cylinder, and injectivity makes its subset unique. Taking the least-level pair is a definable operation on set-coded representatives.

Let $\mathbb{B}$ be the class of canonical pairs. To perform a finite Boolean operation, lift the arguments to a common level, operate in the power set there, and take the canonical pair; the result is independent of the common level by (17.1). Thus $\mathbb{B}$ is a class Boolean algebra whose elements are sets.

Theorem 17.1 (Set-complete dense completion). *$\mathbb{B}$ is complete for every set-sized family of its elements. The map*

$$e : P \longrightarrow \mathbb{B} \setminus \{0\}, \quad e(p) = [(b(p), \{p\})],$$

is an injective dense order embedding; in particular $e(p) \leq_{\mathbb{B}} e(q) \Leftrightarrow q \subseteq p$.

Proof. Let S be a set of canonical pairs and choose an ordinal γ at least every level occurring in S . Lift all of them to B_γ and take the union; its canonical pair is their join, since any upper bound can be moved to a common later level where the power-set union lies below it, and cylinder embeddings preserve that union. Complements give meets; the empty join and meet are 0 and 1.

A condition p determines its nonempty cylinder of extensions. If p extends q , its cylinder lies in that of q ; if they disagree, the cylinders are disjoint; if p is a proper prefix of q , the cylinder of p is not contained in that of q (extend p with the opposite sign at the next coordinate of q). This gives the order equivalence and injectivity. Finally, if $b \neq 0$ is represented by (α, A) with $A \neq \emptyset$ and $p \in A$, then $e(p) \leq b$, so the embedding is dense. $\square$

The construction answers the existence question when “complete” means set-complete, the size convention in which a class Boolean algebra has joins of its set-sized subsets, as in Holy–Krapf–Lücke–Njegomir–Schlicht [16]. General existence and uniqueness questions for class forcing are subtle; the construction works directly for this binary tree. It is not claimed that every class partial order admits the same construction, nor that this completion is the unique class completion.

17.2 Why joins of all subclasses are impossible

For the negative statement only, work in GBC with Global Choice. A class Boolean algebra is *class-complete* if every subclass of its carrier, including proper subclasses, has a supremum in the carrier. This is strictly stronger than set-completeness.

Lemma 17.2 (Diagonal obstruction). *In GBC with Global Choice, a class Boolean algebra with an ordinal-indexed proper-class antichain of nonzero elements is not class-complete.*

Proof. Let $(a_\alpha)_{\alpha \in \text{Ord}}$ be such an antichain. The carrier is a proper class, and Global Choice gives a class surjection $h : \text{Ord} \rightarrow \mathbb{B}$ (restrict a set-like global well-order). Let $A = \{\alpha : a_\alpha \not\leq h(\alpha)\}$. If $\{a_\alpha : \alpha \in A\}$ had a supremum b , then

$$a_\alpha \leq b \iff \alpha \in A. \quad (17.2)$$

Indeed, for $\alpha \notin A$, a_α is disjoint from every member of the subclass, so $\neg a_\alpha$ is an upper bound and $b \leq \neg a_\alpha$; since $a_\alpha \neq 0$, $a_\alpha \not\leq b$. The other direction is the definition of supremum. Choose γ with $h(\gamma) = b$; then $\gamma \in A \Leftrightarrow a_\gamma \not\leq b \Leftrightarrow \gamma \notin A$, a contradiction. All classes used are legitimate comprehension instances; no collection of all classes is formed. $\square$

Theorem 17.3 (No all-subclass completion). *In GBC with Global Choice there is no class-complete Boolean algebra with a dense order embedding of $P = {}^{<\text{Ord}}2$ into its nonzero part.*

Proof. The signs $p_\alpha = +^\alpha \frown -$, $\alpha \in \text{Ord}$, are pairwise incompatible. Under a dense embedding their images are pairwise disjoint, since a nonzero meet would contain the image of a condition, a common extension of incompatible signs. They form an ordinal-indexed proper-class antichain; apply lemma 17.2. $\square$

The two answers are consistent: a set-sized family of cylinders has a common level at which its join is computed, and a proper-class family need not. Set-completeness neither implies nor purports to imply all-subclass completeness. This also prevents the misleading claim that Question 2 requires an inconsistency of ordinary set or class theory.

Part VII

Scope, questions and limits

18 Scope and significance

18.1 What the new information consists of

The usual statement that the full surreal class realizes every set-sized cut is internal to a universe. The main theorem explains exactly how it changes when the elements are held fixed while the allowed families of parameters change. Its most distinctive features are these:

1. a cut with one actual ground side is filled even if its other side is a new and arbitrarily large set;
2. a first new sequence can be encoded by a cut of the same cardinal length, however large its ordinal range;
3. every set-presented failure of the cut principle comes from a unique fresh sign, and is symmetric;
4. removing the distinguished real structure by passing to the pure surcomplex language erases the obstruction entirely.

Together they identify a concrete set-theoretic invariant inside the external order of the old surreal field, and show that “the model theory of surcomplex numbers” has no language-independent saturation answer.

The interval method (source 01) uses only set-sized interpolation inside the ground order, enough room to split an interval into any prescribed set of pairwise disjoint subintervals, and a ground-set containment property for external parameter sets. Surreal numbers supply all three explicitly. Other definable universal orders with the same properties may be worth studying; no classification of such orders is claimed.

18.2 Limits of the assertions

The argument does not address exponential or differential expansions of **No**, the Berarducci–Mantova derivation, or expansions naming simplicity. Their types are not classified by real closed cuts, so the sufficiency proof cannot simply be copied after adding those operations; even absoluteness of the individual old operations would not give a type-realization theorem in the expanded language. An elementary reduct can be well behaved while an expansion has additional omitted types [8, 7].

The results do not assert that every class function or arbitrary proper-class cut is available in ZFC. Set-sized data, the ground predicate, and the stated uses of class principles are hypotheses, not conventions to be supplied later. The report is not a consistency proof for ZFC, GBC, measurable cardinals or the model configurations; the Prikry examples are conditional on a measurable cardinal. No countable set model whose ordinals stop at an external height is used; the same-ordinals hypothesis is essential to the birthday cover.

No theorem here is Lean-checked. The collection’s existing formal proofs and the source manuscripts are separate evidence. Successful typesetting is not mathematical verification, and the finite Python checks (section C) verify no transfinite claim.

18.3 A formalization route without class-sized type objects

These are proposed proof tasks, not existing declarations. A first target is the branch-decoder theorem: a set ordinal, a set alphabet, a table of intervals, and an abstract predicate for old sequences; the obligations are sibling uniqueness, limit nesting, totality of the default decoder and its agreement with an externally coded branch (source 01). Source 03 proposes starting instead with a generic ordered sign tree and an initial subclass predicate, proving lemmas 8.2 and 8.3 and theorem 8.4 as ordinal-combinatorial statements. The one-old-side theorem needs a bounded-stage cover and an internal cut operation, and should quantify over actual set-sized sides. For a universe-polymorphic proof assistant, the fragment theorem (theorem 12.1) is the cleanest end-to-end target: all structures can live in a large universe while the two set models remain distinct; full-class formulations become metatheorems or universe-relative schemes, not an illicit same-universe type of all surreals. The class back-and-forth should be formalized with set-stage maps and a class-theory interface. The surcomplex results separate into two reusable lemmas: external type realization for a proper-class algebraically closed field with ground-set parameter covers, and interpretation of the ordered base field through fixed squares in the quadratic complexification, which avoids repeating the collection’s quadratic-algebra development. The relation of proposition 14.1 to the Lean form of `found.thm:discrete` is recorded in remark 14.4.

19 Questions, and which of them the merge answered

The sources’ questions are recorded here with their status. They are directions suggested by the results, not claims that a particular published or catalogued question remains open.

Question 19.1 (External spectra of expansions; sources 01, 08; open). Determine the external saturation spectrum of old surreal exponential fields, and more generally of canonical expansions (surreal exponentiation, a specified derivation). What extra obstruction appears after naming them?

The main theorem gives a necessary condition, since the ordered-field reduct must be saturated; no sufficiency theorem for an expanded language is proved.

Question 19.2 (The cut-cofinality spectrum above δ ; source 01; asymmetric part answered by source 03). Source 01 asked to classify more of the external cut-cofinality spectrum than its first threshold, and to decide which asymmetrical pairs can occur.

Status. The asymmetric part is answered by source 03: every set-presented gap is symmetric, and the spectrum is exactly $\{(\text{cf}^N \mathbf{b}(s), \text{cf}^N \mathbf{b}(s)) : s \in \text{Fresh}(M, N)\}$ (corollary 8.6); source 01’s “set-character gaps” are source 03’s “set-presented gaps”. What stays open is which symmetric pairs above δ occur for a specified forcing, that is, computing the outer cofinalities of the lengths of fresh signs; proposition 13.2 computes only the first gap. This is related to, and must not be conflated with, the fresh-function spectra of Fischer–Koelbing–Wohofsky [14], whose spectrum-realization questions are not resolved here. *Status (batch 37, 28 September 2026): answered for the following forcings, open in general.* Part VIII reduces every case to the fresh-function spectrum of the pair: the characters are the outer cofinalities of the ground regular cardinals that carry fresh functions (theorem 22.7). It computes the full spectrum for one Cohen subset or one collapse of a regular κ , with any ground alphabet and no cardinal arithmetic ($\{(\kappa, \kappa)\}$, theorem 23.6); for any nonzero number of Cohen reals (proposition 23.12); for GCH Easton products (the Easton closure, theorem 24.3); for Prikry forcing ($\{(\omega, \omega)\}$) and for Namba forcing under CH ($\{(\omega, \omega), (\omega_1, \omega_1)\}$) (corollaries 28.8 and 28.10);

and for the intermediate grounds of sparse products and of countable GCH products (theorems 25.8 and 25.15). What stays open is the spectrum of an arbitrary forcing, of intermediate grounds of products with uncountable index sets at their accumulation points, and of these constructions without GCH (questions 31.10 to 31.12); the Fischer–Koelbing–Wohofsky realization questions remain unresolved.

Question 19.3 (Class isomorphisms of the pure surcomplex fields; source 01; settled under Global Choice). Under which weak class-theoretic assumptions is there a class isomorphism $SC^M \cong SC^N$?

Status. Under GBC with Global Choice there is one, even fixing any N -set of old parameters: theorem 15.2 (sources 03, 04, 08), by the collection’s class back-and-forth (`saut:thm:acfhomogeneity`). Only the question whether an assumption *weaker* than Global Choice suffices remains open. Set-saturation (theorem 11.2) gives local realizations but is not itself a global construction.

Question 19.4 (Completeness of the invariant; sources 03, 08; open, one part answered). For same-ordinal inner models M_0, M_1 of a fixed N , which invariants beyond the least omitted-cut size distinguish $\mathbf{No}^{M_0}$ and $\mathbf{No}^{M_1}$ as class orders or real closed fields in N ? Under what hypotheses does equality of $\text{GapSpec}_N(\mathbf{No}^{M_0})$ and $\text{GapSpec}_N(\mathbf{No}^{M_1})$ imply isomorphism?

Status. Source 08’s remark that its construction “provides one family of gap witnesses, but not a complete spectrum of all omitted cut characters” is answered for a fixed pair by source 03’s corollary 8.6. The classification question stays open: the spectrum records which cofinalities occur, not how many gaps of each kind occur or how they are arranged, and a classification may need an enriched incidence structure of cuts. *Status (batch 37, 28 September 2026): answered negatively for real closed fields; open for pure orders and under extra hypotheses.* Equality of the full gap spectra does not imply isomorphism as real closed fields, nor as pure fields: theorem 27.4 (source 12; ZFC and Cohen forcing only, with equal saturation at every cardinal and field embeddability exactly inclusion of index sets), theorem 28.5 (source 09; continuum many fields in one universe, CH sufficing for an instance) and theorem 29.2 (source 11; a measurable cardinal). Invariants beyond the least omitted-cut size that do distinguish old fields are the higher characters (sources 10, 11, 13, 14), the real trace $\text{Tr}_{\mathbb{R}}(\mathbf{No}^M) = \mathbb{R}^M$ (sources 12, 09) and ω -saturation (source 11); counting how often a character occurs does not help, since every occurring character occurs in every interval a proper class of times (proposition 26.4). Within the countable coordinate family the full spectrum does classify (proposition 25.17). Still open: whether equal full gap spectra *and* equal real traces imply isomorphism (question 31.2); the pure-order version, where no construction above applies (question 31.4); and positive classification theorems under further hypotheses.

Question 19.5 (Recovering more than outer cofinality; source 03; open). Can an intrinsic invariant of the real closed field recover the least *length* of a fresh sign, rather than only outer cofinalities of such lengths?

Birthday length is not part of the field language. The least length of a fresh sign is $\beta(M, N)$ (remark 10.4), but β is defined through the ambient sign tree, so this does not answer the question.

Question 19.6 (Additional compatible structure; source 03; open). Which transported real forms can also be required to preserve a prescribed valuation structure, a surreal exponential, or a specified derivation on an old set-sized substructure?

The pure-field extension lemma fixes arbitrary set-sized field data but does not extend isomorphisms in these stronger languages. Homogeneous-universal results for suitable surreal H -fields [7] have additional hypotheses and should not be substituted without checking them. *Status (batch 32): related, not answered.* Among involutions of $\mathbf{No}[i]$ required to preserve the valuation, Hahn summation and $\mathbf{Oz}[i]$, the automorphism report classifies the conjugacy classes by the coefficient involution (`saut:fs:thm:main-involutions`, `saut:fs:prop:conjugacy`; see the note after remark 16.4). It does not decide which transported real forms preserve such a structure.

Question 19.7 (Infinite families and relative real forms; sources 03, 08; open, finite version answered). Which external saturation spectra, and which infinite collections of regular thresholds or entire gap spectra, can be realized simultaneously by fixed fields of involutions of one SC^N that agree on $\mathbb{C}^N$ and have no set-sized cofinal subset?

Status. The finite-threshold version is answered by corollary 16.7 (source 03), and spectra arising from given inner models are realized by theorems 16.2 and 16.6. A classification of arbitrary real forms, or of infinite families, needs genuine iteration and spectrum analysis. This is also where the results continue the automorphism report’s open real-form classification (remark 16.4). *Status (batch 32).* Read in a single universe, with the threshold of a real form taken to be the least cofinality of a set-presented gap of its fixed field (the invariant of corollary 6.3), the clause on infinite collections of regular thresholds is answered without forcing by the report docs/surcomplex/first-kappa-coefficients (fkc:sb:thm:involutions, fkc:sb:rem:univ). Under Global Choice, for every class C of infinite regular cardinals it gives pairwise nonconjugate involutions of $\text{No}[i]$ that agree with conjugation on $\mathbb{C}$, whose fixed fields have no set-sized cofinal subset and have thresholds exactly the members of C ; their saturation spectra are initial segments of the cardinals and each gap spectrum is a single pair. Arbitrary saturation spectra, gap spectra with more than one character, and families of old fields No^M remain open, and no real form is classified. *Status (batch 37, 28 September 2026): answered for multicharacter spectra and for families of old fields; open for arbitrary prescribed spectra.* Part VIII realizes, under Global Choice, arbitrarily large set-indexed families of pairwise nonconjugate involutions of one SC^N whose fixed fields are isomorphic to actual old fields No^{M_A} , share one saturation threshold, have distinct multicharacter gap spectra and no set-sized cofinal subset, and agree with c_N on $\mathbb{C}^N$, or on any prescribed set-sized conjugation-stable core (theorems 30.5, 30.7, 30.9 and 30.12); also continuum many with one *identical* full spectrum, agreeing only on $\mathbb{C}^M$, since agreement on $\mathbb{C}^N$ is impossible there (theorem 30.3 and proposition 30.4), and a ZFC version for one set field (theorem 30.11). Still open: which arbitrary saturation spectra or gap spectra, prescribed in advance, can be realized simultaneously; infinitely many real forms with one identical full spectrum agreeing on $\mathbb{C}^N$ (question 31.7); and any classification of real forms.

Berenbeim’s Question 2 (2020 notes) is answered in the precise set-complete interpretation of theorem 17.1, with the class-complete interpretation refuted under Global Choice (theorem 17.3). It is not an open question of the collection.

20 What this report does not claim

Every limitation, disclaimer and priority caveat of the four sources is kept. They are listed per source; items written for the merge are marked M. The non-claims of sources 09–14 are listed in section 32.

Source 01

- 01.1 The principal spectrum and decoding results are proposed contributions with proofs, not certified historical priority; no Lean verification and no independent referee review.
- 01.2 Classical surreal facts, quantifier elimination and standard forcing facts are not claimed; several consequences are deliberately labelled consequences.
- 01.3 Berenbeim’s Question 2 comes from 2020 research notes, not a peer-reviewed conjecture. The construction may be folklore in class forcing; it is not a claim that Boolean completions were previously unavailable, and general class-forcing completion theory predates the notes. The notes’ broader suggestions about embeddings of universes are not settled.

- 01.4 External means computed in N , not in a larger metatheory; countable transitive models must be read internally. No unrestricted class truth predicate is assumed.
- 01.5 Field elementarity does not assert $M \preccurlyeq N$. The ground-bound lemma is not a covering property of a forcing, and the containing set may be much larger than the external set.
- 01.6 The decoder does not infer that an external branch is old because its nodes are old. Interval widths are not claimed to tend to zero.
- 01.7 Regularity of δ is an elementary minimal-length fact, not a new characterization of regular cardinals. The gap invariant excludes cuts with proper-class cofinality on one side.
- 01.8 Set-saturation of the pure field does not produce a class isomorphism; the nonisomorphism corollary requires set-sized images; proposition 12.3 is a consequence, not a new field theorem.
- 01.9 The distributivity equivalence is standard; statements are pair by pair, not about every generic extension of a poset unless so quantified; no preservation of larger cardinals is asserted.
- 01.10 A topology looking only at selected scales is a different structure and is not used.
- 01.11 The Boolean completion is not claimed for every class partial order nor claimed unique; set-completeness does not imply class-completeness; the negative result assumes Global Choice, with no optimality claim for that assumption.
- 01.12 No classification of other orders with the interval-method properties is claimed.
- 01.13 Exponential and differential expansions are not addressed. Arbitrary class functions and proper-class cuts are not claimed available in ZFC. Not a consistency proof for ZFC, GBC, measurables or the model configurations; Prikry is conditional on a measurable. Nothing is Lean-checked; typesetting is not verification.
- 01.14 Its three questions are recorded with their current status in section 19.
- 01.15 Its repository review was limited: the foundations report was read to line 180, the automorphism report only in its opening text, and code searches used a default-branch index. The repository was not modified or built. Literature searches were not exhaustive. Its novelty claim was: candidate contributions not located in the consulted literature or the stated repository review, subject to independent checking and further priority research.

Source 03

- 03.1 Historical novelty is not certified by an exhaustive literature review, peer review or formal verification; no named published conjecture is claimed solved.
- 03.2 The forcing theory of fresh functions is not renamed as a new invariant; quantifier elimination is standard; the collection's class back-and-forth, conjugacy/fixed-field criterion and countably cofinal real form are not claimed.
- 03.3 No set of all surreals, no class-object of all automorphisms, no class Zorn lemma and no general class-valued recursion; sizes are relative to N ; proper-class size in N is not proper-class size in every outside universe.
- 03.4 Principal cuts and cuts needing a proper-class presentation are not considered.
- 03.5 The invariant recovers a threshold and cofinality data only, not the inner model, the forcing or the sign lengths; equality of spectra is not proved to classify the fields.

- 03.6 $\kappa = \omega$ must not be substituted into the field-saturation clause. The invariant belongs to a pair of universes, not to a poset.
- 03.7 The $\text{Add}(\kappa, 1)$ computation does not compute the full gap spectrum or exclude larger gaps, and asserts nothing about preservation of higher cardinals. No Prikry property is reproved.
- 03.8 Set-saturation of a proper-class algebraically closed field is an established mechanism, not a novelty claim; class back-and-forth already exists in the collection; the inclusion is not the isomorphism Φ ; the conjugacy criterion is the transport mechanism, not the new invariant.
- 03.9 The conjugation structures are elementarily equivalent; Φ need not preserve simplicity, the natural valuation, the real exponential or a derivation, and no such compatibility is claimed.
- 03.10 Finite towers only: no infinite iteration, and no arbitrary proper class of invariants realized simultaneously.
- 03.11 Set-discreteness is classical and already in the collection; no convergence of an ordinary sequence to a missing point is asserted.
- 03.12 Tempting shortcuts are false or unproved: replacing a new ordinal-valued sequence by a binary one of the same length without a coding proof; interchanging ground and outer cofinalities; letting global cofinality control bounded gaps; inferring type preservation over new parameter sets from elementary equivalence; assuming a pure-field isomorphism preserves the real axis; inferring a full-class property from one set-sized workspace.
- 03.13 Not a Lean formalization; the repository was not rebuilt or axiom-audited; the Python program cannot test freshness, forcing, uncountable regularity or the class arguments; its proposed formalization order names proof tasks, not existing declarations.
- 03.14 Its questions are directions, not assertions that a published question remains open.
- 03.15 The repository comparison was focused, not exhaustive. Fischer–Koelbing–Wohofsky’s spectrum-realization questions are not resolved, and their convention for spectra of a forcing must not be conflated with cofinalities in a particular outer universe. The novelty statement is narrow; a specialist may identify an earlier theorem subsuming part or all of the arguments.
- 03.16 Its shipped `03-fresh-sign-gaps-PROOF-STATUS.md` lists what was established by argument, the established ingredients and seven boundaries, and states that no Lean proof, theorem-prover verification, forcing simulation or repository build was performed and that independent specialist review is still appropriate.

Source 04

- 04.1 Proposed contributions; no exhaustive priority claim, no solution of a named published problem, no Lean certification, no peer review.
- 04.2 Ehrlich’s absolutely saturated models and the universal H -fields of van den Dries–Ehrlich are precedents. External means internal to the specified outer universe.
- 04.3 Definability of M is a convenience. No claims about arbitrary externally countable models; class isomorphisms add Global Choice.
- 04.4 Not new: cut realization in **No**, quantifier elimination, saturation of algebraically closed fields with enough independent elements, conjugation naming a real closed fixed field, and many surcomplex automorphisms and real axes in the collection.

- 04.5 Birthday minimization over old candidates is not claimed absolute; “initial” means under simplicity, not numerically; the cover lemma is not the κ -cover property; no equivalence between $\eta_{\aleph_0}$ and $\aleph_0$ -saturation of a real closed field; no class-valued recursion.
- 04.6 No regularity assumption on κ ; cardinals are computed in the outer universe; its name for the first new length was local notation, not a claim to a new forcing invariant; regularity is a set-theoretic fact.
- 04.7 In the pure-field section the contrast is the contribution, not a classification of types.
- 04.8 Forcing statements quantify over every generic extension; one extension is not enough. The comparison table does not alter internal saturation.
- 04.9 The class back-and-forth is noncanonical and preserves no coordinates, valuation, real axis, simplicity or exponential. The several-spectra corollary concerns a given finite family; arbitrary spectra are not claimed producible.
- 04.10 Full-class fine-topology results are already in the collection.
- 04.11 The full text of Ehrlich’s 1989 paper was not accessible to it. No extension to exponential or differential languages; the Rangel–Mariano and Bertram questions are not resolved. Its novelty claim: proposed original results, with priority remaining to be independently established.
- 04.12 A code-search response marked incomplete was not treated as an exhaustive negative; the nonduplication statement is not a claim about later commits.
- 04.13 Its sensitive proof points (old elements are not an old set; a minimal fresh length is a limit; ordinal labels cannot be replaced by binary digits; class recursion is not silently assumed; external cardinal bounds belong to the outer universe; finite parameters need not give a finite type; forcing preservation quantifies over all generics; a pure field does not remember a chosen real form) and its finite-check limits are kept; no verified priority or solved named problem.

Source 08

- 08.1 Proposed contributions; classical theory credited; no Lean, no peer review, no named problem solved, no exhaustive priority claim.
- 08.2 The algebraically closed field statement is not a new principle; class back-and-forth is classical and the isomorphism is a corollary; exponentials, derivations and unrestricted automorphisms are not classified.
- 08.3 Global Choice is used only for the class isomorphisms. No countable set model and no set-valued field object is used.
- 08.4 The cover lemma is not a covering or approximation property; δ is undefined when $M = N$; ordinal sequences must not be replaced by binary ones.
- 08.5 The algebraically closed field argument is a classical instance; conjugation is not definable in the pure field.
- 08.6 Distributivity needs the uniform quantifier (lottery sum). Closure facts are standard; no GCH; no claim about preserving higher cardinals; Prikry properties are imported, not reproved.
- 08.7 Its real-form corollary does not classify involutions; the collection’s back-and-forth, conjugacy criterion and countably cofinal real form are prior work.

- 08.8 An omitted cut is not a Cauchy sequence, and no topological completion theorem is claimed. The ordered exponential field, the Berarducci–Mantova derivation and simplicity expansions are not treated. Nonisomorphism compares inner and ambient universes only; nothing is claimed about $\mathbf{No}^{M_1} \cong \mathbf{No}^{M_2}$.
- 08.9 Its questions are not claims about previously catalogued open questions.
- 08.10 No repository file was modified and no independent build was done; the finite checks do not establish transfinite claims.
- 08.11 It inspected a scan of Ehrlich’s 1989 paper; its repository and literature searches were limited, and absence of a search hit is weak evidence. Its warranted conclusion: a specific relative-universe theorem package not found in the inspected literature or repository material, with historical novelty subject to broader expert checking.

Added by the merge

- M.1 The merge adds no mathematical claim beyond the observation of remark 10.4; its other additions relate statements and reports, or restate a source’s statement in another source’s notation.
- M.2 The real-form results continue, and do not settle, the real-form classification question of the automorphism report.
- M.3 The relation to the Lean form of `found.thm:discrete` (remark 14.4) is information for formalization; nothing in this report is formalized.
- M.4 The sibling report is cited by directory and by what it is described to prove; this report makes no claim about its numbering or its other statements.
- M.5 The source manuscripts were not rebuilt for the merge; their page counts are as delivered.

This gives 15, 16, 13 and 11 items from sources 01, 03, 04 and 08, and 5 added by the merge: 60 in all.

Part VIII

Full gap spectra, relative grounds and classification (sources 09–14)

21 How Part VIII was assembled

This part was added on 28 September 2026. It merges six manuscripts of that date, called *sources 09–14* here, into one addition to the report. The numbers continue the report’s own local sequence (its sources 01, 03, 04 and 08 are the manuscripts of Parts I–VII), and they are the file prefixes of the delivered code and data. Parts I–VII, their numbering and their labels are unchanged; the only edits outside this part are dated status pointers (on the title page, in section 1.6, after theorem 9.2, proposition 13.2, and corollary 16.7, in the statuses of questions 19.2, 19.4 and 19.7, and at the head of sections C and 20), five new bibliography entries, and the notation macros of the preamble.

21.1 Six manuscripts, one addition

All six manuscripts were written against commit `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9` of VladimirReshetnikov/ProveIt (short `e21766d`, 25 September 2026), at which this report was the four-source merge of Parts I–VII, unchanged since. They arrived in commit `37e61c1fd` and were placed, as additions to this report, in commit `0e53d1034` (batch 37 of the collection’s intake, manuscripts 01–06). They do not know one another. No source manuscript, source README or source PDF is shipped; their code, data and audit files are, with the prefixes listed in the README.

Source	Manuscript (batch-37 number, archive, pages)	Contribution here
09	<i>Full Gap Spectra of Old Surreal Fields: forcing transfer, Easton realization, and continuum many nonisomorphic fields with identical gap spectra</i> (01, <code>proveit_gap_spectra</code> , 23 pages)	Transfer with fixed-width blocks; the weak-approximation reading; Easton realization (base); pointwise versus union spectra; common-small-forcing invariance by canonical names (section 28); the rational-cut invariant (second route to the real trace); Cohen amplification: continuum many old fields with one full spectrum, pairwise nonisomorphic as pure fields; exact Prikry and Namba spectra; same-spectrum involutions agreeing on $\mathbb{C}^M$ and the obstruction to agreement on $\mathbb{C}^N$. Files prefixed <code>09-cohen-amplification-</code> .
10	<i>Exact Gap Spectra of Ground-Model Surreal Fields: fresh functions, forcing, intermediate grounds, and many nonconjugate surcomplex real forms</i> (02, <code>ProveIt_Exact_Surreal_Gap_Spectra</code> , 22 pages)	The delimiter code; the filter-base bound (base of the density bound); the initial-segment route to the Cohen and collapse singletons; Easton examples; isolated-coordinate localization; the 2^X Boolean family with one threshold (theorem 25.21); involutions agreeing on any set-sized conjugation-stable core, elementarily equivalent over it. Files prefixed <code>10-isolated-coordinates-</code> .
11	<i>Beyond the First Omitted Cut: full gap spectra of old surreal fields, Cohen collapse, and surcomplex real forms</i> (03, <code>ProveIt_Full_Gap_Spectra</code> , 25 pages)	Base of the transfer; the lower barrier; every omitted set cut is a genuine gap; the external density bound; the generalized Cohen theorem for every ground alphabet (base); relative finite-complement spectra of sparse products; same-saturation families of any length; the Cohen–Prikry tower (section 29); the tower bound. Files prefixed <code>11-finite-complements-</code> .
12	<i>Beyond Gap Spectra: real traces and embeddability of old surreal fields</i> (04, <code>proveit_beyond_gap_spectra</code> , 22 pages)	The real trace and its functoriality (base); the Boolean family with constant spectrum $\{(\omega, \omega)\}$ and embeddability exactly inclusion, with no large cardinal, CH or GCH (section 27); realization of every ground set-sized partial order; the square-chain-condition bound; the generalized Cohen theorem by explicit collapse; equivariant embeddings of involutions. Files prefixed <code>12-real-traces-</code> .
13	<i>Cofinality Compression and Coordinate Spectra of Old Surreal Fields</i> (05, <code>Surreal_Gap_Spectra_Research</code> , 22 pages)	The transfer for arbitrary same-ordinal pairs; the Cohen and single-collapse singletons by substring enumeration; name localization and intersections of mutually generic extensions (base); the exact spectra of <i>all</i> coordinate grounds of a countable GCH product (base, theorem 25.15); binary intersections; local ubiquity (base); the continuum of equal-threshold real forms. Files prefixed <code>13-coordinate-spectra-</code> .

Source	Manuscript (batch-37 number, archive, pages)	Contribution here
14	<i>Beyond the First Gap: Exact Spectra and Boolean Families of Old Surreal Fields</i> (06, Surreal_Gap_Spectra_Research(1), 26 pages)	Localization under distributivity only; finite Boolean cubes without sparseness; the countable theorem at $\kappa_n = \aleph_{n+1}$ with its own preservation proof (second route); exact gap pairs of set-sized birthday cutoffs, including asymmetric boundary pairs (section 26); a ZFC theorem on real forms of one set field; the class version. Files prefixed 14-relative-cutoffs-.

Why one addition. Each manuscript answers one or more of this report's named questions questions 19.2, 19.4 and 19.7, and all six prove the same spine: the outer-cofinality pushforward from fresh functions to gap characters. Printed side by side, they would repeat the fresh-sign correspondence (already theorem 8.4) six times, the transfer six times, the Cohen singleton five times and the class back-and-forth (already theorem 15.1) six times. Otherwise they are complementary, and several answer each other's questions (section 31).

Bases, and what is printed once. Where two or more sources prove the same statement it is printed once, with every proving source named at the statement, from the *base* named below; genuinely different proofs are kept as marked second routes.

- Fresh-sign correspondence: all six reprove theorems 8.4 and 8.5; it is not reprinted (remark 22.2 records their variants).
- Transfer: base source 11, whose proof is explicitly collapse-sensitive; the three block codes of the sources are all kept.
- Density bound: base source 10 (an arbitrary cofinal base of the actual generic filter, not required to lie in M); sources 11, 13, 14 (a ground dense set) and 09 (the whole poset) are special cases, and source 12's square chain condition is a second route.
- Generalized Cohen spectrum: base source 11 (every ground alphabet with at least two letters); second routes from sources 10 (initial-segment bases) and 12 (explicit collapse); sources 13 and 14 use source 11's route.
- Easton realization: base source 09; sources 10 and 11 prove the same.
- Name localization: statement from source 14 (distributivity suffices), proof also by closure from sources 10, 11, 13 as a second route; intersections: base source 13.
- Countable coordinate spectra: base source 13 (any strictly increasing uncountable regular κ_n); source 14 proves the case $\kappa_n = \aleph_{n+1}$, with its own preservation proof as a second route.
- Real trace: base source 12; source 09's rational-cut invariant is the same set, with a second proof.
- Equal full spectra without isomorphism: three constructions, all kept (sources 12, 09, 11); source 12 has the weakest hypotheses.
- Local ubiquity: base source 13; source 14's affine route kept.
- Class back-and-forth: all six reprove theorem 15.1; not reprinted (remark 30.1).

Where the merge had to choose. (a) Notation, fixed in section 21.3; no normalization changes. (b) Index direction: subscripts on intermediate grounds always list *retained* coordinates; three sources indexed by omitted ones, and their statements are rewritten (section 21.3). (c) Questions answered inside the cluster are

recorded as answered (section 31). (d) Two sentences of the sources overstate what this report said; they are corrected where they are printed (remarks 25.25 and 28.11). Text written for the merge is confined to this section, the framing remarks that relate sources, the question statuses, the non-claims marked M and section 33; it adds no mathematical claim beyond the comparisons stated as merge observations.

21.2 What the addition answers

- **Transfer** (theorem 22.7; all six). For every same-ordinal pair $M \subseteq N$, the characters of the set-presented gaps of $\mathbf{No}^M$ are the outer cofinalities of the ground regular cardinals that carry fresh ordinal-valued functions: $\text{Char}_N(\mathbf{No}^M) = \{\text{cf}^N \lambda : \lambda \in \mathfrak{F}(M, N)\}$.
- **Question 19.2, computed families.** One Cohen subset, or one collapse, of an infinite regular κ gives exactly $\{(\kappa, \kappa)\}$, with no cardinal arithmetic (theorem 23.6); GCH Easton products give exactly the Easton closure of their coordinates (theorem 24.3); Prikry gives $\{(\omega, \omega)\}$ and Namba under CH gives $\{(\omega, \omega), (\omega_1, \omega_1)\}$ (corollaries 28.8 and 28.10); every coordinate ground of a countable GCH product has an explicit spectrum (theorem 25.15).
- **Question 19.4, negative for fields.** Equality of the entire gap spectrum does not imply isomorphism of old real closed fields, even with equal saturation at every cardinal: theorem 27.4 (source 12, ZFC), theorem 28.5 (source 09) and theorem 29.2 (source 11, a measurable cardinal). The least threshold and even the whole saturation spectrum do not classify (theorem 25.21, source 10, and others).
- **Question 19.7, families of old fields.** Arbitrarily large set-indexed families of pairwise nonconjugate involutions of one SC^N , with one saturation threshold and distinct multicharacter spectra, agreeing with c_N on $\mathbb{C}^N$ or on any prescribed set-sized conjugation-stable core (section 30).

What stays open is recorded in the dated statuses of questions 19.2, 19.4 and 19.7 and in section 31. In particular it is open whether equal real traces *and* equal full gap spectra imply isomorphism (question 31.2), and whether equal full spectra imply isomorphism of the pure orders (question 31.4).

21.3 Conventions for Part VIII

The standing hypothesis (H) and the class discipline of section 1.2 apply throughout. Every forcing is a set forcing; forcing statements are relative, in the usual ground/extension sense, and none asserts that ZFC proves the existence of a transitive model of itself or of a generic filter over the universe. The real-form theorems use (GC), except theorem 30.11, which is a ZFC theorem about a set field. “Set-presented gap” has the meaning of definition 6.1; all six sources count exactly these, identify different presentations of one partition, and exclude cuts with a proper-class side. For an ordered class F of N write

$$\text{Char}_N(F) = \{\mu : (\mu, \mu) \in \text{GapSpec}_N(F)\},$$

the *scalar* (diagonal) spectrum. For an old field $\mathbf{No}^M$ no information is lost, since every set-presented gap is symmetric (corollary 8.6). For the set-sized cutoff fields of section 26, where asymmetric pairs occur, the full pair set GapSpec_N is used.

The sources used different letters, and several letters also carry a meaning elsewhere in this report. One symbol per notion is fixed here.

Notion	Symbol here	In sources 09–14
Scalar gap spectrum	$\text{Char}_N(\mathbf{No}^M)$	$G(M, N)$ (09); $\mathcal{G} = \text{Gap}_N(M)$ (10); $g(M, N)$ (11); Char_N (13); Γ_M^N , DiagGap (14)
Pair spectrum	GapSpec_N	GapSpec (09); GapSpec written Gap (12); Gap_N (13); GapPairs (14)
Fresh-function spectrum of the pair	$\mathfrak{F}(M, N)$	F (09); $\mathcal{F}_{M,N}$ (10); $F(M, N)$ (11); $\mathfrak{F}$ (12); $\text{Fresh}(M, N)$ (13); $\mathcal{F}$ (14)
“Some $f : \gamma \rightarrow \text{Ord}$ is fresh”	$\text{Fr}_\gamma(M, N)$	$\text{Fresh}_\delta(M, N)$ (09)
Union-over-generics spectrum of a poset	$\text{FRESH}(\mathbb{P})$	same (09, 12, 14; after [14])
Easton closure	$\text{Ecl}(A)$	EC (09); Ecl (10, 13); $E(A)$ (11)
Real trace	$\text{Tr}_{\mathbb{R}}(F)$	$\text{RCut}(K)$ (09); $\text{Tr}_{\mathbb{R}}$ (12)
Fixed-width one-plus block code	$\text{bcode}_\theta(f)$	unnamed (09, 11, 12, 14)
Delimiter code $\frown_\xi(+^{f(\xi)} \frown -)$	$\text{dcode}(f)$	$\text{Code}(f)$ (10)
Terminal-plus block code	$\text{code}^+(f)$	s_f (13); as in proposition 9.1
Retained-coordinate ground	M_A (A retained)	M_S (09), M_A (12, 14); retained; M_S, M_B, M_F (10, 11, 13): omitted
Old class fields	$\mathbf{No}^{M_A}$ written out	K_S (09); F_S (10); K, K_ξ (11); F_A (12); K_F (13); K_A (14)
Old set-sized cutoff fields	$(\mathbf{No}_{<\theta})^{M_A}$	F_A (14)
Surcomplex class	SC (roman)	$\mathbf{SC}$ (10–14)
Final universe	N	N (09–13); W (14)
Intermediate universe	W	W (09, 10); B, U (13, 14 in proofs)
Coordinate Cohen reals	y_n, y_i	c_n (09); c_i (12); c (11)
Transported involutions	j_A ; isomorphisms Φ_A	j_S, Φ_S (09); j_S, ϕ_S (10); j_ξ, φ_ξ (11); j_A, ϕ_A (12, 14); j_F, ϕ_F (13); J_A (14)
Generic filter of a small or Cohen factor	$G_{\mathbb{Q}}, G_{\mathbb{C}}$	H (09, 13)
Index cardinal of a family $(2^{<\kappa})^M$	χ	θ (10, 11)
	σ_κ	θ (12)
Bounding cardinals of sparse recursions	$\mathcal{S}i, \mathcal{S}\xi$	σ_i (10); δ_ξ (11)
Chain-condition cardinal	μ	χ (12)
Least omitted coordinate	n_*	b (14)
Range bound of a fresh function	θ	ρ (09 Lemma 3.3, 14 Theorem 4.1); χ (11); θ (12)

Index direction. A subscript on an intermediate ground always lists the coordinates it *retains*: $M_A = V[G \restriction A]$. Sources 09, 12 and 14 used this convention. Sources 10, 11 and 13 indexed by the *omitted* coordinates; their statements are rewritten. Thus source 10’s M_S is M_{K_S} with $K_S = \{\kappa_i : i \in \chi \setminus S\}$, source 11’s M_B is $M_{A \setminus B}$, and source 13’s M_F is $M_{\omega \setminus F}$. Omitted sets are written B and never appear as a subscript. Source 13’s subfamily “ $0 \in F$ ” and source 14’s “ $0 \notin A$ ” are the same subfamily.

Letters reserved elsewhere in this report keep their meaning here: $\delta = \delta(M, N)$ is the first new length and $\beta = \beta(M, N)$ the first new birthday (never a code length); θ is an ordinal bound on a range or a set of birthdays (so source 14’s cutoff λ^{++} keeps the letter θ , and source 13’s collapse alphabet θ is a range bound); ρ is the measurable cardinal of Prikry forcing; K is not used for proper-class fields; c_M, c_N are conjugations, so Cohen reals are renamed; $H = \Phi(\mathbf{No}^M)$ keeps its meaning in theorem 16.2, so generic filters are renamed; $g_N(F)$ is still the least omitted-pair size (for an old field with $M \neq N$, $g_N(\mathbf{No}^M) = \delta = \min \text{Char}_N(\mathbf{No}^M)$), while g also names generic functions, as in section 13.4; $\text{Fresh}(M, N)$ is still the class of fresh *signs*, and

fresh-function spectra are written $\mathfrak{F}(M, N)$. The letter ν is not used. Source-local letters for generic ordinals were renamed silently where they clashed with these reservations.

Tempting false readings. $\mathfrak{F}(M, N)$ is a set of *ground* regular cardinals, while $\text{Char}_N(\mathbf{No}^M)$ is a set of regular cardinals *of* N ; they are compared only through $\lambda \mapsto \text{cf}^N \lambda$. $\mathfrak{F}(M, N)$ is the spectrum of one actual pair of universes, not $\text{FRESH}(\mathbb{P})$, which quantifies over possible generic extensions. An equality “ $\text{Char}_N = \mathfrak{F}$ ” is an identification of the common regular cardinals of cofinality-preserving extensions only. “Isomorphic” for class fields always means a class isomorphism taking sets to sets. A statement about the *inclusion* order of old fields is not a statement about their intersections or composita.

22 The cofinality transfer

22.1 Fresh-function spectra of a pair

Definition 22.1 (Sources 09–14). For a same-ordinal pair $M \subseteq N$ and an ordinal γ , write $\text{Fr}_\gamma(M, N)$ if some $f : \gamma \rightarrow \text{Ord}$ in N is fresh over M (definition 4.1). The *fresh-function spectrum* of the pair is

$$\mathfrak{F}(M, N) = \{\lambda \in \text{Reg}^M : \text{Fr}_\lambda(M, N)\},$$

where Reg^M is the class of infinite regular cardinals of M .

The range of a fresh function is bounded by an ordinal, even though the notation permits arbitrary ordinal values; this makes all codes and name collections below sets (source 09). The notion is Fischer–Koelbing–Wohofsky’s [14], specialized to one actual pair. For a ground poset $\mathbb{P}$, their $\text{FRESH}(\mathbb{P})$ collects the ground regular cardinals at which *some* condition forces a fresh function; the pointwise spectrum of one generic G is $\mathfrak{F}(M, M[G])$, and for homogeneous forcing the two agree [14, Proposition 4.3], homogeneity meaning that for any two conditions an automorphism moves one to be compatible with the other. The distinction prevents a false realization argument: a lottery sum can have a union spectrum with many cardinals, although one generic chooses one summand, and the old surreal field of one universe sees the pointwise spectrum (sources 09, 11, 12, 14). For example (source 13), under GCH a lottery choice between adding a Cohen real and adding a Cohen subset of ω_2 has both thresholds across its generics, while a particular generic chooses one summand; the lottery-sum caution of section 13 is the same point.

Remark 22.2 (The fresh-sign correspondence, six times; merge note). All six sources reprove theorems 8.4 and 8.5 and credit them to Part II: source 09 Theorem 2.5, source 10 Theorem 3.3, source 11 Theorem 3.2 and Corollary 3.3, source 12 Theorem 3.4 (citing it as “Theorems 8.4–8.5”), source 13 Theorem 3.4 (“now established without a forcing hypothesis”) and source 14 Theorem 3.3. It is not reprinted. Their surjectivity arguments take a separator x of a presentation in N and use its *least new prefix* (sources 09, 11, 12), or the simplest separator and the minimality of its birthday (sources 10, 13, 14), where Part II uses convexity of $\text{Cone}(y)$; these are one argument in three wordings. Source 09 adds the endpoint argument of lemma 22.3, which is specific to old surreal fields: for arbitrary dense orders an omitted set pair need not define an endpoint-free gap. Source 11 records that the root of a fresh cone is the simplest separator of its gap, the unique member of least birthday, so birthday information is available in the ambient sign tree although it is not automatically preserved by an ordered-field isomorphism. Source 10 records that the correspondence classifies cuts, not Cauchy limits: a fresh sign may have very large birthday even when its gap has countable character, a whole cone rather than one point fills the cut, and nothing says the old field is dense in the new one. Source 14 warns that the correspondence cannot be used blindly at a birthday cutoff (section 26).

The old ordinal bounds of lemma 3.3 are likewise reproved by all six. *Second route to the positive lower bound* (source 11). If X is an N -set of positive old numbers, choose an old ordinal γ_0 above all their birthdays

and let ε have a plus at position 0 and minus signs at every later position through γ_0 . Every positive x begins with a plus, and its first difference from ε is a further plus or the end of x , both greater than a minus; so $0 < \varepsilon < x$. The argument does not require $X \in M$, and it is not a covering property asserting that X lies in an old set of its own size.

Lemma 22.3 (Every omitted set cut is a genuine gap; sources 11, 09). *Let A, B be N -sets of elements of $\mathbf{No}^M$ with $A < B$ and no $x \in \mathbf{No}^M$ satisfying $A < x < B$. Then both sides are nonempty, A has no largest and B no least element, and their downward and upward closures partition $\mathbf{No}^M$ into a set-presented gap.*

Proof. If one side is empty, the ordinal bounds of lemma 3.3 give a separator. If A has a largest element a , apply the positive lower bound to $\{b - a : b \in B\}$ and choose an old $0 < \varepsilon < b - a$ for all $b \in B$; then $a + \varepsilon$ separates the cut. A least element of B is analogous. Let $L = \{x : (\exists a \in A) x \leq a\}$ and $R = \{x : (\exists b \in B) b \leq x\}$. They are disjoint, and a point in neither would be a strict separator, so they partition $\mathbf{No}^M$; the endpoint observations and the cofinality of A, B finish the proof. $\square$

Source 09 gives the same argument from the reciprocal form of the bounds, and stresses that different presentations of one partition are one gap. The lemma is why an omitted pair is not counted merely because a chosen point of $\mathbf{No}^N$ is absent from $\mathbf{No}^M$ (source 11). Source 12 records that a countable rational cut around a new real can describe a larger interval in $\mathbf{No}^N$, but describes a genuine partition with no separator in $\mathbf{No}^M$ when that real is new over M .

22.2 Compression and three block codes

Lemma 22.4 (Compression to a ground regular domain; sources 09–14). *Let $f : \alpha \rightarrow \text{Ord}$ be fresh over M in N , and put $\lambda = \text{cf}^M \alpha$. Then there is a fresh $h : \lambda \rightarrow \text{Ord}$, so $\lambda \in \mathfrak{F}(M, N)$, and $\text{cf}^N \lambda = \text{cf}^N \alpha$.*

Proof. Freshness makes α a nonzero limit ordinal. Choose in M a strictly increasing cofinal $e : \lambda \rightarrow \alpha$, and an ordinal θ bounding the range of f ; it is an ordinal of M . The ground set $T = \bigcup_{\xi < \alpha} (\xi \theta)^M$ has in M an injective ordinal code $c : T \rightarrow \eta$. Put $h(i) = c(f \upharpoonright e(i))$ for $i < \lambda$. For $j < \lambda$, the positions $e(i)$, $i < j$, are bounded below α , since e is increasing of regular order type λ ; so $h \upharpoonright j$ is definable in M from one old proper restriction of f and the old maps e, c , and is old. If h were old, decoding its values and taking their union would put f in M . So h is fresh.

The map e remains strictly increasing and cofinal in N . A cofinal subset of λ maps to a cofinal subset of α , and the least indices of e above a cofinal subset of α are cofinal in λ ; so the two cofinalities agree in N . No preservation of λ as a regular cardinal of N is assumed. $\square$

Sources 09 and 10 state the lemma for ordinal-valued f , sources 11–14 for a fresh sign; the existence part is a version of [14, Proposition 5.2]. Recording *both* the ground and the outer cofinality is what makes it useful for old surreal fields (source 09).

A fresh ordinal-valued function need not give a fresh *binary* function of the same length: ordinal-valued and binary functions have different first appearances in Prikry extensions (remark 4.3, source 11). The sources use three codes, each of which lets the birthday grow while preserving the outer cofinality. The terminal-plus code $\text{code}^+(f)$ of proposition 9.1 is source 13's s_f ; its freshness and cofinality proof is the one printed there.

Lemma 22.5 (Fixed-width block code; sources 09, 11, 12, 14). *Let $f : \lambda \rightarrow \text{Ord}$ be fresh over M , λ an infinite limit ordinal, and let $\theta \geq 2$ be an ordinal strictly above every value of f . Put*

$$\text{bcode}_\theta(f)(\theta \cdot \xi + \zeta) = \begin{cases} +, & \zeta = f(\xi), \\ -, & \zeta \neq f(\xi), \end{cases} \quad (\xi < \lambda, \zeta < \theta).$$

Then $\text{bcode}_\theta(f)$ is a fresh sign of length $\theta \cdot \lambda$, and $\text{cf}^N(\theta \cdot \lambda) = \text{cf}^N \lambda$.

Proof. The blocks $[\theta \cdot \xi, \theta \cdot (\xi + 1))$ are disjoint and consecutive, so this is one sign. Every proper prefix ends within, or at the start of, some block $\xi < \lambda$ and is determined by θ , its length and $f \upharpoonright (\xi + 1)$, which is old since $\xi + 1 < \lambda$. If the whole sign were old, reading the unique plus of each block would recover f in M . The map $\xi \mapsto \theta \cdot \xi$ is strictly increasing, continuous and cofinal, so the cofinalities agree. $\square$

The four sources choose θ differently: $\theta = \theta_0 + 2$ with $\theta_0 > \sup \text{ran } f$ (source 09, so that every block contains a minus and both signs occur cofinally), an infinite cardinal of M above the range (source 11), an infinite ordinal above the range (source 12), or any $\theta \geq 2$ above the range (source 14). The code may increase the birthday greatly, and no optimal-birthday assertion is made; an unbounded ordinal alphabet may need uncountable blocks, which is why fresh ordinal functions need not be fresh reals (source 09).

Lemma 22.6 (Delimiter code; source 10). *If $f : \alpha \rightarrow \text{Ord}$ is fresh over M , the sign*

$$\text{dcode}(f) = \bigcap_{\xi < \alpha} (+^{f(\xi)} -), \quad \text{b}(\text{dcode}(f)) = \eta = \sum_{\xi < \alpha} (f(\xi) + 1),$$

is fresh, and $\text{cf}^N \eta = \text{cf}^N \alpha$. The sums are ordinal sums.

Proof. Let $b_0 = 0$, $b_{\xi+1} = b_\xi + (f(\xi) + 1)$ and $b_\zeta = \sup_{\xi < \zeta} b_\xi$ at limits; put plus signs on $[b_\xi, b_\xi + f(\xi))$ and a minus at its end. The sequence $(b_\xi)_{\xi \leq \alpha}$ is strictly increasing and continuous with supremum $\eta = b_\alpha$, so $\text{cf}^N \eta = \text{cf}^N \alpha$. A proper prefix ends before some b_ζ , $\zeta < \alpha$, so it is a restriction of the code of the old $f \upharpoonright \zeta$. To decode, enumerate the minus positions in increasing order; the order type of the plus interval between consecutive boundaries is the value of f , at a limit boundary one takes the supremum of the earlier ones, and consecutive minus signs encode zero values. This set recursion works in either universe, so an old code would put f in M . $\square$

The assertion concerns cofinality, not equality of lengths: large ordinal entries are not squeezed into a bounded number of binary coordinates, and the delimiters prevent the loss of information that ordinal sums alone would cause (source 10). Unlike $\text{code}(f)$ and $\text{code}^+(f)$ of Part I, $\text{dcode}(f)$ has no uniformly specified left endpoint in a block with a limit value; it is used here only for freshness and cofinality.

22.3 The transfer theorem

Theorem 22.7 (Cofinality transfer; sources 11 (base), 09, 10, 12, 13, 14). *For every same-ordinal pair $M \subseteq N$ under (H) ,*

$$\text{Char}_N(\mathbf{No}^M) = \{\text{cf}^N \lambda : \lambda \in \mathfrak{F}(M, N)\}, \quad (22.1)$$

$$\text{GapSpec}_N(\mathbf{No}^M) = \{(\text{cf}^N \lambda, \text{cf}^N \lambda) : \lambda \in \mathfrak{F}(M, N)\}. \quad (22.2)$$

If M and N have the same cofinalities, then $\text{Char}_N(\mathbf{No}^M) = \mathfrak{F}(M, N)$, identifying the common regular cardinals. In general the right side is a many-to-one image of the ground spectrum.

Proof. A set-presented gap comes from a fresh sign s (theorem 8.4), with character $(\text{cf}^N \text{b}(s), \text{cf}^N \text{b}(s))$ (lemma 8.3). Applying lemma 22.4 to s gives $\lambda = \text{cf}^M \text{b}(s) \in \mathfrak{F}(M, N)$ with $\text{cf}^N \lambda = \text{cf}^N \text{b}(s)$. Conversely, a fresh function on $\lambda \in \mathfrak{F}(M, N)$ gives, by any of lemmas 22.5 and 22.6 or proposition 9.1, a fresh sign whose length has outer cofinality $\text{cf}^N \lambda$, and hence a gap of that character. Symmetry gives the pair form. If cofinalities are preserved, each ground regular λ remains regular with the same value. $\square$

Source 13 stresses that the transfer holds for every same-ordinal inner-model pair, whether or not N is a set-forcing extension of M . Source 09 notes that a spectrum may be a proper class for a general pair; in the set-forcing examples of this part it is a set (theorem 23.2). The coding lemmas are closely related to [14, Propositions 5.2–5.3]; the transfer makes their interaction with the fresh-sign correspondence explicit, and it is not claimed as a new forcing-spectrum theorem (sources 09, 12).

Corollary 22.8 (What a forcing calculation must compute; source 10). *In a cofinality-preserving extension, excluding a gap at a regular μ is equivalent to excluding fresh ordinal-valued functions on μ . Without preservation, all ground regular domains of outer cofinality μ must be considered together.*

Remark 22.9 (Two sources of information, three measurements; sources 09, 11). Equation (22.1) separates two operations: determine which ground regulars carry fresh functions, then compute their cofinalities in the actual outer universe. The second operation can identify distinct members of the first spectrum, and forgetting it can invent uncountable gaps that do not exist; computing only the least new length discards possibly noninterval behaviour higher in the spectrum (source 09). The length of a fresh ordinal function, the birthday of a fresh sign encoding it, and the character of its gap can all differ. A ground regular cardinal can become singular, and then the last measurement is $\text{cf}^N \lambda$, not λ . Neither a ground cardinality bound on a forcing nor its ground fresh-function spectrum can be relabelled as an external gap spectrum without this step (source 11). An apparently large ground-domain spectrum can give a singleton external spectrum (source 10; theorem 23.6).

A weak-approximation reading (source 09). Following the terminology discussed in [14], the weak λ -approximation property for (M, N) is the absence of a fresh ordinal-valued function on λ . When ground regular cofinalities are preserved, the gap characters of $\mathbf{No}^M$ are exactly the failures of these weak approximation properties at ground regular cardinals; in general, they are the outer-cofinality image of those failures. This concerns *weak* approximation, not the stronger approximation property involving every small intersection of an arbitrary new set.

Corollary 22.10 (Lower barrier; sources 11, 10, 13, 14). *If N adds no ordinal-valued sequences over M of length less than an infinite N -cardinal κ , then $\text{Char}_N(\mathbf{No}^M) \cap \kappa = \emptyset$. Conversely, if $\text{Char}_N(\mathbf{No}^M)$ has no member below an uncountable κ , then no ordinal-valued sequence of length below κ is new (source 13).*

Proof. The hypothesis says $\kappa \leq \delta(M, N)$ when $M \neq N$, and theorem 6.2 bounds every character below by δ . The sources' direct proof: if a fresh sign s had $\mu = \text{cf}^N \text{b}(s) < \kappa$, an increasing cofinal $e : \mu \rightarrow \text{b}(s)$ in N would be old by hypothesis, and the old ordinal codes of the prefixes $s \upharpoonright e(i)$, $i < \mu$, would form an old sequence whose decoded union is s . For the converse, the shortest new restriction of a new sequence of length $\alpha < \kappa$ is fresh of limit length $\gamma \leq \alpha$, and a block code gives a gap of character $\text{cf}^N \gamma \leq |\alpha|^N < \kappa$; this does not need cofinality preservation (source 13). Both directions are the content of corollary 6.3. $\square$

Corollary 22.11 (Uncountable saturation is read off the least character; sources 11, 13, 14). *For every uncountable cardinal λ of N , $\mathbf{No}^M$ is λ -saturated as an ordered field if and only if $\text{Char}_N(\mathbf{No}^M) \cap \lambda = \emptyset$. If $\text{Char}_N(\mathbf{No}^M)$ has least element μ , the uncountable saturation spectrum is exactly $\{\lambda : \lambda \leq \mu\}$.*

Proof. This is corollary 6.4 with $\mu = \delta(M, N)$ (theorem 9.2). Source 11's direct proof: a gap of character (μ, μ) with $\mu < \lambda$ gives a finitely satisfiable omitted type over fewer than λ parameters; conversely every omitted cut presentation of size below λ would, by lemma 22.3 and symmetry, give such a gap, and the real closure argument of lemma 2.3(i) converts filled small cuts into saturation. $\square$

No regularity of λ is required; the size estimate uses one parameter set and its countable algebraic closure, not an unbounded union of λ smaller sets (source 11). The finite-parameter boundary is different: $\mathbf{No}^M$

is ω -saturated exactly when $\mathbb{R}^M = \mathbb{R}^N$ (theorem 7.1; reproved as source 09 Proposition 6.3 and source 11 Theorem 9.3). A new rational cut can fail to have an old realization although its gap character alone does not reveal whether the cut came from a new real or from a much later fresh sign (source 11). Pure dense orders are ω -saturated in this finite-parameter sense, since a finite parameter set specifies only an interval, an endpoint or an exterior ray; the criterion is a *field-language* phenomenon and does not prove that the underlying pure orders of the examples below are nonisomorphic (source 09). Source 09 recalls the criterion because it identifies a particularly simple pair in the family of theorem 28.5; quantifier elimination and the internal saturation argument are classical [20, 9], and no truth predicate for all formulas over a class field is used.

23 External density and single-coordinate spectra

23.1 A ceiling measured in the extension

Definition 23.1 (Source 10). For a generic filter $G \subseteq \mathbb{P} \in M$, a *cofinal filter base* is a set $B \subseteq G$ such that every $p \in G$ has some $b \in B$ with $b \leq p$ (smaller conditions are stronger). Let $\mathfrak{b}_N(G)$ be the least N -cardinality of such a base. This is notation local to this part, not the bounding number of cardinal-invariant theory.

Theorem 23.2 (Filter-base ceiling; source 10 (base), sources 09, 11, 13, 14). *Let $N = M[G]$ for a set forcing $\mathbb{P} \in M$, and let $B \subseteq G$ be a cofinal filter base. If $f : \alpha \rightarrow \text{Ord}$ is fresh over M , then $\text{cf}^N \alpha \leq |B|^N$. Consequently every $\mu \in \text{Char}_N(\mathbf{No}^M)$ satisfies*

$$\mu \leq \mathfrak{b}_N(G) \leq |\mathbb{P}|^N,$$

and $\mu \leq |D|^N$ for every dense $D \subseteq \mathbb{P}$ with $D \in M$. In particular $\text{Char}_N(\mathbf{No}^M)$ is a set.

Proof. Fix a ground name $\dot{f}$ and strengthen inside G to a condition forcing that $\dot{f}$ is a function on α with range in a specified ordinal. For each $\xi < \alpha$ the old restriction $\dot{f} \restriction \xi$ has a canonical ground name, and the truth lemma gives $p_\xi \in G$ forcing $\dot{f} \restriction \xi = (\dot{f} \restriction \xi)$. By directedness of G and cofinality of B choose $b_\xi \in B$ below p_ξ and the fixed condition. If $\text{cf}^N \alpha > |B|^N$, some $b \in B$ equals b_ξ for an unbounded set of $\xi < \alpha$, since otherwise α would be a union of fewer than $\text{cf}^N \alpha$ bounded sets. Then for every $\zeta < \alpha$ the single ground condition b decides the value of $\dot{f}(\zeta)$, and in M the forcing relation and Replacement construct the function sending ζ to that value. In N it is f , contradicting freshness. Apply this to the fresh sign of each gap (theorem 8.4). The generic G is a base inside the ground set $\mathbb{P}$, and $G \cap D$ is a base for every dense $D \in M$. $\square$

The base need not belong to M : a poset may be large in M while its actual generic filter has a much smaller cofinal chain in N , and the argument reconstructs f from one ground condition, not from an allegedly ground sequence of conditions (source 10). The use of $|D|^N$ rather than $|D|^M$ is the point: a forcing with a large ground presentation may enumerate it by a much shorter generic sequence, and the sharper bound is invisible if ground cardinalities are substituted throughout (source 13). If a forcing collapses its own old family of conditions to size κ , the ceiling rules out every character above κ , even when the ground forcing had fresh functions on several larger ground cardinals; their outer cofinalities must fall below the ceiling (source 11). Sources 11, 13 and 14 state the dense-set form (source 11 Theorem 5.1, source 13 Theorem 5.1, source 14 Lemma 5.1), and source 09 the case $D = \mathbb{P}$ with outer cofinality (source 09 Lemma 5.3, where $\mathbb{P}$ is a small forcing $\mathbb{Q}$ over a universe V and $U = V[G_{\mathbb{Q}}]$): if $\text{cf}^U \gamma > |\mathbb{Q}|^U$ then $\text{Fr}_\gamma(V, U)$ fails. For ground regular cardinals this is the familiar size bound recalled in [14, Corollary 2.3]; the outer-cofinality form is convenient when another forcing has already changed cofinalities. Source 14 calls it the usual repeated-condition

argument. For every nontrivial set-forcing extension, the character spectrum is therefore a set, although the field and the class of gaps are proper classes; this separates three issues, namely large birthdays, many fresh ground domains and large externally visible characters, of which only the last is the cut invariant (source 13).

Corollary 23.3 (Two-sided singleton criterion; sources 10, 11, 13). *Suppose $N = M[G]$ adds no ordinal sequence shorter than κ , has a cofinal filter base (for instance $G \cap D$ for a ground dense D) of N -size at most κ , and contains a fresh function whose domain has N -cofinality κ . Then $\text{Char}_N(\mathbf{No}^M) = \{\kappa\}$.*

Proof. Corollary 22.10 and theorem 23.2 and the given fresh function. $\square$

23.2 Generalized Cohen forcing

Fix an infinite regular cardinal κ of M and a set $A \in M$ with at least two elements, and let $\mathbb{P} = \text{Fn}_{<\kappa}(\kappa, A)^M$ be the partial functions $\kappa \rightarrow A$ of ground domain size below κ , ordered by extension. The superscript matters: this is one fixed ground poset. Let $g = \bigcup G : \kappa \rightarrow A$.

Lemma 23.4 (Universal occurrence; source 11). *For every ground word $w : \eta \rightarrow A$ with $\eta < \kappa$ there is $\gamma < \kappa$ with $\gamma + \eta < \kappa$ and $g(\gamma + \xi) = w(\xi)$ for all $\xi < \eta$. Consequently the ground set $T = (\bigcup_{\eta < \kappa} {}^\eta A)^M$ has cardinality at most κ in N .*

Proof. Conditions placing w on some interval are dense: regularity bounds $\text{dom}(p)$ below κ , choose γ beyond that bound, note $\gamma + \eta < \kappa$ since κ is an infinite initial ordinal, and add the prescribed values; the domain still has size below κ . Each such dense set belongs to M , so genericity gives the first assertion for every ground word. In N there are only κ pairs (γ, η) of positions and lengths, each determining at most one word by reading g , and every member of T appears; so $|T|^N \leq \kappa$. $\square$

Lemma 23.5 (The old poset becomes small; source 11). *In N , $|\mathbb{P}| = \kappa$, and $\mathbb{P}$ adds no ordinal-valued sequence of length below κ ; in particular κ stays regular.*

Proof. Choose a ground finite $m \geq 1$ and an injection $\iota : A \sqcup \{\square\} \rightarrow A^m$ (for finite A take m large; for infinite A , Choice gives $m = 1$), the extra symbol recording an undefined coordinate. For $p \in \mathbb{P}$ let $\eta = \sup\{\xi + 1 : \xi \in \text{dom } p\} < \kappa$, write the word of length η with ξ -th symbol $p(\xi)$ or $\square$, and replace each symbol by its m -letter code. The resulting old word of length $m \cdot \eta < \kappa$ determines p by fixed-size block decoding, so $\mathbb{P}$ injects in M into T and $|\mathbb{P}|^N \leq \kappa$ by lemma 23.4. The single-coordinate conditions with a fixed letter give κ distinct conditions. The forcing is $< \kappa$ -closed in M (a union of fewer than κ descending conditions is a condition, by regularity), so the usual recursive decision argument adds no ordinal-valued sequence of length below κ and preserves the regularity of κ ; these are therefore inequalities of N -cardinals. $\square$

The conclusion is not that the generic has listed all *new* short words over an arbitrary alphabet; only the specified ground set is counted, which is exactly the set the ceiling needs (source 11).

Theorem 23.6 (Generalized Cohen spectrum; source 11 (base), sources 10, 12, 13, 14). *For every infinite regular κ of M and every $A \in M$ with at least two elements, if $N = M[G]$ for G generic over $\text{Fn}_{<\kappa}(\kappa, A)^M$, then κ remains regular,*

$$\text{Char}_N(\mathbf{No}^M) = \{\kappa\}, \quad \text{GapSpec}_N(\mathbf{No}^M) = \{(\kappa, \kappa)\},$$

and the least birthday of a new surreal number is κ . No assumption on $2^{<\kappa}$ or $\kappa^{<\kappa}$ is made, and no assertion that cardinals above κ are preserved.

Proof. By lemma 23.5 no ordinal-valued sequence of length below κ is added, so all shorter binary signs are old and corollary 22.10 excludes characters below κ . Fix a ground surjection $\pi : A \rightarrow \{-, +\}$ and put $s = \pi \circ g$. Every proper prefix of g is old (conditions whose domain contains a given $\eta < \kappa$ are dense), so every proper prefix of s is old. For each old $r : \kappa \rightarrow \{-, +\}$ it is dense to make $\pi \circ g$ differ from r at a coordinate outside the condition's domain, using a letter that projects to the opposite sign. So s is fresh of length κ , giving a gap of character (κ, κ) and a new surreal of birthday κ . Finally lemma 23.5 and theorem 23.2 with $D = \mathbb{P}$ exclude every character above κ . $\square$

This answers the single-coordinate case of question 19.2 at the level of the complete external spectrum (source 10); proposition 13.2 computed only the first gap and $\delta = \beta = \kappa$, and its remark that larger gaps were not excluded is now settled for $\text{Add}(\kappa, 1)$. The upper bound concerns *all* gaps, not only a chosen generic cut, and applies when the alphabet is so large that the forcing collapses many ground cardinals (source 11). The theorem does not say that κ is the only ground regular domain supporting a fresh function; the transfer, not an unqualified equality of ground and outer indices, controls that issue (sources 10, 12, 14). Indeed the ground-indexed spectrum of $\text{Add}(\kappa, 1)$ is $\text{FRESH}(\text{Add}(\kappa, 1)) = [\kappa, (2^{<\kappa})^M] \cap \text{Reg}^M$ [14, Proposition 6.1]; the theorem says that all these ground regulars have outer cofinality κ (lemma 23.14), which resolves the single-Cohen case left open beyond the first gap (source 14). Fresh signs of larger birthdays may exist, but their outer cofinalities cannot give larger characters (source 13).

Example 23.7 (Binary Cohen forcing and large alphabets; source 11). Taking $A = 2$ gives the complete spectrum of $\text{Add}(\kappa, 1) = \text{Fn}_{<\kappa}(\kappa, 2)$ at every infinite regular κ ; an assumption such as $2^{<\kappa} = \kappa$ is not needed for the external gap calculation, and the proof does not assert that the ground fresh-function spectrum of this forcing is a singleton. If instead A is a ground infinite cardinal $\tau \geq \kappa$, the generic function maps onto A (requiring each $a \in A$ in its range is dense), so the forcing collapses τ to cardinality at most κ ; its old surreal gap spectrum is still exactly $\{\kappa\}$. This covers the partial-function presentation of the corresponding collapse forcing. Different ground regulars that contribute fresh functions are merged by their new cofinalities, as (22.1) requires.

A constructive certificate (source 11). The lower bound supplies an explicit sign representative, not merely an existential type: its cofinal and cointial witnesses are the prefixes ending immediately before a plus or a minus, of external size κ , and no smaller subfamily suffices by the first-disagreement argument. At the other end, universal occurrence compresses all ground conditions into κ actual intervals of the generic word; a hypothetical larger fresh sign would force one deciding condition to repeat cofinally, turning the supposedly new sign into a ground set.

Second route: initial-segment bases (source 10). For $A = 2$ (source 10 Theorem 5.5), the set $B = \{g \upharpoonright \alpha : \alpha < \kappa\}$ is a cofinal filter base of size κ : every condition has bounded domain, and a sufficiently long prefix of g extends it; each $g \upharpoonright \alpha$ is a ground condition in G , since conditions whose domain contains α are dense. Corollary 23.3 applies directly, with no counting of the ground tree. Theorem 23.2 applied to g also shows that no cofinal base has size below κ . For $\kappa = \omega$, the closure assertions mean finite closure and the unchanged regularity of ω .

Corollary 23.8 (The same singleton after an ordinal collapse; source 10). *Let $\tau \geq 2$ be an ordinal and N a generic extension by $\text{Coll}(\kappa, \tau) = \text{Fn}_{<\kappa}(\kappa, \tau)^M$. Then $\text{Char}_N(\mathbf{No}^M) = \{\kappa\}$. If $\tau \geq \kappa$, the generic map is onto τ , so arbitrarily large ground cardinals can be collapsed while the external spectrum stays a singleton.*

Proof. Union gives $<\kappa$ -closure; the generic g has all proper prefixes in M , is new by the disagreement dense sets, and its initial-segment conditions form a cofinal base of size κ . Apply corollary 23.3. For $\zeta < \tau$, putting ζ into the range is dense. $\square$

The comparison proves that the external gap spectrum does not remember which cardinals the forcing collapsed. It does *not* prove that the resulting old fields are isomorphic, nor does it give two nonisomorphic fields with equal full spectrum; that classification issue is separate (source 10, and sections 27 to 29).

Third wording: substring enumeration (sources 13, 14). Source 13 (alphabet $A = 2$ or an ordinal $A \geq \kappa$, its Theorem 1.2) and source 14 ($A = 2$, its Theorem 5.3) prove theorem 23.6 by the argument of lemma 23.4: initial-segment conditions are dense, so the old tree $T = ({}^{<\kappa}A)^M$ is a dense presentation of the forcing; every old $t \in T$ occurs in g as a contiguous block $\langle g(\gamma + \xi) : \xi < \eta \rangle$; there are at most κ pairs (γ, η) in the extension, all blocks are old by closure, and κ distinct constant strings give $|T|^N = \kappa$. For $\kappa = \omega$ the argument uses only finite closure and Cohen-style genericity; for $\kappa > \omega$ it does not assume $\kappa^{<\kappa} = \kappa$; for $A \geq \kappa$ the union takes every ground value, which explains the single-collapse interpretation; and the result concerns the entire spectrum, not only gaps whose separator has birthday κ (source 13).

23.3 The square chain condition

Lemma 23.9 (Mitchell-type square-chain-condition lemma; source 12). *Work in M . Let $\mu \leq \lambda$ be infinite regular cardinals, and suppose $\mathbb{P} \times \mathbb{P}$ has the μ -chain condition. Then no $\mathbb{P}$ -generic extension contains a fresh function on λ over M .*

Proof. Suppose p forces that $\dot{f} : \lambda \rightarrow \text{Ord}$ is fresh. Below (p, p) in the square, let D be the set of pairs whose coordinates decide different values of $\dot{f}$ at some common position. D is dense: otherwise below some (r, s) no disagreement is possible; for each $\alpha < \lambda$ choose an extension of s deciding $\dot{f}(\alpha) = a_\alpha$; every extension of r deciding that coordinate gives the same value, so r forces $\dot{f}$ to be the ground function $\alpha \mapsto a_\alpha$, contradicting freshness. Choose a maximal antichain $\mathcal{A} \subseteq D$ and for each member a disagreement position α_a . The chain condition gives $|\mathcal{A}| < \mu \leq \lambda$, and regularity gives $\beta = \sup\{\alpha_a + 1 : a \in \mathcal{A}\} < \lambda$. Every member of $\mathcal{A}$ forces that the two copies of $\dot{f} \restriction \beta$ differ, so by maximality (p, p) does. But freshness gives $q \leq p$ deciding $\dot{f} \restriction \beta$ as one old sequence, and (q, q) forces the copies to agree. $\square$

This is [14, Proposition 2.2], which credits Mitchell and Unger antecedents; the proof is reproduced for transparency. The *square* is essential: a c.c.c. Suslin-tree forcing can add a fresh function on ω_1 , so the hypothesis is not replaced by the c.c.c. of $\mathbb{P}$ alone (source 12).

Corollary 23.10 (Source 12). *If a forcing has a dense subset of infinite cardinality τ in M , it adds no fresh function on a ground regular cardinal greater than τ .*

Proof. Pass to the dense subset; its square has size at most τ , hence the τ^+ -chain condition. Apply lemma 23.9. $\square$

This is a second route to the ground-size case of theorem 23.2, through the chain condition instead of the generic filter; it bounds ground regular indices, not outer cofinalities.

Proposition 23.11 (Real-adding square-c.c.c. extensions; source 12). *If $\mathbb{P} \times \mathbb{P}$ is c.c.c. in M and $N = M[G]$ contains a new real, then $\text{GapSpec}_N(\mathbf{No}^M) = \{(\omega, \omega)\}$.*

Proof. A new binary real is fresh on ω . Lemma 23.9 excludes fresh functions on uncountable ground regulars. The c.c.c. preserves cardinals and cofinalities: each coordinate of a name for an ordinal sequence has only countably many possible values on a maximal antichain, so a name for a sequence of length below an uncountable regular λ has range in a ground set of size below λ and is not cofinal in it. Apply theorem 22.7. $\square$

Proposition 23.12 (Any nonzero number of Cohen reals; source 12). *For every nonempty set $J \in M$, if $N = M[G]$ for $\text{Add}(\omega, J)^M$, then $\mathfrak{F}(M, N) = \{\omega\}$ and $\text{GapSpec}_N(\mathbf{No}^M) = \{(\omega, \omega)\}$.*

Proof. Conditions are finite partial binary functions on $J \times \omega$. An uncountable family of their domains has an uncountable Δ -system subfamily; after fixing the finitely many values on its root, any two members are compatible. This is the Knaster property. The square is itself a Cohen forcing on two disjoint copies of the coordinates, hence c.c.c. by the same argument. Each coordinate adds a fresh binary real, so proposition 23.11 applies. $\square$

The fresh-spectrum equality is noted after [14, Corollary 2.4]. What section 27 uses is that it remains true in *every intermediate coordinate model*, where the quotient is again a Cohen forcing; no generic filter is required to meet dense sets from an inappropriate universe (source 12).

Fourth route to theorem 23.6 for $A = 2$: explicit collapse (source 12, its Theorem 10.1). It needs only a size bound, closure and an explicit collapse, and applies to each actual generic.

Lemma 23.13 (Dense tree and closure; source 12). *Write $\sigma_\kappa = (2^{<\kappa})^M$. The forcing $\text{Fn}(\kappa, 2, <\kappa)^M$ has a dense subset of cardinality σ_κ , namely the binary sequences of length below κ . It is $<\kappa$ -closed and adds no ordinal-valued sequence of length below κ .*

Proof. A condition has bounded domain by regularity, so it extends to a binary function on an initial segment; there are $2^{<\kappa}$ of these. A union of a decreasing sequence of fewer than κ conditions is a condition. To decide a name for a sequence of length $\mu < \kappa$, decide each coordinate recursively, taking lower bounds at limits and at the end. This also prevents κ from acquiring a shorter cofinal sequence or being collapsed. $\square$

Lemma 23.14 (Explicit collapse of the relevant ground cardinals; source 12). *If λ is regular in M and $\kappa \leq \lambda \leq \sigma_\kappa$, then $\text{cf}^N \lambda = \kappa$.*

Proof. For each nonzero cardinal $\mu < \kappa$ of M , partition κ by a ground bijection into κ labelled blocks of size μ . For a fixed $a \in (2^\mu)^M$ it is dense that the generic function equal a on some block: a condition meets fewer than κ blocks, and filling an unused block still gives a condition. So every old binary μ -sequence appears on some block, each block readout is old by lemma 23.13, and $|(2^\mu)^M|^N \leq \kappa$. The claim is immediate for $\lambda = \kappa$. If $\kappa < \lambda < \sigma_\kappa$, some $\mu < \kappa$ has $\lambda \leq (2^\mu)^M$, so $|\lambda|^N = \kappa$. If $\lambda = \sigma_\kappa > \kappa$ is regular in M , the supremum defining σ_κ is attained, since otherwise its cofinality in M would be at most $\kappa < \lambda$; the same conclusion follows. So $\text{cf}^N \lambda \leq \kappa$, and it cannot be smaller, since a shorter cofinal sequence would be a new ordinal sequence of length below κ . $\square$

Proof of theorem 23.6 for $A = 2$ along this route. Closure adds no fresh function on a ground regular below κ ; corollary 23.10 and the dense tree exclude ground regulars above σ_κ . So $\mathfrak{F}(M, N) \subseteq [\kappa, \sigma_\kappa] \cap \text{Reg}^M$, and lemma 23.14 sends every member to outer cofinality κ . The generic binary sequence on κ is fresh, so $\kappa \in \mathfrak{F}(M, N)$; apply theorem 22.7. $\square$

This route illustrates a second kind of information loss: the field spectrum forgets not only which rational cuts are present (section 27) but also which ground regular lengths merged to the same cofinality after forcing. An analysis in which “regular cardinal” and “cofinality” are silently read in different universes would miss this collapse (source 12).

24 Easton products

Work in $V \models \text{GCH}$. For a regular α write $\mathbb{C}_\alpha = \text{Add}(\alpha, 1)^V$, and for a set $A \subseteq \text{Reg}^V$ let $\mathbb{P}_A = \prod_{\alpha \in A}^E \mathbb{C}_\alpha$ be the Easton product: coordinatewise order, supports bounded below every inaccessible cardinal and unrestricted at other limits [14, Definition 4.7]. Under GCH the regular limit cardinals relevant to the product are inaccessible. All posets and supports are ground objects.

Definition 24.1 (Fischer–Koelbing–Wohofsky; sources 09, 10, 11). A set A of infinite regular cardinals is *Easton closed* if every limit point μ of A satisfies: if μ is regular and not Mahlo, then $\mu \in A$; if μ is singular, then $\mu^+ \in A$. The least Easton-closed superset of A is $\text{Ecl}(A)$.

The Mahlo exception is part of the definition and cannot be dropped; Mahlo has its usual meaning (inaccessible, with stationarily many inaccessibles below). The final published formulation is used, not a simplified rule that closes at every regular limit, nor the more restrictive wording of earlier talk notes (sources 09, 10, 11).

Theorem 24.2 (Published input [14]; imported). *Assume $V \models \text{GCH}$. Then $\mathbb{P}_A$ preserves cardinals and cofinalities, is homogeneous, and for every V -generic G has $\mathfrak{F}(V, V[G]) = \text{Ecl}^V(A)$.*

Preservation is [14, Proposition 4.10]; the spectrum is their Theorem 4.24 with the realization statement in Corollary 4.25; the passage to the actual generic is their homogeneity discussion, Proposition 4.3. The proof includes substantive forcing and pcf arguments, which are imported and not claimed (sources 09, 10, 11).

Theorem 24.3 (Exact Easton realization; source 09 (base), sources 10, 11). *If G is $\mathbb{P}_A$ -generic over $V \models \text{GCH}$ and $N = V[G]$, then*

$$\text{Char}_N(\mathbf{No}^V) = \text{Ecl}^V(A), \quad \text{GapSpec}_N(\mathbf{No}^V) = \{(\mu, \mu) : \mu \in \text{Ecl}^V(A)\}.$$

Every ground Easton-closed set of regular cardinals is the exact full set-presented gap-character set of an old surreal field in a cofinality-preserving set-forcing extension.

Proof. Homogeneity identifies the published spectrum with the pointwise spectrum of this G , and cofinality preservation makes $\lambda \mapsto \text{cf}^N \lambda$ the identity on Reg^V . Apply theorems 22.7 and 24.2; for the last assertion take A Easton closed. $\square$

This computes the entire spectrum for the specified family of forcings, a broad family within question 19.2. It is not a characterization of the spectra of all cofinality-preserving forcings, does not assert that Easton closure is necessary for arbitrary forcing spectra, and is not a proof of a more general spectrum-realization conjecture (sources 09, 10, 11).

Example 24.4 (A genuine hole; sources 09, 10). For $A = \{\aleph_1, \aleph_3\}$ there are no limit points, so $\text{Ecl}(A) = A$. The old field has exactly the characters $\aleph_1$ and $\aleph_3$ and no set-presented gap of character $\aleph_2$. The smallest character does not determine the full spectrum even in very elementary product examples; the example is relative to a GCH ground, and all three cardinals are preserved.

Example 24.5 (A character with no coordinate; sources 09, 10, 11). For $A = \{\aleph_{2n+1} : n < \omega\}$ the only limit point is the singular $\aleph_\omega$, so $\text{Ecl}(A) = A \cup \{\aleph_{\omega+1}\}$ (adding the successor creates no new limit point), and

$$\text{Char}_{V[G]}(\mathbf{No}^V) = \{\aleph_{2n+1} : n < \omega\} \cup \{\aleph_{\omega+1}\}.$$

The character $\aleph_{\omega+1}$ occurs although no coordinate sits there, and there are no others; the rule “one character per coordinate” is false for the full spectrum. The extra character is the surreal image of the published singular-limit phenomenon, not the result of a finite-support heuristic. The product is $< \aleph_1$ -closed and adds no reals, so multicharacter behaviour here is not a new-real phenomenon.

Example 24.6 (Same first character, different full spectra; source 09). The products indexed by $\{\aleph_1\}$ and $\{\aleph_1, \aleph_3\}$ have the same first omitted character but different entire spectra. This compares the two extensions; by itself it is not a simultaneous construction of two old fields in one universe, which needs the further arguments of section 25.

Example 24.7 (Why the Mahlo clause matters; source 10). At a regular non-Mahlo limit point the closure adds that cardinal; at a Mahlo limit point it imposes no such requirement by that clause alone. Whether a cardinal belongs to the closure must be computed from the stated operator, including the original set, not from the slogan “close under regular limits”.

At the saturation threshold the higher characters carry strictly more information: for uncountable κ , external saturation is controlled by the least new length (theorem 1.1), which in the nonempty Easton examples is $\min A$, while sections 25 and 27 show that even *all* characters together do not classify the fields (source 09).

Why quotients need a new argument (sources 10, 11). Suppose some coordinates have been added, producing an intermediate W . A remaining factor is the old set $\text{Add}^V(\kappa, 1)$, not automatically $\text{Add}^W(\kappa, 1)$: if W has new short binary sequences, it has new partial functions allowed by the latter poset, and the closure of the old set can fail for newly available descending sequences. So theorem 24.2 cannot simply be applied in every intermediate model. The next section proves exactly the distributivity needed, placing the closed high forcing before the small forcing, instead of assuming that closure survives every change of universe.

25 Relative spectra of intermediate grounds

This section keeps the final universe fixed and varies the intermediate ground. The object computed is not the spectrum of a forcing over its original ground but the spectrum of each inclusion $M_A \subseteq N$ inside one product extension. A ground Cohen poset, viewed after another coordinate has been added, need not be the Cohen poset defined afresh there, and its ground closure proof cannot simply be reused (sources 10, 11, 13, 14).

25.1 Forcing tools

Lemma 25.1 (Short-name localization; source 14 (statement), sources 10, 11, 13). *In a ground V let $\mathbb{P}, \mathbb{Q}$ be set forcings and μ an infinite cardinal with $|\mathbb{P}| \leq \mu$, and suppose that $\mathbb{Q}$ adds no ordinal-valued sequences of length μ (for instance, $\mathbb{Q}$ is μ -closed in V). For mutually generic $G_{\mathbb{P}}, G_{\mathbb{Q}}$,*

$${}^\mu\text{Ord} \cap V[G_{\mathbb{P}}, G_{\mathbb{Q}}] = {}^\mu\text{Ord} \cap V[G_{\mathbb{P}}].$$

More generally, if $\mathbb{Q}$ adds no sequences of length below a cardinal κ , the same holds for each length η with $\max(|\mathbb{P}|, |\eta|, \aleph_0) < \kappa$.

Proof. Reorder the product as $\mathbb{Q} * \check{\mathbb{P}}$. In $V[G_{\mathbb{Q}}]$, a $\mathbb{P}$ -name for a function on μ is represented below a condition in $G_{\mathbb{P}}$ by a maximal deciding antichain for each coordinate together with the ordinal assigned to each member. A ground enumeration of $\mathbb{P}$ has length at most μ , so the whole array of deciding conditions and values is coded by an ordinal sequence of length at most μ . By the hypothesis on $\mathbb{Q}$ the array belongs to V ; predensity and incompatibility of its antichains are absolute, since the elements and order of $\mathbb{P}$ are unchanged. So the array is a name in V whose interpretation by $G_{\mathbb{P}}$ is the given function. Replacing μ by a cardinal below κ that is large enough to code the array proves the last assertion. $\square$

It is enough that $\mathbb{Q}$ be $< \kappa$ -closed in the original ground: its original distributivity shows that a *name array* is old, and no claim is made that $\check{\mathbb{Q}}$ remains closed after forcing with $\mathbb{P}$. This is the key safeguard when intermediate grounds contain only some coordinates (source 14).

Second route, by closure (sources 13, 10, 11). Assume $\mathbb{Q}$ is $\leq \mu$ -closed in V . Fix a product name $\dot{f}$ and (p_0, q_0) forcing $\dot{f} : \mu \rightarrow \text{Ord}$. For fixed $\alpha < \mu$, enumerate $\mathbb{P} \restriction p_0$ and build an antichain A_α below p_0 while strengthening the $\mathbb{Q}$ -coordinate: at an enumerated p compatible with a chosen member do nothing; otherwise choose $a \leq p$ and a stronger $\mathbb{Q}$ -condition so that the pair decides $\dot{f}(\alpha)$, add a and record the value. Closure gives lower bounds at limits and after the enumeration, and A_α is maximal below p_0 . Repeating for all $\alpha < \mu$ with decreasing $\mathbb{Q}$ -conditions gives at most μ choices at each of at most μ stages; a decreasing chain indexed by an ordinal of cardinality at most μ has a cofinal subchain of length at most μ , so the stated closure suffices even if the recursion has length $\mu \cdot \mu$. A common lower bound q_* and the recorded antichains define a $\mathbb{P}$ -name $\dot{h}$ with $(p_0, q_*) \Vdash \dot{f} = \dot{h}$, and such representations are dense (source 13, its Lemma 6.1). Source 10 (its Lemma 7.1) states the case $|\mathbb{P}| \leq \kappa$, $\mathbb{Q} \leq \kappa$ -closed, lengths at most κ ; source 11 (its Lemma 8.2) the case $|\mathbb{P}| < \mu$, $\mathbb{Q} < \mu$ -closed, concluding that $\mathbb{Q}$ is $< \mu$ -distributive over the $\mathbb{P}$ -extension, building one maximal antichain per coordinate in at most $|\mathbb{P}|$ steps. Both are small-poset instances of Easton's lemma: if $\mathbb{P}$ is κ^+ -c.c. and $\mathbb{Q}$ is $\leq \kappa$ -closed, then after $\mathbb{P}$ the ground poset $\mathbb{Q}$ is $\leq \kappa$ -distributive [22, Lemma 3.8]. The assertion is distributivity, not closure in the intermediate model, and that distinction is why it is useful (source 11). A poset closed in V need not retain the same closure in $V[G_{\mathbb{P}}]$ (source 13).

Lemma 25.2 (Product restrictions and high tails; sources 10, 11). *An Easton product over a set of coordinates factors as a product over any finite partition of the coordinates. If every coordinate of a factor exceeds an infinite cardinal μ , that factor is closed under descending sequences of length at most μ in V , and adds no ordinal-valued sequences of length at most μ .*

Proof. Restriction to a subset of coordinates preserves the support condition, and a finite union of supports bounded below an inaccessible is bounded there; restriction and union are inverse product maps. For closure, take the coordinatewise union of at most μ descending conditions. At a coordinate $\alpha > \mu$, regularity keeps the domain below α . Below an inaccessible $\tau' \leq \mu$ the high factor has empty support; below an inaccessible $\tau' > \mu$ the union of at most μ bounded subsets is bounded. The recursive decision proof, with a lower bound after μ steps, gives the last assertion. $\square$

Lemma 25.3 (Intersections of mutually generic extensions; source 13 (base), source 14). *If $G_{\mathbb{P}} \times G_{\mathbb{Q}}$ is generic for a product of two set forcings over V , every set of ordinals, and every ordinal-valued function, belonging to both $V[G_{\mathbb{P}}]$ and $V[G_{\mathbb{Q}}]$ belongs to V . Consequently, for coordinate subextensions of a product and ground sets A, A' of coordinates, $V[G \restriction A] \cap V[G \restriction A'] = V[G \restriction (A \cap A')]$ when restricted to sets of ordinals or ordinal-valued functions.*

Proof. Let $x \subseteq \gamma$ lie in both, with a $\mathbb{P}$ -name $\dot{x}$, a $\mathbb{Q}$ -name $\dot{y}$ and (p, q) in the generic forcing $\dot{x} = \dot{y} \subseteq \gamma$. For each $\xi < \gamma$, p decides $\xi \in \dot{x}$: otherwise $p_1, p_2 \leq p$ force opposite answers, a strengthening of q decides $\xi \in \dot{y}$, and one of (p_1, q') , (p_2, q') forces a contradiction. Separation in V collects the positive decisions, which form x . Functions are coded by sets of ordinals using a ground range bound and pairing. For the coordinate form, factor out the common coordinates first; the two remaining groups are mutually generic over that subextension, and unused coordinates do not affect the intersection. $\square$

Source 14 gives the same proof strengthening q first. Source 09 proves a different intersection statement for a common small forcing over two nested universes (lemma 28.2).

Corollary 25.4 (Freshness survives an independent base extension; sources 13, 14). *If $f \in V[G_{\mathbb{P}}]$ is fresh over V and $G_{\mathbb{P}}, G_{\mathbb{Q}}$ are mutually generic, then f is fresh over $V[G_{\mathbb{Q}}]$ in $V[G_{\mathbb{P}}, G_{\mathbb{Q}}]$.*

Proof. Its proper prefixes lie in $V \subseteq V[G_{\mathbb{Q}}]$, and lemma 25.3 prevents $f \in V[G_{\mathbb{Q}}]$. $\square$

Ordinary persistence of freshness to a larger ambient universe keeps the *ground* fixed; this corollary changes the ground, which is the extra ingredient needed here (source 14).

25.2 Isolated coordinates (source 10)

Fix $A \subseteq \text{Reg}^V$, assume $V \models \text{GCH}$, let G be $\mathbb{P}_A$ -generic and $N = V[G]$. For $K \subseteq A$ in V put $M_K = V[G \restriction K]$: the coordinates in K are retained, those in $A \setminus K$ are missing.

Theorem 25.5 (Isolated-coordinate localization; source 10). *Suppose $\kappa \in A$ and $|\mathbb{P}_{A \cap \kappa}|^V < \kappa$. Then*

$$\kappa \in \text{Char}_N(\mathbf{No}^{M_K}) \iff \kappa \notin K.$$

Proof. The full product preserves cardinals and cofinalities (theorem 24.2), so every intermediate model has the same cardinals and cofinalities as V and N : a collapsing bijection or shorter cofinal map in an intermediate model would persist to N .

A missing coordinate gives a gap. If $\kappa \notin K$, the coordinate generic g_κ has all proper initial segments in $V \subseteq M_K$. By product factorization it is generic over M_K for the *old* Cohen poset, and for any $h : \kappa \rightarrow 2$ in M_K the conditions disagreeing with h are dense there: extend a ground condition at one unused ordinal by the opposite bit, which is still a ground condition even when h is new over V . So g_κ is fresh over M_K ; apply theorem 22.7.

A kept coordinate gives no gap. If $\kappa \in K$, put

$$W = V[G \restriction (K \setminus (\kappa + 1))], \quad \mathbb{R} = \mathbb{P}_{A \cap (\kappa + 1)}, \quad \mathbb{Q} = \mathbb{P}_{(A \setminus K) \setminus (\kappa + 1)}, \quad \mathbb{L} = \mathbb{P}_{(A \setminus K) \cap \kappa},$$

all ground posets, with corresponding restrictions $G_{\mathbb{R}}, G_{\mathbb{Q}}, G_{\mathbb{L}}$ of G . Factorization gives $N = W[G_{\mathbb{R}}][G_{\mathbb{Q}}]$ and $W[G_{\mathbb{R}}] = M_K[G_{\mathbb{L}}]$, the latter because $\kappa \in K$ puts all missing low coordinates strictly below κ . The forcing from V to W is $\leq \kappa$ -closed (lemma 25.2), so it adds no sequences of length at most κ , and the old set $\mathbb{Q}$ remains $\leq \kappa$ -closed in W : a short descending sequence of its old conditions is coded, through a ground well-order of $\mathbb{Q}$, by an old ordinal sequence, and has a ground lower bound. Under GCH, $|\text{Add}^V(\kappa, 1)| = \kappa$, so $|\mathbb{R}|^V \leq \kappa$, also in W . By lemma 25.1 the final high forcing adds no functions on κ over $W[G_{\mathbb{R}}]$. A fresh $f : \kappa \rightarrow \text{Ord}$ over M_K in N would therefore lie in $W[G_{\mathbb{R}}] = M_K[G_{\mathbb{L}}]$; but $|\mathbb{L}|^{M_K} < \kappa$ and κ is regular there, so theorem 23.2 excludes it. Cofinalities are preserved between M_K and N , so theorem 22.7 excludes the gap. $\square$

The theorem computes one isolated character of a quotient extension, not its whole spectrum. It uses the smallness of the *full* low product, not merely of one coordinate; at accumulation points such as the singular limit of example 24.5 the hypothesis fails and extra characters can occur. No step identifies $\text{Add}^V(\lambda, 1)$ with a Cohen poset recomputed over M_K (source 10). The low product used with the localization lemma *includes* the kept coordinate at κ , so its size may be κ ; only the subsequent missing-low factor has size strictly below κ , and confusing the two bounds would invalidate the argument (source 10).

25.3 Sparse products and finite complements (source 11)

Definition 25.6 (Source 11). Under $V \models \text{GCH}$, a set A of infinite regular cardinals is *sparse* if $|\prod_{\alpha \in A \cap \mu} \mathbb{C}_\alpha|^V < \mu$ for every $\mu \in A$, the product being the full-support product (so the bound holds for every Easton subproduct below μ).

“Sparse” is local terminology for this hypothesis, not a claim that every Easton set or increasing sequence has it; the strong form makes preservation at intermediate grounds transparent.

Lemma 25.7 (Missing upper coordinates remain distributive; source 11). *Let A be sparse, $B \subseteq A$ finite, $N = V[G]$ the full Easton extension and $M_{A \setminus B} = V[G \restriction (A \setminus B)]$. If $\mu \in B$ and $T = \prod_{\alpha \in B, \alpha \geq \mu} \mathbb{C}_\alpha$ is the finite ground product of the missing coordinates at or above μ , then T is $< \mu$ -distributive over $M_{A \setminus B}$.*

Proof. Factor the retained coordinates $A \setminus B$ into the high part H' above μ and the low part L below μ (there is no retained coordinate at μ), so $M_{A \setminus B} = V[G_{H'}][G_L]$. In V the finite product T is $< \mu$ -closed; by lemma 25.2, H' adds no sequences of ground objects of length below μ , so every such descending sequence in the old set T in $V[G_{H'}]$ was already in V and has its ground lower bound, and T is still $< \mu$ -closed in $V[G_{H'}]$. Sparseness gives $|L|^V < \mu$, which persists; all subproducts preserve cardinals and cofinalities (theorem 24.2), so μ is regular at this stage. Apply lemma 25.1 in $V[G_{H'}]$. $\square$

Theorem 25.8 (Exact finite-complement spectrum; source 11). *Assume $V \models \text{ZFC} + \text{GCH}$ and A sparse, and let $N = V[G]$ be generic for $\mathbb{P}_A$. For each finite nonempty $B \subseteq A$,*

$$\mathfrak{F}(M_{A \setminus B}, N) = \text{Char}_N(\mathbf{No}^{M_{A \setminus B}}) = B,$$

identifying cardinals preserved between the models.

Proof. Removing finitely many coordinates does not affect the support restriction, so $\mathbb{P}_A \cong \mathbb{P}_{A \setminus B} \times \prod_{\alpha \in B} \mathbb{C}_\alpha$, and the missing-coordinate generic is generic over $M_{A \setminus B}$ for the finite product of the old posets. All intermediate models have the cardinals and cofinalities of V and N .

Lower inclusion. For $\alpha \in B$, the coordinate function $g_\alpha : \alpha \rightarrow 2$ is new over $M_{A \setminus B}$ (disagreement with any prescribed $h : \alpha \rightarrow 2$ is dense over it, a condition leaving an unused coordinate), and each bounded prefix lies in V , because in the original coordinate it is dense to decide the whole prefix by one ground condition. So g_α is fresh and $\alpha \in \mathfrak{F}(M_{A \setminus B}, N)$.

Upper inclusion. Let $\lambda \notin B$ be infinite regular, and split the finite missing product into $S = \prod_{\alpha \in B, \alpha < \lambda} \mathbb{C}_\alpha$ and $T = \prod_{\alpha \in B, \alpha > \lambda} \mathbb{C}_\alpha$. If S is nontrivial, its ground size is below λ , since under GCH $|\mathbb{C}_\alpha| = \alpha$ and a finite product has the maximum of its factors' sizes; the bound persists. Suppose a fresh $f : \lambda \rightarrow \text{Ord}$ over $M_{A \setminus B}$ lies in N , and force first with T , writing $U = M_{A \setminus B}[G_T]$. If $f \notin U$, it is fresh over U and $N = U[G_S]$ with $|S|^N < \lambda = \text{cf}^N \lambda$, contradicting theorem 23.2; so $f \in U$. If T is empty, $f \in M_{A \setminus B}$, a contradiction. Otherwise let $\mu = \min(B \setminus (\lambda + 1))$; then $\lambda < \mu$ and lemma 25.7 says T adds no sequences of length λ over $M_{A \setminus B}$, again a contradiction. So $\mathfrak{F}(M_{A \setminus B}, N) = B$, and cofinality preservation and theorem 22.7 give the gap spectrum. $\square$

The ambient spectrum $\text{Char}_N(\mathbf{No}^V)$ can have additional limit-generated characters; they do not automatically remain in the relative spectrum over $M_{A \setminus B}$, because the generic information responsible for them may already be in $M_{A \setminus B}$. The upper-inclusion proof, not a subtraction of ground spectra, establishes their absence (source 11). The central point is that the missing coordinates are *ground posets*, not Cohen posets recomputed in the intermediate model, so distributivity must be proved there.

Proposition 25.9 (Sparse sets of any set length; source 11). *Assume $V \models \text{GCH}$. Given an uncountable regular μ and an infinite cardinal χ , there are increasing regular cardinals $(\kappa_\xi)_{\xi < \chi}$ above μ such that $\{\mu\} \cup \{\kappa_\xi : \xi < \chi\}$ is sparse.*

Proof. At stage ξ choose an infinite cardinal ς_ξ at least μ , $|\xi|$ and every earlier κ_η , and set $\kappa_\xi = (2^{\varsigma_\xi})^+$, a regular successor. Each earlier coordinate is at most ς_ξ and its Cohen poset has that size under GCH; there are at most ς_ξ of them, so the product below κ_ξ has size at most $\varsigma_\xi^{\varsigma_\xi} = 2^{\varsigma_\xi} < \kappa_\xi$. At μ the lower product is trivial. The recursion has set length χ . $\square$

25.4 Finite Boolean cubes (source 14)

Theorem 25.10 (Finite relative spectrum; source 14). *Let $M \models \text{ZFC} + \text{GCH}$, let $\kappa_0 < \dots < \kappa_d$ be uncountable regular cardinals of M , force with $\mathbb{P} = \prod_{i \leq d} \text{Add}(\kappa_i, 1)^M$, and put $N = M[G]$ and $M_A = M[G_i : i \in A]$ for $A \subseteq \{0, \dots, d\}$. Then*

$$\text{Char}_N(\mathbf{No}^{M_A}) = \{\kappa_i : i \notin A\}.$$

If $n_ = \min(\{0, \dots, d\} \setminus A)$ exists, the first new ordinal sequence and the first new sign over M_A both have length κ_{n_*} ; that is, $\delta(M_A, N) = \beta(M_A, N) = \kappa_{n_*}$.*

Proof. Under GCH each coordinate has size κ_i . For preservation, factor at any regular μ into the coordinates at most μ and those above: the lower factor has size at most μ , the upper adds no sequences of length μ , and lemma 25.1 shows the upper factor cannot add short cofinal maps after the lower. If a coordinate sits at μ it is $< \mu$ -closed and the still lower finite product has size below μ ; if not, the lower factor is small. Either way the regularity of μ is preserved, hence all cofinalities and cardinals, also for every subproduct.

For $i \notin A$, g_i is fresh over M and stays fresh over M_A (corollary 25.4). Conversely, let μ be regular and not a missing coordinate, and suppose $f : \mu \rightarrow \text{Ord}$ is fresh over M_A . Put $U = M[G_i : \kappa_i > \mu]$ and $V_* = U[G_i : i \in A, \kappa_i \leq \mu]$. The remaining factor $\mathbb{R}$ consists of the missing lower coordinates; it has size strictly below μ , since no missing coordinate equals μ and finitely many lie below it; and $M_A \subseteq V_*$, $N = V_*[G_{\mathbb{R}}]$. If $f \notin V_*$, it is fresh over V_* and added by a forcing of size below μ , contradicting theorem 23.2; so $f \in V_*$. The high product adds no μ -sequences over M and the retained lower product has size at most μ , so lemma 25.1 gives $f \in M[G_i : i \in A, \kappa_i \leq \mu] \subseteq M_A$, a contradiction. Cofinality preservation and theorem 22.7 finish. For the last assertion, all factors below κ_{n_*} are retained; factor into these and the $< \kappa_{n_*}$ -closed tail, and the general form of lemma 25.1 puts every sequence of length below κ_{n_*} into M_A , while g_{n_*} is a fresh sign of length κ_{n_*} . $\square$

Example 25.11 (Four fields, one threshold; source 14). With three coordinates $\kappa_0 < \kappa_1 < \kappa_2$ and $0 \notin A$, the four spectra are $\{\kappa_0\}$, $\{\kappa_0, \kappa_1\}$, $\{\kappa_0, \kappa_2\}$ and $\{\kappa_0, \kappa_1, \kappa_2\}$: four old fields in one universe with the same first gap and the same uncountable saturation spectrum, pairwise nonisomorphic. With $d + 1$ coordinates one obtains 2^d such fields. The countable construction does not simply replace d by ω : an additional singular-successor character appears.

25.5 All coordinate grounds of a countable product

Fix in $M \models \text{GCH}$ a strictly increasing sequence $\langle \kappa_n : n < \omega \rangle$ of uncountable regular cardinals, put $\lambda = \sup_n \kappa_n$, and let

$$\mathbb{P} = \prod_{n < \omega}^{\text{full}} \text{Add}(\kappa_n, 1)^M, \quad N = M[G], \quad M_A = M[G \restriction A] \quad (A \subseteq \omega), \quad (25.1)$$

with full support and ground conditions throughout. For $A \subseteq \omega$ write $B = \omega \setminus A$ for the omitted coordinates. Source 13 treats this generality; source 14 treats $\kappa_n = \aleph_{n+1}^M$, $\lambda = \aleph_\omega^M$ (its final universe is called W there).

Proposition 25.12 (Published product input; imported; sources 13, 14). *Assume GCH and let $\langle \kappa_n \rangle$ be strictly increasing infinite regular cardinals with supremum λ . The full product of the Cohen forcings at the κ_n preserves all cardinals and cofinalities, and every infinite coordinate subproduct adds a fresh function on λ^+ ; more precisely, the full product over an unbounded set of regulars below the singular λ adds a λ^+ -Cohen subset over the ground.*

Source, not a new proof. Preservation is [14, Proposition 4.10]; these full products are Easton products, since only finitely many coordinates lie below any ordinal below λ . The witness is [14, Corollary 4.21],

obtained there from Shelah's Cohen-algebra embedding theorem [24]. Such a λ^+ -Cohen subset is fresh over the original ground: its own Cohen extension is $< \lambda^+$ -closed and adds no shorter sequences, even though the full product does. The full support and the singular limit are indispensable. Neither assertion is reproved, and every novelty claim excludes them (sources 13, 14). The finite theorem, the single-Cohen theorem and every exclusion argument below are independent of this input; it is used only for the presence of λ^+ when B is infinite (source 14).

Lemma 25.13 (Sizes and unchanged reals; source 13). *Let $\mathbb{P}_F$ be the subproduct on $F \subseteq \omega$, formed in M . Then $|\mathbb{P}_F|^M$ is at most 1 if $F = \emptyset$, at most $\max_{n \in F} \kappa_n$ if F is finite and nonempty, and at most λ^+ if F is infinite; the same bounds hold in every intermediate extension and in N . Moreover $\mathcal{P}(\omega)^N = \mathcal{P}(\omega)^M$.*

Proof. Each coordinate has size κ_n under GCH, and for infinite F , $|\mathbb{P}_F| \leq \lambda^{\aleph_0} \leq (2^\lambda)^{\aleph_0} = 2^\lambda = \lambda^+$. Ground enumerations persist, and by proposition 25.12 the cardinal bounds are preserved; preservation in the final model gives it in intermediate models, since a collapse or a shortened cofinality cannot be undone. Since κ_0 is uncountable, a descending countable sequence of product conditions has a coordinatewise union, so the product is countably closed and adds no reals. $\square$

Lemma 25.14 (Preservation and localization for the full product; source 14, second route). *For $\kappa_n = \aleph_{n+1}^M$: (i) $\mathbb{P}$ adds no countable ordinal sequences; (ii) every κ_n , and λ and λ^+ , remain cardinals with their original cofinalities; (iii) all cardinals and cofinalities are preserved; (iv) for every infinite cardinal $\mu < \lambda$, some finite head extension $M[G_i : i \leq n]$ contains every μ -sequence of ordinals of N . All intermediate subproduct models have the cardinals and cofinalities of M and N .*

Proof. (i) Full support permits coordinatewise unions of countable descending chains, every κ_n being uncountable regular. (iv) The head through n has size κ_n and the tail after n is $< \kappa_{n+1}$ -closed in M , so lemma 25.1 gives ${}^\mu\text{Ord} \cap N = {}^\mu\text{Ord} \cap M[G_i : i \leq n]$ for $\mu \leq \kappa_n$. To preserve κ_n , split immediately before coordinate n : the head has size below κ_n and the tail from n on is $< \kappa_n$ -closed, so every η -sequence, $\eta < \kappa_n$, lies in the head extension, and a forcing of size below a ground regular cannot change its cofinality. The limit λ remains a cardinal (collapsing it would collapse a preserved cardinal below it) and its ground countable cofinal sequence stays cofinal. GCH gives $|\mathbb{P}|^M \leq \lambda^{\aleph_0} \leq (2^\lambda)^{\aleph_0} = \lambda^+$. If λ^+ acquired cofinality $\mu < \lambda^+$, then $\mu < \lambda$ (an infinite cofinality is regular, λ is singular, and forcing creates no cardinals), so choose n with $\mu \leq \kappa_n$; by (iv) the cofinal sequence lies in a head extension by a forcing of size $\kappa_n < \lambda^+$, which preserves the regularity of λ^+ . This also rules out its collapse. The λ^{++} -chain condition preserves the cofinalities of ground regulars at least λ^{++} . So all ground regulars keep their cofinalities; a change in the cofinality of another ordinal would change that of a ground regular, and no cardinal collapses while all ground regulars are preserved. Intermediate models follow by the same factorization, or because a collapse there would persist. $\square$

This is a self-contained second route to the preservation half of proposition 25.12 in the case $\kappa_n = \aleph_{n+1}^M$. Every $A \subseteq \omega$ of N is in M , so all the M_A are defined by ground subproducts; $\mathbb{R}^{M_A} = \mathbb{R}^M = \mathbb{R}^N$; and $N \models 2^{\aleph_0} = \aleph_1$ (source 14, its Corollary 8.2).

Theorem 25.15 (Exact spectra of all coordinate grounds; source 13 (base), source 14). *In (25.1), for every $A \subseteq \omega$ with $B = \omega \setminus A$,*

$$\text{Char}_N(\mathbf{No}^{M_A}) = \{\kappa_n : n \in B\} \cup \begin{cases} \emptyset, & B \text{ finite,} \\ \{\lambda^+\}, & B \text{ infinite.} \end{cases}$$

All cardinals in this formula, including λ^+ , have the same meaning in M , every M_A and N . There are no asymmetric set-presented gaps. Every character on the right has a fresh binary witness whose length is that character (source 14).

Proof. All relevant cofinalities are preserved, so by theorem 22.7 it suffices to compute $\mathfrak{F}(M_A, N)$.

Lower inclusion. For $n \in B$, the coordinate union $g_n : \kappa_n \rightarrow 2$ is fresh over M in the one-coordinate extension (closure makes proper prefixes old, genericity makes it new), and the retained coordinates are independent of it, so it is fresh over M_A by corollary 25.4. If B is infinite, it is unbounded in ω , so its coordinate cardinals still have supremum λ ; by proposition 25.12 the subextension $M[G \restriction B]$ contains a λ^+ -Cohen subset r , fresh over M , and corollary 25.4 makes it fresh over M_A .

Nothing unexpected below λ (source 13). Let $\mu < \lambda$ be infinite regular with $\mu \notin \{\kappa_n : n \in B\}$, and suppose $f : \mu \rightarrow \text{Ord}$ is fresh over M_A . Let $J = \{n : \kappa_n \leq \mu\}$, a finite head of size at most μ in M ; the tail is $\leq \mu$ -closed in M , so lemma 25.1, applied to the whole original product before changing the base model, gives $f \in L := M[G \restriction J]$. Let $W_0 = M[G \restriction (J \cap A)]$. For $\alpha < \mu$, $f \restriction \alpha$ lies in M_A and in L , so in W_0 by lemma 25.3; and $f \notin W_0 \subseteq M_A$. So f is fresh over W_0 in L . But L is obtained from W_0 by the finite old product on $J \cap B$, whose coordinates are all *strictly* below μ (equality would put $\mu \in \{\kappa_n : n \in B\}$); it has size below μ in W_0 and L . If it is empty there is no extension; otherwise theorem 23.2 excludes a fresh function on the still-regular μ .

The ceiling. If $B = \emptyset$ then $M_A = N$ and the internal cut theorem leaves no set-presented gap. Otherwise N/M_A is a forcing extension by the old poset $\mathbb{P}_B$, whether or not it keeps its former closure. If B is finite, lemma 25.13 and theorem 23.2 exclude every character above $\max_{n \in B} \kappa_n$, and the rest is below λ . If B is infinite, the ceiling excludes every character above λ^+ ; λ remains singular and no cardinal lies strictly between λ and λ^+ , so λ^+ is the only possible character not below λ . Below λ use the previous paragraph. $\square$

Second route to the exclusion of retained coordinates (source 14). Fix $n \in A$, $\mu = \kappa_n$, and a hypothetical fresh $f : \mu \rightarrow \text{Ord}$ over M_A . Put $U = M[G_i : i > n]$ and $V_* = U[G_i : i \in A, i \leq n]$; then $M_A \subseteq V_*$, since V_* contains every retained coordinate and all missing high coordinates, and N/V_* is the finite ground product $\mathbb{R} = \prod_{i \in B, i \leq n} \text{Add}(\kappa_i, 1)^M$, all of whose factors have index below n , so $|\mathbb{R}| < \mu$. If $f \notin V_*$ it is fresh over V_* , contradicting theorem 23.2; so $f \in V_*$. Reordering, the high factor is $< \kappa_{n+1}$ -closed in M and adds no μ -sequences, and the retained low factor has size at most μ , so lemma 25.1 gives $f \in M[G_i : i \in A, i \leq n] \subseteq M_A$. This works even when A and B are both infinite. The remaining characters are excluded as follows: no countable sequence is new, so ω is out; every regular below λ is ω or some κ_n , and λ is singular; above λ^+ the quotient $\mathbb{P}_B$ has size at most λ^+ , and if B is finite at most its largest coordinate, which excludes λ^+ too. The argument deliberately allows M_A to contain new sequences of old conditions: closure is invoked only in M , at the head–tail factorization where it is known, and fresh prefixes are pulled back to the common base by mutual genericity. This is the relative step that a direct appeal to the ground product’s spectrum would not supply (sources 13, 14).

Example 25.16 (A concrete spectrum table; sources 13, 14). For $\kappa_n = \aleph_{n+1}$, $\lambda = \aleph_\omega$, the following are full character spectra in one extension, not lower bounds.

Omitted coordinates B	$\text{Char}_N(\mathbf{No}^{M_{\omega \setminus B}})$
$\emptyset$	$\emptyset$
$\{0\}$	$\{\aleph_1\}$
$\{0, 2\}$	$\{\aleph_1, \aleph_3\}$
$\{2m : m < \omega\}$	$\{\aleph_{2m+1} : m < \omega\} \cup \{\aleph_{\omega+1}\}$
ω	$\{\aleph_{n+1} : n < \omega\} \cup \{\aleph_{\omega+1}\}$

The two infinite rows show the successor-of-singular contribution, which listing the generic coordinate lengths would miss. All nonempty rows have least omitted coordinate 0 and hence the same saturation spectrum; the isolated character $\aleph_{\omega+1}$ detects an infinite complement.

Proposition 25.17 (An exact coordinate invariant; source 13, with source 14). *For $A, A' \subseteq \omega$ with omitted sets B, B' :*

$$\begin{aligned} \mathbf{No}^{M_A} \subseteq \mathbf{No}^{M_{A'}} &\iff A \subseteq A', \\ \mathbf{No}^{M_A} \cap \mathbf{No}^{M_{A'}} &= \mathbf{No}^{M_{A \cap A'}}, \\ \text{Char}_N(\mathbf{No}^{M_A}) \subseteq \text{Char}_N(\mathbf{No}^{M_{A'}}) &\iff B \subseteq B', \end{aligned}$$

and $A = \{n < \omega : \kappa_n \notin \text{Char}_N(\mathbf{No}^{M_A})\}$. Consequently $\mathbf{No}^{M_A} \cong \mathbf{No}^{M_{A'}}$ as class real closed fields (or as orders with a set-respecting class isomorphism) if and only if $A = A'$.

Proof. If $A \subseteq A'$ then $M_A \subseteq M_{A'}$. If $n \in A \setminus A'$, the generic sign g_n lies in $\mathbf{No}^{M_A}$ but not in $\mathbf{No}^{M_{A'}}$, by mutual genericity. A sign in both fields has an ordinal code in both models, and lemma 25.3 puts it in $M_{A \cap A'}$. The coordinate cardinals recover A , since λ^+ differs from all of them, and the λ^+ clause respects inclusion. An isomorphism of real closed fields preserves the order (positivity is being a nonzero square), and as a class map preserves set presentations and their sizes, hence the spectrum. $\square$

The fields form a poset anti-isomorphic to $\mathcal{P}(\omega)$ under omitted sets. The intersection assertion is a *binary* one; field composita are not identified, and it is not asserted that arbitrary intersections commute with unions of omitted sets (source 13). Within this displayed family the gap spectrum is complete, which is a completeness result for the indexed family only, not a theorem that equal spectra classify arbitrary old fields, arbitrary real closed class fields or arbitrary involutions (sources 13, 14). The models do not recover the first new birthday from the pure field language in every forcing extension; cofinality compression is specifically a loss of ordinal-length information (source 13).

Proposition 25.18 (Exact saturation in the coordinate family; sources 13, 14). *Let $B \neq \emptyset$ and $n_* = \min B$. Then $\delta(M_A, N) = \beta(M_A, N) = \kappa_{n_*}$; $\mathbf{No}^{M_A}$ is ω -saturated, and for uncountable τ it is τ -saturated in N exactly when $\tau \leq \kappa_{n_*}$. If $B = \emptyset$, $\mathbf{No}^{M_A} = \mathbf{No}^N$ realizes all N -set-sized types.*

Proof. By theorem 25.15 there is no character below κ_{n_*} and one at κ_{n_*} , so by corollary 22.10 no ordinal sequence shorter than κ_{n_*} is new, and theorem 1.1 and corollary 22.11 give the uncountable clause; no new reals and theorem 7.1 give ω -saturation. The fresh sign g_{n_*} has length κ_{n_*} , so $\beta(M_A, N) \leq \kappa_{n_*}$ and $\delta \leq \beta$ (proposition 10.2) gives equality. *Source 14's route* proves the short-sequence clause directly: all coordinates below n_* are retained, the head below n_* has size below κ_{n_*} , the tail from n_* on is $< \kappa_{n_*}$ -closed, so every sequence of length below κ_{n_*} lies in the retained head extension, while g_{n_*} is a fresh binary sequence of length κ_{n_*} (its Lemma 10.1). *Source 13's route* proves saturation directly: an N -set $A_0 \subseteq \mathbf{No}^{M_A}$ of size below τ lies in a bounded old set of signs, so an ordinal-coded enumeration of it is old; a complete one-type over A_0 has at most $\max(\aleph_0, |A_0|)$ formulas and is old by the same coding; finite satisfiability is absolute by quantifier elimination; and inside M_A the real closure of the field generated by A_0 and the surreal cut theorem realize it, including end cuts and infinitesimal sides (its Lemma 8.2). Either way, a gap of character κ_{n_*} gives a finitely satisfiable omitted type over κ_{n_*} parameters, so every $\tau > \kappa_{n_*}$ fails. $\square$

The threshold is proved directly in the setting at hand and does not depend on formal verification of the general external saturation theorem (source 13). The fields with one least omitted coordinate are indistinguishable by this threshold but distinguishable by their spectra.

Corollary 25.19 (One threshold, many nonisomorphic fields; sources 14, 13). *The fields $\mathbf{No}^{M_A}$ with $0 \notin A$ are $2^{\aleph_0}$ in number in N , pairwise nonisomorphic as class ordered fields under set-respecting class isomorphisms, contain the same real field, agree on the common subfield $(\mathbf{No}_{<\omega_1})^M = (\mathbf{No}_{<\omega_1})^N$, and are all κ_0 -saturated but not κ_0^+ -saturated, with the same entire saturation spectrum and ω -saturated. If $A \subseteq A'$*

then $\mathbf{No}^{M_A} \subseteq \mathbf{No}^{M_{A'}} \subseteq \mathbf{No}^N$ are elementary ordered-field inclusions, and all these fields have the same first-order theory. For $\kappa_n = \aleph_{n+1}$ they are ω_1 -saturated but not ω_2 -saturated.

No reals are added, so here $2^{\aleph_0} = \aleph_1$ in N ; “continuum many” does not assert an arbitrarily large continuum (sources 13, 14). One may restrict further to infinite B without reducing the size of the family, since only countably many B are finite.

25.6 Families of any size with one threshold

Source 10’s Boolean family. Fix an infinite cardinal χ in a GCH ground V and an uncountable regular $\kappa_* > \chi$. For $i < \chi$ let $\varsigma_i = \sup(\{\kappa_*\} \cup \{\kappa_j : j < i\})$ and $\kappa_i = (2^{\varsigma_i})^+$ (cardinal arithmetic in V), so $\kappa_i > \varsigma_i$ and the sequence increases. Let $I = \{\kappa_i : i < \chi\}$, $A = \{\kappa_*\} \cup I$ and $N = V[G]$ for G generic over $\mathbb{P}_A$.

Lemma 25.20 (Every designated coordinate is isolated enough; source 10). *For every $i < \chi$, $|\mathbb{P}_{A \cap \kappa_i}|^V < \kappa_i$. The full product is $< \kappa_*$ -closed, preserves all cardinals and cofinalities, and adds no subsets of χ and no reals.*

Proof. Every factor below κ_i has size at most ς_i , and there are at most $\chi + 1 \leq \varsigma_i$ of them, so even the full-support product has size at most $\varsigma_i^{\varsigma_i} = 2^{\varsigma_i} < \kappa_i$. For a descending chain of length below κ_* , coordinatewise union preserves condition sizes and the bounds below inaccessibles, as in lemma 25.2. Preservation is theorem 24.2. Closure adds no ordinal sequences of length below $\kappa_* > \chi, \omega$. $\square$

For $S \subseteq \chi$ let $K_S = \{\kappa_i : i \in \chi \setminus S\}$: the fields $\mathbf{No}^{M_{K_S}}$ retain the coordinates outside S , omit the test coordinates in S , and omit the common coordinate κ_* always. Since no subset of χ is added, all $S \in \mathcal{P}(\chi)^N$ are ground sets.

Theorem 25.21 (Boolean-indexed nonisomorphic old fields with one threshold; source 10). *In the single universe N , the fields $\mathbf{No}^{M_{K_S}}$, $S \subseteq \chi$, satisfy:*

- (i) *their isolated-character profiles are exact:* $\text{Char}_N(\mathbf{No}^{M_{K_S}}) \cap A = \{\kappa_*\} \cup \{\kappa_i : i \in S\}$;
- (ii) $\mathbf{No}^{M_{K_S}} \subseteq \mathbf{No}^{M_{K_T}} \iff T \subseteq S$, *and every such inclusion is elementary in the ordered-field language;*
- (iii) *distinct indices give nonisomorphic ordered fields and nonisomorphic pure fields; there are 2^χ of them, computed in N ;*
- (iv) *all have the same real field $\mathbb{R}^V = \mathbb{R}^N$, no set-sized cofinal or cointial subset, least gap size κ_* and first new birthday κ_* relative to N ;*
- (v) *for every infinite cardinal τ of N , each is τ -saturated exactly when $\tau \leq \kappa_*$.*

Proof. (i) Apply theorem 25.5 with lemma 25.20 at each κ_i : $\kappa_i \notin K_S$ iff $i \in S$. The generic g_{κ_*} is fresh over every M_{K_S} by the missing-coordinate argument. (ii) If $T \subseteq S$, then $K_S \subseteq K_T$. If $i \in T \setminus S$, the sign obtained from g_{κ_i} lies in $\mathbf{No}^{M_{K_S}}$ but is fresh over M_{K_T} . Quantifier elimination makes inclusions of real closed fields elementary. (iii) Distinct indices give different spectra by (i); an order isomorphism preserves characters, and a pure-field isomorphism of real closed fields is an order isomorphism. There are $|\mathcal{P}(\chi)^N| = 2^\chi$ indices. (iv) No ordinal sequence of length below κ_* is added from V , hence from any M_{K_S} , so corollary 22.10 excludes smaller characters; g_{κ_*} has birthday κ_* while every shorter sign is in V ; no reals are added; old ordinals bound every N -set (lemma 3.3). (v) Source 10 proves saturation without importing a preservation theorem: a parameter set A_0 of size below κ_* has bounded birthdays, an N -enumeration of it coded through a well-order of $(\mathbf{No}_{<\gamma})^{M_{K_S}}$ is a short ordinal sequence, hence in V and in M_{K_S} ; a complete finite-variable type over A_0 is coded by a binary sequence of length at most $\max(|A_0|, \aleph_0) < \kappa_*$, old for the same reason, and

its finite satisfiability is absolute; internally the surreal field realizes it (real closure of the parameters, cut filling, successive variables). The gap of g_{κ_*} gives a finitely satisfiable omitted type over κ_* parameters, so κ_*^+ -saturation fails; monotonicity gives the rest, including ω . $\square$

The least gap has sides of size κ_* , and saturation at κ_* only tests parameter sets of size *strictly* less than κ_* , so it is not contradicted by that gap: the first failure is at κ_*^+ (source 10).

Corollary 25.22 (The endpoint members; source 10). $\text{Char}_N(\mathbf{No}^{M_I}) = \{\kappa_*\}$ and $\text{Char}_N(\mathbf{No}^{M_\omega}) = \text{Char}_N(\mathbf{No}^V) = \text{Ecl}^V(A)$.

Proof. With no coordinate retained, $M_\emptyset = V$ and theorem 24.3 applies. With all test coordinates retained, the only missing factor is the Cohen forcing at κ_* ; the retained product is $\leq \kappa_*$ -closed in V , so adds no sequences of length at most κ_* and no new Cohen conditions at κ_* , and the remaining old poset is exactly $\text{Add}^{M_I}(\kappa_*, 1)$. Apply theorem 23.6. This equality of posets is justified here and not assumed for arbitrary intermediate members. $\square$

Example 25.23 (Continuum many fields with the same first gap; source 10). Take $\chi = \omega$, $\kappa_* = \aleph_1$ in a GCH ground; the recursion gives $\kappa_i = \aleph_{2i+3}$. The final universe has $2^{\aleph_0}$ pairwise nonisomorphic old fields, all $\aleph_1$ -saturated but not $\aleph_2$ -saturated, with $\aleph_{2i+3} \in \text{Char}_N(\mathbf{No}^{M_{\aleph_S}}) \iff i \in S$, the same reals and least new birthday $\aleph_1$. The largest in inclusion has spectrum $\{\aleph_1\}$; the smallest is $\mathbf{No}^V$ with spectrum $\{\aleph_{2n+1} : n < \omega\} \cup \{\aleph_{\omega+1}\}$. Source 10 does not infer the remaining full spectra from the isolated profiles and leaves room for accumulation characters. *Merge note:* $\kappa_* < \kappa_0 < \kappa_1 < \dots$ is a strictly increasing ω -sequence of uncountable regulars and, having no inaccessible limit point, its Easton product is the full product; so theorem 25.15 (source 13) computes every member: $\text{Char}_N(\mathbf{No}^{M_{\aleph_S}}) = \{\aleph_1\} \cup \{\aleph_{2i+3} : i \in S\} \cup \{\aleph_{\omega+1} : S \text{ infinite}\}$. This answers source 10's own question for $\chi = \omega$ (question 31.11).

Corollary 25.24 (The least threshold is not a classification invariant; source 10). *Even after fixing the real field, the first new birthday, the least omitted-cut character and the entire saturation spectrum, there can be arbitrarily large set-indexed families of pairwise nonisomorphic old surreal fields in one universe, and their elementary-subfield inclusion order can contain the opposite of $\mathcal{P}(\chi)$.*

The last assertion is about inclusion order only; it does not identify intersections or field composita with Boolean operations (source 10).

Remark 25.25 (A corrected overstatement of source 10). Source 10 presents this corollary as “a concrete negative answer to the suggestion that the least gap size, or equivalently the saturation spectrum in this family, could classify the fields”. This report made no such suggestion: question 19.4 asks which invariants *beyond* the least omitted-cut size distinguish old fields, and section 9 states that different grounds can have the same threshold. The corollary is correctly read as showing that the least threshold, and even the whole saturation spectrum, is not a complete invariant, which that question presupposes; source 10 itself states that it does not answer the other half of the question, whether the *full* gap spectrum is complete (answered negatively by sources 12, 09, 11: sections 27 to 29).

Source 11's family of any length. Here every family member omits exactly two coordinates.

Theorem 25.26 (A large same-saturation family in one universe; source 11). *Assume $V \models \text{ZFC} + \text{GCH}$, μ uncountable regular and χ an infinite cardinal. There are a cardinal- and cofinality-preserving set-forcing extension N and intermediate grounds M_ξ , $\xi < \chi$, with $\text{Char}_N(\mathbf{No}^{M_\xi}) = \{\mu, \kappa_\xi\}$ for pairwise distinct regular $\kappa_\xi > \mu$. All the $\mathbf{No}^{M_\xi}$ are ω -saturated and, for uncountable λ , λ -saturated exactly when $\lambda \leq \mu$; they are pairwise nonisomorphic as ordered fields, have no set-sized cofinal subsets, and all contain every real of N .*

Proof. Take $A = \{\mu\} \cup \{\kappa_\xi : \xi < \chi\}$ sparse by proposition 25.9, force with $\mathbb{P}_A$, and let $M_\xi = M_{A \setminus \{\mu, \kappa_\xi\}}$. By theorem 25.8 the spectrum is exactly $\{\mu, \kappa_\xi\}$. The full product is $< \mu$ -closed (no coordinate below μ ; bounded supports at inaccessibles above μ stay bounded under unions of fewer than μ), so no reals are added. Apply theorem 7.1, corollary 22.11, and lemma 3.3. A class order isomorphism with Replacement for its graph preserves the spectrum. $\square$

Corollary 25.27 (Saturation is strictly coarser than the full spectrum; source 11). *Even among old fields containing all reals of the ambient universe and having no set-sized cofinal subset, the complete saturation spectrum determines neither the ordered-field isomorphism type nor the complete gap spectrum.*

This cannot be obtained by assigning distinct least thresholds: the threshold is held fixed, and the second omitted character carries the distinguishing information (source 11). An infinite family obtained by changing only the least character would not establish theorem 25.26.

26 Set-sized cutoffs and local ubiquity

26.1 The exact birthday boundary (source 14)

For $\mathbf{No}^M$ every proper prefix of an old sign is in the field. For a birthday cutoff $(\mathbf{No}_{<\theta})^M$ a missing separator can instead be an *old* sign of length θ , and such signs can have asymmetric characters; so the correspondence of theorem 8.4 cannot be used blindly at a cutoff (source 14). The nonendpoint clause of definition 6.1 is essential here: strict inequalities $x > a$ with a coinital family above a can fail to have a realization without describing a genuine gap, since the cut may have the existing endpoint a ; only genuine gaps are counted, not all omitted systems of strict inequalities.

Lemma 26.1 (Cutoff gap decomposition; source 14). *Suppose $M \subseteq N$ have the same cofinalities, θ is a common uncountable regular cardinal and $F_\theta = (\mathbf{No}_{<\theta})^M$. Put $H_\theta(M, N) = \{\text{cf}^N \alpha : \alpha < \theta, \text{ some fresh sign has length } \alpha\}$. Then*

$$\text{GapSpec}_N(F_\theta) = \{(\mu, \mu) : \mu \in H_\theta(M, N)\} \cup \{(\mu, \theta), (\theta, \mu) : \mu \leq \theta \text{ infinite regular in } N\}.$$

Proof. Take a genuine gap of the set F_θ ; its two sides form a set presentation, and their simplest separator s in $\mathbf{No}^N$ has $\text{b}(s) \leq \theta$ by (2.1). If $\text{b}(s) < \theta$, s is not old (it would lie in F_θ); its shortest new prefix has the same comparisons with old elements and separates the cut, so minimality makes s fresh; its proper prefixes lie in F_θ , and the canonical-prefix argument gives the character $(\text{cf}^N \text{b}(s), \text{cf}^N \text{b}(s))$. If $\text{b}(s) = \theta$ and s is new, it is fresh by the same argument, both sign sets are cofinal in θ , and the character is (θ, θ) . If s is old of length θ , its plus and minus positions P, Q index a strictly increasing cofinal chain on the left and a strictly decreasing coinital chain on the right; neither is empty or has a last index, as the gap is genuine; $P \cup Q = \theta$ and θ is regular, so one of them has cofinality θ and the other some infinite regular cofinality at most θ . Conversely, a fresh sign of length below θ gives its usual gap in F_θ . For regular $\mu < \theta$ the old sign $+^\mu \frown -^\theta$ has length θ and a genuine gap of character (μ, θ) : its left canonical chain has cofinality μ , its right one θ , and no sign shorter than θ extends it. Negation gives (θ, μ) , and an old sign of length θ with plus at limit and minus at successor positions gives (θ, θ) . $\square$

With $\mu = \omega$ the lemma gives an (ω, θ) gap in the full internal cutoff $\mathbf{No}_{<\theta}$, which is not a fresh-sign gap across universes: “all genuine gaps of a birthday cutoff are symmetric” is false even without forcing. The full-class theorem avoids this because the old sign of length θ is then an available separator (source 14). The symmetry of corollary 8.6 is a statement about the class $\mathbf{No}^M$ only. The lemma is the gap-pair counterpart, across universes, of theorem 12.1, which gives the saturation of such fragments.

Theorem 26.2 (Exact gap pairs at a common cutoff; source 14). *In the setting of (25.1) with $\kappa_n = \aleph_{n+1}$, put $\theta = \lambda^{++} = \aleph_{\omega+2}$ and $S_A = \text{Char}_N(\mathbf{No}^{M_A})$. Then*

$$\text{GapSpec}_N((\mathbf{No}_{<\theta})^{M_A}) = \{(\mu, \mu) : \mu \in S_A\} \cup \{(\mu, \theta), (\theta, \mu) : \mu \leq \theta \text{ infinite regular}\},$$

so $\text{Char}_N((\mathbf{No}_{<\theta})^{M_A}) = S_A \cup \{\theta\}$. Every $(\mathbf{No}_{<\theta})^{M_A}$ is a real closed field of cardinality exactly θ , and distinct indices give nonisomorphic fields of this same cardinality.

Proof. The fields are real closed since θ is uncountable regular in every model. Each character in S_A has a fresh binary witness of length equal to it (theorem 25.15), at most $\lambda^+ < \theta$; conversely a fresh sign of length below θ is a fresh sign for the full old field. So $H_\theta(M_A, N) = S_A$ and lemma 26.1 applies. Each field contains every ordinal below θ . For the upper bound count names in M : $|\mathbb{P}| \leq \lambda^+$; for $\eta < \theta$ a binary η -sequence has a name coded by a subset of $\mathbb{P} \times \eta$, and $|\eta|^M \leq \lambda^+$ with $M \models 2^{\lambda^+} = \theta$, so there are at most θ such names, and θ choices of η ; so $|(\mathbf{No}_{<\theta})^N|^N \leq \theta$. Distinct A give different S_A at coordinate cardinals strictly below θ , which the common boundary character does not erase, and field isomorphisms preserve order and gap pairs. $\square$

Corollary 26.3 (Equal-sized real closed counterexamples to classification by saturation; source 14). *The fields $(\mathbf{No}_{<\theta})^{M_A}$, $0 \notin A \subseteq \omega$, are $2^{\aleph_0}$ pairwise nonisomorphic real closed fields of cardinality θ . They are elementary subfields of $(\mathbf{No}_{<\theta})^N$, contain the same real field and the same birthday fragment below ω_1 , and have exactly the same saturation spectrum: ω - and ω_1 -saturated, not τ -saturated for any uncountable $\tau > \omega_1$.*

Proof. For fewer than κ_{n_*} endpoints, the coding argument of proposition 25.18 makes the endpoint sets old; their birthday supremum is below θ by regularity, so the internal simplest separator stays in the cutoff. The fresh gap at κ_{n_*} lies below the cutoff. A countable type uses countably many elements of bounded birthday, so ω -saturation stays within the cutoff. Take $n_* = 0$ and apply theorem 26.2. $\square$

26.2 Local ubiquity

Proposition 26.4 (Every character in every interval; source 13 (base), source 14). *Let F be a real closed class field and $a < b$ in F . There is a parameter-definable order isomorphism from F onto $(a, b)_F$. Consequently every set-presented gap character of F is represented inside every such interval, and if F contains all ordinals as an ordered subclass, every occurring character has a proper class of distinct gap realizations, in pairwise disjoint subintervals.*

Proof. With positive square roots put $u(x) = x/\sqrt{1+x^2}$ and $v(y) = y/\sqrt{1-y^2}$ for $-1 < y < 1$; direct substitution gives $u(v(y)) = y$ and $v(u(x)) = x$. The map u preserves signs, and for $0 \leq x < z$, $x^2/(1+x^2) < z^2/(1+z^2)$ by cross-multiplication; oddness gives the negative half, so u is strictly increasing and $x \mapsto a + \frac{b-a}{2}(1+u(x))$ is the required isomorphism. Transport a set presentation of a gap through it and add the parts of F below and above the interval to the lower and upper classes; the transported cofinal and coinital sets keep their exact sizes. Doing this for one fixed gap into the disjoint intervals $(\alpha, \alpha+1)$, α an ordinal, gives proper-class many distinct gaps. $\square$

Second route (source 14, its Proposition 10.4), for old fields. Choose old endpoints a, b bounding a fixed gap. A positive affine map with old coefficients sends a, b to any two old interior points of a target interval; it is an order automorphism of the old field and moves the gap into the interval with unchanged character. Inside $(0, 1)$ the old intervals $I_\alpha = (\omega^{-(\alpha+1)}, 2\omega^{-(\alpha+1)})$, $\alpha \in \text{Ord}$, are pairwise disjoint, a later exponent giving a scale at least a factor ω smaller; an affine change of variables places the family inside any interval,

and transporting the same presented gap into each I_α by explicitly given affine maps needs no selection of a proper class of witnesses.

Recording merely whether a character has finitely many, set many or proper-class many occurrences, or whether it occurs in a chosen interval, therefore adds no information: every occurring character has proper-class multiplicity, even locally. This does *not* rule out finer incidence, dependence or algebraic information about families of cuts (sources 13, 14). In the coordinate family the exact *list* of characters already distinguishes the fields; when two other fields have the same list, a subtler invariant is needed.

27 The real trace and a Boolean family with one spectrum

A gap spectrum records the sizes of cofinal witnesses but not the rational cuts already realized. The latter is an intrinsic invariant of the field, even when no copy of the real numbers is named in its language. This section is source 12's, with source 09's rational-cut invariant as a second route.

27.1 The real trace

For an ordered subfield $F \subseteq \mathbf{No}^N$ put

$$O_F = \{x \in F : |x| < n \text{ for some integer } n \geq 1\}, \quad \mathfrak{m}_F = \{x \in F : |x| < 1/n \text{ for every integer } n \geq 1\},$$

the finite valuation ring and its infinitesimal ideal. The quantifiers over positive integers are external, set-indexed quantifiers; no first-order definability of these classes in a real closed field is claimed. Every finite surreal has a unique standard part in $\mathbb{R}^N$, and one useful cut formula, avoiding a possible greatest rational in a lower cut, is

$$C_x = \{q \in \mathbb{Q} : q + 1/n < x \text{ for some } n \geq 1\}.$$

If $r = \text{st}_N(x)$ then $C_x = \{q \in \mathbb{Q} : q < r\}$; the extra $1/n$ matters when x lies infinitesimally above a rational, since the unmodified set $\{q : q < x\}$ would contain that rational.

Definition 27.1 (Real trace; source 12, source 09). The *real trace* of F in N is $\text{Tr}_{\mathbb{R}}(F) = \{\text{st}_N(x) : x \in O_F\} \subseteq \mathbb{R}^N$, equivalently the set of real Dedekind cuts represented by C_x for finite $x \in F$. Source 09's *rational-cut invariant* $\text{RCut}(F)$ is the set of $r \in \mathbb{R}^N$ such that some $x \in F$ satisfies $a < x < b$ whenever $a, b \in \mathbb{Q}$ and $a < r < b$; it is the same set.

“Intrinsic” refers to invariance under field embeddings; the canonical realization inside $\mathbb{R}^N$ uses the rational prime field. It is stronger than remembering the cardinality of an abstract residue field (source 12). At a rational boundary, the definition deliberately forgets whether that rational lies in the strict lower cut and records its real supremum; the two comparisons are made in their own fields, and no subtraction of an element of F and an external real is needed. It is an isomorphism invariant, not an assertion that “Archimedean-finite” is first-order definable by one formula (source 09).

Proposition 27.2 (Residue realization and functoriality; source 12). O_F is a ring with maximal ideal $\mathfrak{m}_F$, and standard part is a surjective ring homomorphism $\text{st}_N : O_F \rightarrow \text{Tr}_{\mathbb{R}}(F)$ with kernel $\mathfrak{m}_F$, whose image is an Archimedean ordered subfield of $\mathbb{R}^N$. If $j : F \hookrightarrow E$ is an order-preserving field embedding, then $\text{st}_N(j(x)) = \text{st}_N(x)$ for $x \in O_F$, and $\text{Tr}_{\mathbb{R}}(F) \subseteq \text{Tr}_{\mathbb{R}}(E)$.

Proof. Sums and products of finite elements are finite, and a finite multiple of an infinitesimal is infinitesimal; a finite non-infinitesimal x has $|x| \geq 1/n$ for some n , so x^{-1} is finite. So $\mathfrak{m}_F$ is the nonunit ideal. Writing

$x = r + \epsilon$, $y = s + \epsilon'$ with $r, s \in \mathbb{R}^N$ and ϵ, ϵ' infinitesimal gives $\text{st}_N(x + y) = r + s$ and $\text{st}_N(xy) = rs$; inversion of a finite element with nonzero standard part gives inverses. The embedding j fixes every rational, preserves every rational inequality, and preserves and reflects finiteness and infinitesimality; so $C_{j(x)} = C_x$, and two reals with the same lower rational cut are equal. $\square$

The notation $\mathcal{O}_F/\mathfrak{m}_F$ is harmless only if its meaning is specified: for a proper-class field an equivalence class modulo infinitesimals can itself be a proper class and cannot be an element of a set or of a class of sets. The residue field is *represented by the set of real cuts* $\text{Tr}_{\mathbb{R}}(F)$, with the displayed standard-part map; it has the usual quotient universal property without treating proper classes as elements, and “residue field” always has this meaning here (source 12).

Theorem 27.3 (Exact trace of an old surreal field; source 12 (base), source 09). *For every same-ordinal pair $M \subseteq N$, $\text{Tr}_{\mathbb{R}}(\mathbf{No}^M) = \mathbb{R}^M \subseteq \mathbb{R}^N$. If $M_0, M_1 \subseteq N$ and there is any unital field embedding $\mathbf{No}^{M_0} \hookrightarrow \mathbf{No}^{M_1}$, then $\mathbb{R}^{M_0} \subseteq \mathbb{R}^{M_1}$. Consequently an isomorphism forces equality of the real traces as actual subfields of the ambient real line, and $\mathbb{R}^{M_0} \neq \mathbb{R}^{M_1}$ rules out an isomorphism of ordered fields and of pure fields.*

Proof. Let $x \in \mathbf{No}^M$ be finite. Its cut C_x is a set in M , defined from the old parameter x by rational inequalities whose truth is absolute; completeness of the reals *inside* M gives $r \in \mathbb{R}^M$ representing it, and in N it is the same cut, so $r = \text{st}_N(x)$. Every old real is an old finite surreal with itself as standard part. A unital embedding between real closed fields preserves their unique orders ($a < b$ iff $b - a$ is a nonzero square), so proposition 27.2 applies; an isomorphism and its inverse give equality. $\square$

Second route (source 09, its Lemma 6.2). Every $r \in \mathbb{R}^M$ realizes its own rational cut. Conversely, an old surreal x bounded by an integer has, inside M , a real supremum $r = \text{st}(x) \in \mathbb{R}^M$ of its rational cut; equivalently $x - r$ is smaller in absolute value than every positive rational, and these individual rational inequalities are absolute, so the same r is determined by x in N . An ordered-field isomorphism fixes $\mathbb{Q}$ and hence the real label of every rational cut it realizes, and for real closed fields a pure-field isomorphism is order preserving.

An embedding is *not* asserted to fix every real element pointwise as a surreal; it could move a transcendental real by an infinitesimal amount. The rigid conclusion is that its *residue* has the same rational cut, which is exactly what the nonembedding proof needs (source 12).

The trace as a spectrum of countable types (source 12). For $r \in \mathbb{R}^N$ let

$$p_r(X) = \{q < X : q \in \mathbb{Q}, q < r\} \cup \{X < q : q \in \mathbb{Q}, r < q\},$$

rational constants abbreviating terms, or inequalities with cleared integer denominators. Every finite subset is satisfiable in $\mathbb{Q}$, and p_r is realized in an ordered subfield $F \subseteq \mathbf{No}^N$ exactly when $r \in \text{Tr}_{\mathbb{R}}(F)$: a realization is finite with standard part r , and conversely. So p_r is realized in $\mathbf{No}^M$ iff $r \in \mathbb{R}^M$ (compare (7.1)). The sentence $\Phi_r = \exists X \wedge p_r(X)$ is an $L_{\omega_1, \omega}$ sentence of the ordered-field language and, by replacing order with nonzero-square formulas, of the pure field language; it is not a finitary sentence. The real r occurs only in the *external code of its rational inequalities*, not as a named parameter, so there is no conflict with the elementary equivalence of all real closed fields.

27.2 The Boolean family

Theorem 27.4 (Boolean embeddability with a constant full gap spectrum; source 12). *Let V be a ground universe, $I \in V$ a set and $*$ $\notin I$ a reserved coordinate. Let $G \subseteq \text{Add}(\omega, I \cup \{*\})^V = \text{Fn}((I \cup \{*\}) \times \omega, 2, < \omega)^V$ be generic and $N = V[G]$. For $A \in \mathcal{P}(I)^V$ let G_A be the restriction to A and $M_A = V[G_A]$. Then, in N :*

- (i) $\text{GapSpec}_N(\mathbf{No}^{M_A}) = \{(\omega, \omega)\}$ for every A ;
- (ii) for arbitrary unital class field embeddings, $\mathbf{No}^{M_A} \hookrightarrow \mathbf{No}^{M_{A'}} \iff A \subseteq A'$;
- (iii) $\mathbf{No}^{M_A} \cong \mathbf{No}^{M_{A'}}$ if and only if $A = A'$;
- (iv) all $\mathbf{No}^{M_A}$ are elementarily equivalent real closed fields, and none is κ -saturated for any infinite N -cardinal κ ;
- (v) as pure orders, all are ω -saturated and fail κ -saturation for every uncountable κ ;
- (vi) every N -set of elements of $\mathbf{No}^{M_A}$ is bounded above and below in $\mathbf{No}^{M_A}$.

The reserved coordinate is not cosmetic: it makes N add a real over *every* M_A , including M_I , so no endpoint of the family has a different spectrum or saturation status. The family is uniformly class-defined over the set of indices $\mathcal{P}(I)^V$; it is not a set whose elements are proper classes. No large cardinal, CH or GCH is used (source 12).

Let $J = I \cup \{*\}$. For $A \in \mathcal{P}(I)^V$ restriction gives $\text{Add}(\omega, J)^V \cong \text{Add}(\omega, A)^V \times \text{Add}(\omega, J \setminus A)^V$. Finite partial functions on a ground set are the same in an intermediate extension, so in M_A the second factor is precisely $\text{Add}(\omega, J \setminus A)^{M_A}$, not an obsolete ground version of a forcing with new finite conditions missing, and $N = M_A[G_{J \setminus A}]$; the quotient is nontrivial because it contains the reserved coordinate. For $i \in J$ let $y_i : \omega \rightarrow 2$ be the i -th generic sequence and $r_i = \sum_{n < \omega} 2y_i(n)/3^{n+1} \in \mathbb{R}^N$. This ternary coding has a canonical inverse on its image and, unlike the binary expansion map, is injective even at endpoints: at the first different digit the leading difference exceeds the largest possible opposing tail.

Lemma 27.5 (Coordinate detection; source 12). *For $i \in I$ and $A \in \mathcal{P}(I)^V$, $r_i \in \mathbb{R}^{M_A} \iff i \in A$; and $r_* \notin \mathbb{R}^{M_A}$ for every such A .*

Proof. If $i \in A$, $y_i \in M_A$ and so is its code. If $i \notin A$, y_i is Cohen generic over M_A , and for every binary $d \in M_A$ disagreement on the i -coordinate is dense (a finite condition leaves a position free); so $y_i \notin M_A$, and the canonical ternary decoder in M_A would recover y_i from an old r_i . The same holds for $*$. $\square$

Proof of theorem 27.4. (i) Apply proposition 23.12 inside each M_A to its quotient; this excludes *every* uncountable character, not only the first. (ii) If $A \subseteq A'$, inclusion of old signs is an ordered-field embedding. If j is any unital embedding, theorem 27.3 gives $\mathbb{R}^{M_A} \subseteq \mathbb{R}^{M_{A'}}$, and for $i \in A$, $r_i \in \mathbb{R}^{M_A}$ forces $i \in A'$ by lemma 27.5. (iii) Apply (ii) both ways. (iv) Real closed fields share one complete theory. The type p_{r_*} is finitely satisfiable and omitted in every $\mathbf{No}^{M_A}$; for a complete omitted type take the complete type of r_* in $\mathbf{No}^N$ over the empty set, every finite part of which is realized in $\mathbf{No}^{M_A} \preceq \mathbf{No}^N$. So ω -saturation fails, hence every larger saturation. (v) A dense linear order without endpoints is ω -saturated (types over a finite set are realized by named points, the finitely many intervening intervals or the two unbounded ones), and a countably presented omitted gap from (i) gives an omitted type over a countable set. (vi) is lemma 3.3. $\square$

No field-theoretic preservation requirement was imposed beyond a unital embedding: the nonembedding direction allows maps that move ω , change birthdays, fail to preserve simplicity or normal-form monomials; rational-cut rigidity still blocks them (source 12).

Example 27.6 (A four-field diagram; source 12). With $I = \{0, 1\}$ and a reserved coordinate, the four fields embed exactly as $\emptyset \subset \{0\}, \{1\} \subset \{0, 1\}$. The middle two do not embed into one another; each embeds properly into $\mathbf{No}^{M_{\{0,1\}}}$, and $\mathbf{No}^{M_\emptyset}$ into each. All four have only countable-character set-presented gaps and none is ω -saturated as a field. A diagram this small already refutes the implication from equal full spectra to field isomorphism.

Corollary 27.7 (One Cohen real suffices; source 12). *In a one-Cohen-real extension of V there is a family indexed by $\mathcal{P}(\omega)^V$ of pairwise nonisomorphic old surreal fields with all the properties of theorem 27.4. Its size is the continuum of the extension, and each member's real trace has that cardinality.*

Proof. Use $I = \omega$; a ground bijection of $(\omega \cup \{*\}) \times \omega$ with ω identifies the forcing with one-real Cohen forcing. With $\mathfrak{c}^V = (2^{\aleph_0})^V$, the index set has size $\mathfrak{c}^V$, preserved by the c.c.c. A countable forcing has at most $\mathfrak{c}^V$ nice names for reals (each is coded by a subset of a countable set), every real has one, and all ground reals remain; so $\mathfrak{c}^N = \mathfrak{c}^V$. Each M_A contains $\mathbb{R}^V$ and lies in N , so $|\mathbb{R}^{M_A}|^N = \mathfrak{c}^N$. $\square$

The family is indexed by *ground* subsets of ω , not by all subsets in N ; the equality of cardinalities just proved is what licenses “continuum”. So the examples do not exploit different sizes of residue fields: a two-element inclusion chain and a two-element incomparability pattern both occur with the spectrum $\{(\omega, \omega)\}$ (source 12).

Corollary 27.8 (Universality for ground set-sized partial orders; source 12). *For every set-sized partial order $(P, \leq) \in V$, a Cohen extension contains old surreal fields E_p , $p \in P$, with $E_p \hookrightarrow E_q \iff p \leq q$ and $\text{GapSpec}_N(E_p) = \{(\omega, \omega)\}$; distinct indices give nonisomorphic fields with the saturation profiles of theorem 27.4.*

Proof. Take $I = P$ and $A_p = \{q : q \leq p\} \in V$. Then $p \leq q \iff A_p \subseteq A_q$, and $E_p = \mathbf{No}^{M_{A_p}}$; antisymmetry and theorem 27.4(iii) give nonisomorphism. $\square$

“Universality” means exactly this realization of ground set-sized partial orders; it is not a statement about Borel or effective reducibility, nor about a proper-class collection of all partial orders. The order need not be well founded, countable or a lattice, so prescribed antichains, descending chains and mixed diagrams occur inside one gap-spectrum class. The trace does not classify arbitrary old fields; it classifies embeddability *within this family* (source 12).

Theorem 27.9 (Failure of the unrestricted field-classification implication; source 12). *There are same-ordinal $M_0, M_1 \subseteq N$ with $\text{GapSpec}_N(\mathbf{No}^{M_0}) = \text{GapSpec}_N(\mathbf{No}^{M_1})$ and $\mathbf{No}^{M_0} \not\cong \mathbf{No}^{M_1}$ as fields. They can be chosen with the same saturation profile at every infinite cardinal, the same least new birthday ω , and equally large Archimedean residue fields; in one example neither field embeds into the other.*

Proof. Use two incomparable coordinate sets in the countable-coordinate construction; apply theorem 27.4 and corollary 27.7. Finite signs are absolute and the reserved coordinate gives a new sign of length ω , so the first new birthday is ω over both. $\square$

This is a negative answer to the unrestricted *real-closed-field* implication asked about in question 19.4. It does not answer the pure linear-order question: an arbitrary order isomorphism need not preserve 0, 1, addition or the rational copy, so the trace obstruction does not apply to it. Nor does it rule out positive classification under additional hypotheses, such as equality of real traces together with suitable higher compatibility data (source 12).

Proposition 27.10 (The trace is exact on the family; source 12). *For the fields of theorem 27.4 the following are equivalent: (i) $\mathbf{No}^{M_A} \hookrightarrow \mathbf{No}^{M_{A'}}$; (ii) $\text{Tr}_{\mathbb{R}}(\mathbf{No}^{M_A}) \subseteq \text{Tr}_{\mathbb{R}}(\mathbf{No}^{M_{A'}})$; (iii) every type p_r realized in $\mathbf{No}^{M_A}$ is realized in $\mathbf{No}^{M_{A'}}$; (iv) $A \subseteq A'$. Equality of traces is equivalent to $A = A'$, hence to field isomorphism within the family.*

Proof. Theorem 27.3 gives (i) $\implies$ (ii), the type characterization gives (ii) $\iff$ (iii), lemma 27.5 gives (ii) $\implies$ (iv), and inclusion gives (iv) $\implies$ (i). $\square$

The fields do not have different first-order theories; they realize different *sets of countable types* while agreeing on saturation at every cardinal, so appending the entire saturation profile to the gap spectrum does not repair the proposed classification invariant. There is no claim that $\text{Tr}_{\mathbb{R}}(F)$ is efficiently computable or finitary definable; its significance is structural (source 12).

Invariant or property (source 12)	Value throughout the countable-coordinate family
First-order ordered-field theory	RCF
Full set-presented gap spectrum	$\{(\omega, \omega)\}$
Field saturation at infinite cardinals	Fails at every infinite cardinal
Pure-order saturation at infinite cardinals	Holds exactly at ω
Least new birthday over the inner universe	ω
Set-sized cofinal or coinital subset	None
Cardinality of the real trace	$\mathfrak{c}^N$
The actual real trace	$\mathbb{R}^{V[G_A]}$, different for different A
Field embeddability	Exactly inclusion of the coordinate sets

28 Common small forcing and Cohen amplification

This section is source 09's. Its technical ingredient compares different old fields in one ambient universe. It concerns a square, not a two-step iteration with an arbitrarily reinterpreted second forcing:

$$\begin{array}{ccc} M & \subseteq & W \\ \cap & & \cap \\ M[G_{\mathbb{Q}}] & \subseteq & W[G_{\mathbb{Q}}], \end{array}$$

where $G_{\mathbb{Q}}$ is generic over W for the *same set partial order* $\mathbb{Q} \in M$; the outer version of $\mathbb{Q}$ has no additional conditions.

28.1 Two name facts and the base-change theorem

Lemma 28.1 (Atomic forcing is absolute for old names; source 09). *Let $M \subseteq W$ be transitive, $\mathbb{Q} \in M$ the identical partial order in both, and $\sigma, \tau \in M$ $\mathbb{Q}$ -names. For every $q \in \mathbb{Q}$, $q \Vdash \sigma \in \tau$ and $q \Vdash \sigma = \tau$ are absolute between M and W , and so are their forced negations.*

Proof. Induct on the ranks of the names through the recursive definitions of atomic forcing. The quantifiers range over the same conditions, extensions and members of old names, with unchanged compatibility and order. Forced negation is the absence of an extension forcing the positive assertion, again over the same set of conditions. $\square$

Lemma 28.2 (Intersection for ordinal-valued functions; source 09). *Let $G_{\mathbb{Q}}$ be $\mathbb{Q}$ -generic over W as above. An ordinal-valued function belonging to both W and $M[G_{\mathbb{Q}}]$ belongs to M .*

Proof. Let $f : \gamma \rightarrow \Gamma$ be in both, Γ an old ordinal bound. There is an old name τ with $\tau^{G_{\mathbb{Q}}} = f$, and in W the truth lemma gives $q \in G_{\mathbb{Q}}$ forcing $\tau = \check{f}$. For every $(\alpha, \zeta) \in \gamma \times \Gamma$, membership in f is equivalent to $q \Vdash (\alpha, \check{\zeta}) \in \tau$, absolute by lemma 28.1; Separation in M recovers the graph of f from q, τ, γ, Γ . $\square$

Only this function version of the usual intersection argument is needed. A fresh function already in W remains new over $M[G_{\mathbb{Q}}]$, and its prefixes remain old because they were in M to start with.

Theorem 28.3 (Common-small-forcing invariance; source 09). *Let $M \subseteq W$ have the same ordinals, $\mathbb{Q} \in M$ the same set partial order in both, $G_{\mathbb{Q}}$ generic over W and $U = W[G_{\mathbb{Q}}]$. For every limit ordinal γ with*

$$\text{cf}^U \gamma > |\mathbb{Q}|^U \quad (28.1)$$

we have $\text{Fr}_\gamma(M, W) \iff \text{Fr}_\gamma(M[G_{\mathbb{Q}}], U)$. The forward implication does not need (28.1).

Proof. Forward: a fresh $f \in W$ over M has proper prefixes in $M \subseteq M[G_{\mathbb{Q}}]$, and $f \notin M[G_{\mathbb{Q}}]$ by lemma 28.2.

Converse: take $f : \gamma \rightarrow \Gamma$ in U fresh over $M[G_{\mathbb{Q}}]$, a W -name $\dot{f}$ and $q_0 \in G_{\mathbb{Q}}$ forcing that it is a function with this domain and range bound. A *ground set* of names is needed, not an unspecified collection of all names: identify a graph name with a subset of $(\gamma \times \Gamma) \times \mathbb{Q}$, using check names for graph pairs; let T be the set of such names in M and T_α those supported on $(\alpha \times \Gamma) \times \mathbb{Q}$. Restricting a graph name to α is a ground operation, and every subset of $\alpha \times \Gamma$ in $M[G_{\mathbb{Q}}]$ has a name in T_α by the nice-name construction. Fix in M a bijection $e : \tau_0 \rightarrow T$ from an ordinal. For $q \leq q_0$ define in W

$$B_q = \{\alpha < \gamma : \exists \sigma \in T_\alpha \ q \Vdash_W \dot{f} \restriction \alpha = \sigma\}.$$

Every $\alpha < \gamma$ lies in some B_q with $q \in G_{\mathbb{Q}}$, since $f \restriction \alpha \in M[G_{\mathbb{Q}}]$ and the truth lemma supplies a condition. By (28.1) one $q \in G_{\mathbb{Q}}$, $q \leq q_0$, has B_q unbounded in γ , an absolute property of this old set $B_q \in W$. Restricting a witness at a larger member of B_q shows that for every $\alpha < \gamma$ some $\sigma \in T_\alpha$ has $q \Vdash_W \dot{f} \restriction \alpha = \sigma$. Define in W

$$h(\alpha) = \min\{\xi < \tau_0 : e(\xi) \in T_\alpha, q \Vdash_W \dot{f} \restriction \alpha = e(\xi)\}.$$

For $\zeta < \gamma$ fix $\sigma_\zeta \in T_\zeta$ with $q \Vdash_W \dot{f} \restriction \zeta = \sigma_\zeta$; for $\alpha < \zeta$ the minimum is the least index of a $\sigma \in T_\alpha$ with $q \Vdash_W \sigma = \sigma_\zeta \restriction \alpha$, an equality-forcing relation between old names, computed in M by lemma 28.1. So $h \restriction \zeta$ is constructed in M from $q, \sigma_\zeta, e, \zeta$. If $h \in M$, the union $v = \bigcup_{\alpha < \gamma} e(h(\alpha))$ is in M ; each selected name evaluates to $f \restriction \alpha$ since $q \in G_{\mathbb{Q}}$, graph-name union commutes with evaluation, and $v^{G_{\mathbb{Q}}} = f$ as γ is a limit, putting f in $M[G_{\mathbb{Q}}]$. So h is fresh over M . $\square$

Corollary 28.4 (Common dense suborder; source 09). *The theorem holds with $|\mathbb{Q}|^U$ replaced by $|D|^U$ for a dense suborder $D \in M$ of $\mathbb{Q}$ that is the identical dense suborder in M and W .*

Proof. Use the induced D -generic filter; the extensions are unchanged. $\square$

The bound is strict; at the size of the forcing the pigeonhole step can fail. The argument is not replaced by a chain-condition assertion, and the countable boundary is not claimed to behave like uncountable cofinalities; the counterexamples below exploit exactly the difference between countable ordinal sequences and reals. The theorem keeps the underlying poset fixed: a ground forcing that is closed need not remain closed after a small generic is added, since the new universe has new descending sequences of old conditions, and recomputing a forcing from a definition may supply additional conditions; neither operation is the fixed-poset square. So there is no conflict with the small-then-strategically-closed gap-forcing phenomena of Hamkins [21], also discussed in [14, Section 2] (source 09). No assertion of first discovery of the general forcing technique is made.

28.2 Cohen amplification

Theorem 28.5 (Cohen-coordinate amplification; source 09). *Let $M \subseteq W$ be same-ordinal transitive universes with (i) $\mathbb{R}^M = \mathbb{R}^W$ and (ii) some $g : \omega \rightarrow \text{Ord}$ in $W \setminus M$. Let $G_{\mathbb{C}}$ be generic over W for the ground countable forcing $\mathbb{C} = \text{Fn}(\omega \times \omega, 2, < \omega)$ and $N = W[G_{\mathbb{C}}]$. For $S \in \mathcal{P}(\omega)^M$ let G_S be the restriction to $S \times \omega$ and $M_S = M[G_S]$, $W_S = W[G_S]$. Then, in N :*

- (a) $\text{Char}_N(\mathbf{No}^{M_S}) = \text{Char}_W(\mathbf{No}^M)$ for every S : all these fields have the same full gap spectrum, and the cofinality labels on the right are unchanged by the final Cohen forcing;
- (b) for $S \neq T$, $\mathbf{No}^{M_S} \not\cong \mathbf{No}^{M_T}$, even as pure fields;
- (c) the index set $\mathcal{P}(\omega)^M$ has cardinality $2^{\aleph_0}$ in N ;
- (d) $\mathbf{No}^{M_\omega}$ is ω -saturated as a real closed field, and each $\mathbf{No}^{M_S}$ with $S \subsetneq \omega$ is not;
- (e) every $\mathbf{No}^{M_S}$ is a real closed elementary subfield of $\mathbf{No}^N$ with no N -set-sized cofinal subset.

Proof. Step 1: coordinate factorization. For old S the ground poset is $\mathbb{C}_S \times \mathbb{C}_{\omega \setminus S}$ with the expected restriction maps, all countable and identical in M and W . So G_S is generic over W , the complementary filter over W_S , and $M_S \subseteq N$ is an inner universe. Cohen forcing preserves cardinals and cofinalities, so $\text{Reg}^{M_S} = \text{Reg}^M$, and the cofinalities of a fixed ordinal in W , W_S and N agree.

Step 2: agreement above countable cofinality. For $\lambda \in \text{Reg}^M$ with $\text{cf}^N \lambda > \omega$, theorem 28.3 for $\mathbb{C}_S$ gives $\text{Fr}_\lambda(M, W) \iff \text{Fr}_\lambda(M_S, W_S)$ (empty S is trivial). Passing from W_S to N by the countable complementary forcing: a function fresh over M_S in W_S stays fresh in N ; conversely, a function fresh over M_S in N but not in W_S would have its proper prefixes in $M_S \subseteq W_S$ and be fresh over W_S , contrary to theorem 23.2 since $\text{cf}^N \lambda > \omega$. So $\text{Fr}_\lambda(M_S, W_S) \iff \text{Fr}_\lambda(M_S, N)$, and with theorem 22.7 the character sets agree above ω .

Step 3: the countable character occurs everywhere. Finite restrictions of g are old, so g is fresh over M in W ; by lemma 28.2, $g \notin M[G_\mathbb{C}]$, so $g \notin M_S$, and the same prefixes make g fresh over M_S in N . The transfer gives character ω in every field, including $\mathbf{No}^M$ in W . This proves (a).

Step 4: coordinate reals distinguish the fields. Let $y_n : \omega \rightarrow 2$ be the n -th coordinate real, identified through a fixed ground injection with a real of the Cantor set (a coding that preserves membership in each universe). If $n \in S \setminus T$, then $y_n \in M_S$; and $y_n \notin M_T$, since by product factorization over M the n -th coordinate filter is Cohen generic over the extension by all coordinates in T . So $\mathbb{R}^{M_S} \neq \mathbb{R}^{M_T}$, and theorem 27.3 gives (b), without distinguishing saturation levels of two proper subsets.

Step 5: all final reals already appear over M . Every real of $W[G_\mathbb{C}]$ has a nice $\mathbb{C}$ -name in W , coded by a subset of the countable ground set $\omega \times \mathbb{C}$, hence by a real of W , which lies in M by (i); the decoded name is an M -name, so $\mathbb{R}^N = \mathbb{R}^{M[G_\mathbb{C}]}$. This uses the countability of the poset, not merely its chain condition. The old real codes of valid names map onto all final reals (and all old codes do, by a default value), so in N , $2^{\aleph_0} = |\mathbb{R}^N| \leq |\mathcal{P}(\omega)^M| \leq |\mathcal{P}(\omega)^N| = 2^{\aleph_0}$, which is (c), even if W changed the ground cardinal label of the continuum. By theorem 7.1, $\mathbf{No}^{M_\omega}$ is ω -saturated; for proper S choose $n \notin S$, and $y_n \in N \setminus M_S$ shows that $\mathbf{No}^{M_S}$ is not. This is (d).

Step 6. Real closedness and elementarity are proposition 3.1 for $M_S \subseteq N$, and lemma 3.3 excludes set-sized cofinal subsets. $\square$

The theorem holds for *every* character, not only the least, and produces the examples in one fixed universe, where a class-field isomorphism question between them is meaningful. The same-spectrum conclusion rests on theorem 28.3; it does not follow from the mere fact that a countable forcing adds no fresh function over its immediate ground at uncountable cofinality, because the old base has changed from M to M_S and that change must be controlled. Nor is the distinction between the fields a difference of names of embedded reals: a field isomorphism fixes every rational and preserves positivity, hence every rational cut, so the fields are abstractly nonisomorphic, not merely unequal subclasses of $\mathbf{No}^N$ (source 09).

Corollary 28.6 (An immediate obstruction to completeness; source 09). *Whenever the hypotheses of theorem 28.5 are realized, equality of the entire set-presented gap spectrum is not a complete invariant for old surreal real closed fields in one universe; the failure has continuum many pairwise nonisomorphic witnesses with exactly the same invariant.*

28.3 Two exact spectra: Prikry and Namba

Let M have a measurable cardinal ρ with a normal measure $\mathcal{U}$, and let $W = M[G]$ for the usual Prikry forcing $\mathbb{P}_{\mathcal{U}}$ (finite increasing stems and measure-one reservoirs). The standard Prikry properties give no new bounded subsets of ρ , preservation of cardinals, and $\text{cf}^W \rho = \omega$; so no reals are added and a countable ordinal sequence is.

Theorem 28.7 (Published Prikry spectrum; imported). $\text{FRESH}(\mathbb{P}_{\mathcal{U}}) = \{\omega, \rho\}$; both members occur in every Prikry generic extension, and no others occur [14, Theorem 7.21(1)].

The lower witnesses are the Prikry sequence on ω and its range, a fresh subset of ρ all of whose proper initial segments are finite; the upper exclusions are part of the cited theorem, and no GCH is required (source 09). Source 11 recalls the same facts: the actual spectrum bound follows from the cited theorem, and both entries occur in each generic extension by the displayed witnesses, so nothing is inferred from a possible-generic union.

Corollary 28.8 (Prikry’s exact surreal gap spectrum; source 09, source 13). *For $W = M[G]$ as above, $\text{Char}_W(\mathbf{No}^M) = \{\omega\}$. After adjoining countably many Cohen reals over W , the final universe contains continuum many pairwise nonisomorphic old surreal fields, all with $\text{GapSpec} = \{(\omega, \omega)\}$; one of them is ω -saturated as a real closed field, and those indexed by proper coordinate subsets are not.*

Proof. Both ground regulars of theorem 28.7 have cofinality ω in W ; apply theorem 22.7 and then theorem 28.5. $\square$

This refines corollary 13.4, which exhibited (ω, ω) gaps: every gap is of that character. Reporting a (ρ, ρ) gap would use the wrong universe’s cofinality. A measurable cardinal supplies a particularly small common spectrum, but the failure of classification itself needs no large cardinal (source 09).

Let $M \models \text{CH}$ and let $\mathbb{P}_{\text{Nb}}$ be Namba forcing in the form of [14, Section 7.6]: trees on finite sequences from ω_2 in which every node extends to a node with ω_2 immediate successors, ordered by inclusion; let $W = M[G]$. Under CH it adds no reals but adds a countable sequence cofinal in ω_2^M .

Theorem 28.9 (Published Namba spectrum; imported). *Under CH, $\text{FRESH}(\mathbb{P}_{\text{Nb}}) = \{\omega, \omega_1^M, \omega_2^M\}$; all three lower witnesses occur in each generic extension, the upper exclusion holds for all generics, ω_1 is preserved and $\text{cf}^W(\omega_2^M) = \omega$ [14, Theorem 7.23(1)].*

The no-new-reals property and cofinality facts are recalled in the same section of [14]; CH is essential to the no-new-reals assertion. No claim is made that the forcing preserves all cardinals or all other cofinalities; ω_2^M is an ordinal label, not the second uncountable cardinal of W (source 09).

Corollary 28.10 (Namba’s exact surreal gap spectrum; source 09, source 13). *For this Namba extension $\text{Char}_W(\mathbf{No}^M) = \{\omega, \omega_1^W\}$. After adjoining countably many Cohen reals over W , the final universe contains continuum many pairwise nonisomorphic old surreal fields inside $\mathbf{No}^N$, all with $\text{GapSpec}_N = \{(\omega, \omega), (\omega_1^N, \omega_1^N)\}$. This needs CH in the ground but no large-cardinal hypothesis.*

Proof. The outer cofinalities of the three ground regulars are ω , ω_1^W and ω . Apply the transfer; the hypotheses of theorem 28.5 hold, and the final countable Cohen forcing preserves ω_1^W . $\square$

Remark 28.11 (A corrected overstatement of source 09). Source 09 concludes here that “the unconditional implication proposed as a possible classification principle in the repository” cannot be a theorem. This report proposed no such principle: question 19.4 only asks under what hypotheses equality of gap spectra implies

isomorphism, and section 9 states that equality of gap spectra is not proved to classify the fields. Read correctly, the corollary shows that the unconditional implication about which that question asks fails for real closed fields, even over the usual CH-based forcing setting. It is a relative construction in the ordinary sense of forcing, not a proof of the consistency of ZFC.

Extension (sources)	Ground-regular fresh spectrum	Outer characters Char	Hypothesis
One Cohen subset of regular κ (13; theorem 23.6)	$[\kappa, (2^{<\kappa})^M] \cap \text{Reg}^M$ [14, Proposition 6.1]	$\{\kappa\}$	none
Easton Cohen product (09)	$\text{Ecl}^M(A)$	$\text{Ecl}^M(A)$	GCH in M
Prikry at ρ (09, 13)	$\{\omega, \rho\}$	$\{\omega\}$	normal measure on ρ
Namba (09, 13)	$\{\omega, \omega_1^M, \omega_2^M\}$	$\{\omega, \omega_1^W\}$	CH in M
Cohen-amplified old bases M_S (09)	not asserted equal as ground-regular sets at countable outer cofinality	exactly $\text{Char}_W(\mathbf{No}^M)$	no new reals in W , a new countable ordinal sequence

For the Prikry and Namba rows, $\text{cf}^N \rho = \omega$, $\text{cf}^N \omega_2^M = \omega$ and $\omega_1^M = \omega_1^N$; the forcing facts are standard [10] and the fresh-domain lists are credited to [14]. The Cohen row is proved independently of its middle entry (theorem 23.6). The table shows why the transfer must use the map $\lambda \mapsto \text{cf}^N \lambda$, not merely intersect a ground spectrum with the new cardinals (source 13). The last row is intentionally not strengthened to equality of the *ground-regular* fresh spectra: the proof controls their images under outer cofinality, with a separate argument at ω , which is exactly what surreal gap spectra need (source 09). Countable-character gaps do not by themselves say that a new real appeared: Prikry adds no reals, nor does Namba under CH; the gap sees a new arrangement of old parameters, not necessarily a new binary sequence on ω (source 13).

29 A Cohen–Prikry tower

This section is source 11's. It uses two standard inputs: small forcing preserves a measurable cardinal [23], and for Prikry forcing at a normal measure on a measurable ρ no bounded subsets of ρ are added, $\text{cf}(\rho)$ becomes ω , the actual ground-regular spectrum is $\{\omega, \rho\}$ (theorem 28.7), and cardinals and all other regular cofinalities are preserved. These are imported.

Lemma 29.1 (An elementary tower bound; source 11). *Let $M_0 \subseteq M_1 \subseteq N$ have the same ordinals, with $M_0 \subseteq M_1$ preserving regular cardinals. If $f : \lambda \rightarrow \text{Ord}$ is fresh over M_0 and λ is regular in M_0 , then $\lambda \in \mathfrak{F}(M_0, M_1)$ or $\lambda \in \mathfrak{F}(M_1, N)$. Consequently $\mathfrak{F}(M_0, N) \subseteq \mathfrak{F}(M_0, M_1) \cup \mathfrak{F}(M_1, N)$.*

Proof. If $f \in M_1$, its proper prefixes are in M_0 and it witnesses the first alternative; otherwise the same prefixes lie in M_1 and f is fresh over M_1 . Regularity preservation lets λ index both spectra. $\square$

This is an upper inclusion only: freshness over the intermediate model does not imply that all proper prefixes were already in the earlier model.

Theorem 29.2 (Failure of completeness of the full gap spectrum; source 11). *Suppose $V \models \text{ZFC}$ has a measurable cardinal ρ . There is a set-forcing tower $V \subset V[y] \subset N$ with*

$$\text{Char}_N(\mathbf{No}^V) = \text{Char}_N(\mathbf{No}^{V[y]}) = \{\omega\},$$

but $\mathbf{No}^V$ and $\mathbf{No}^{V[y]}$ are not isomorphic as ordered fields in N : only the second is ω -saturated in N .

Proof. Add one Cohen real y . The forcing is countable, so ρ stays measurable in $V[y]$; choose a normal measure there and force with *its* Prikry forcing (genuinely the one of $V[y]$, not an unchanged ground poset), obtaining N . For $V[y] \subseteq N$ the imported spectrum and theorem 22.7 give $\text{Char}_N(\mathbf{No}^{V[y]}) = \{\text{cf}^N \omega, \text{cf}^N \rho\} = \{\omega\}$. For the whole tower: countable Cohen forcing has no fresh functions at uncountable ground regulars (theorem 23.2) and adds a fresh ω -sequence, so $\mathfrak{F}(V, V[y]) = \{\omega\}$; the first step preserves cofinalities, so lemma 29.1 gives $\mathfrak{F}(V, N) \subseteq \{\omega, \rho\}$. Both indices have cofinality ω in N , and y stays new over V with finite prefixes in V , so $\omega \in \mathfrak{F}(V, N)$ and $\text{Char}_N(\mathbf{No}^V) = \{\omega\}$. Prikry forcing adds no reals, so $\mathbb{R}^N = \mathbb{R}^{V[y]} \neq \mathbb{R}^V$; by theorem 7.1, $\mathbf{No}^{V[y]}$ is ω -saturated in N and $\mathbf{No}^V$ is not. Saturation is invariant under a class isomorphism with Replacement for set images. $\square$

Corollary 29.3 (A strict hierarchy of information; source 11). *The full gap spectrum does not determine the ordered-field isomorphism type of an old surreal field: after all its characters are known, the finite-parameter saturation boundary can supply independent information. Conversely, fixing every saturation cardinal does not determine the full gap spectrum (theorem 25.26).*

There is no contradiction between the two directions. At uncountable cardinals the gap criterion controls saturation, but at ω types may contain infinitely many formulas over a *finite* parameter set, and their real codes are information that the cardinality of a full gap forgets (source 11).

Proposition 29.4 (The simplest missing separators have different birthdays; source 11). *In this tower $\beta(V, N) = \omega$ and $\beta(V[y], N) = \rho$, while both full gap spectra are $\{\omega\}$.*

Proof. No finite sign is new, and y is a new sign of length ω over V . Over $V[y]$ no sign of length below ρ is added, since such a sign codes a bounded subset of ρ ; the Prikry generic subset of ρ is new with every bounded prefix old, a fresh sign of birthday ρ and external cofinality ω . $\square$

The full gap spectrum remembers the external cofinality of a root but loses its birthday. The birthday profile is richer in the ambient sign tree, but its invariance under pure ordered-field isomorphism would need a separate argument, so it is not used as an invariant; the nonisomorphism follows from saturation, which is intrinsic (source 11).

29.1 Three constructions of equal spectra without isomorphism (merge comparison)

	Source 12 (theorem 27.4)	Source 09 (theorem 28.5)	Source 11 (theorem 29.2)
Hypotheses	ZFC; Cohen forcing only	a no-new-reals extension adding a countable sequence; instances need a measurable (Prikry) or CH (Namba)	a measurable cardinal
Common spectrum	$\{(\omega, \omega)\}$	any $\text{Char}_W(\mathbf{No}^M)$; $\{(\omega, \omega)\}$ or $\{(\omega, \omega), (\omega_1, \omega_1)\}$	$\{(\omega, \omega)\}$
Number of fields	$ \mathcal{P}(I)^V $ for any ground set I	$2^{\aleph_0}$	two
Saturation profiles	all equal (no field is ω -saturated)	$\mathbf{No}^M$ is ω -saturated, the others not	one ω -saturated, one not

	Source 12 (theorem 27.4)	Source 09 (theorem 28.5)	Source 11 (theorem 29.2)
Distinguishing invariant	real trace; embeddability is exactly inclusion	real trace (rational cuts)	ω -saturation
Equal reals?	no	no	no

In all three the fields have different real traces, so none decides whether equal real traces together with equal full spectra imply isomorphism (question 31.2), and all three distinguish fields, not pure orders (question 31.4). Source 11's example has equal gaps but unequal saturation; source 12's has equal gaps *and* equal saturation at every cardinal, which answers source 11's question to that effect (question 31.1). Source 09's Theorem 7.1(d) and source 12's Theorem 1.1(iv) are consistent: they concern different constructions, and source 12's reserved coordinate is what prevents an ω -saturated member.

30 Surcomplex real forms from families of old fields

Throughout this section assume (GC), except in theorem 30.11. Every family of class maps below is one class relation whose sections are indexed by a set; no set whose members are proper classes, and no choice from a collection of classes of possible isomorphisms, is used. Each theorem is an instance of the transport mechanism of theorem 16.2, corollary 16.5, and lemma 16.1: an isomorphism $\Phi_A : SC^{M_A} \rightarrow SC^N$ fixing a common set-sized subfield turns c_{M_A} into an involution $j_A = \Phi_A c_{M_A} \Phi_A^{-1}$ of SC^N with $\text{Fix}(j_A) = \Phi_A(\mathbf{No}^{M_A})$, and two such involutions are conjugate exactly when their fixed fields are isomorphic. The proofs below print what each instance adds. A field isomorphism of real closed fields preserves the order, and as a class map with Replacement it preserves set presentations, their sizes, and the absence of set-sized cofinal subsets.

Remark 30.1 (The class back-and-forth, six times; merge note). All six sources reprove theorem 15.1 (source 09 Lemma 9.1, source 10 Lemma 9.1, source 11 Lemma 11.1, source 12 Theorem 9.2 with Lemma 9.1, source 13 Lemma 8.4, source 14 Lemma 12.2), by set-valued recursion along Ord with class parameters and a set-like global well-order: at each stage an isomorphism between set-sized (algebraically closed) subfields, extended at a forth step by a root of a transported minimal polynomial or by an element transcendental over the current range, which exists because the algebraic closure of a set-sized subfield is a set (lemma 11.1; source 12 Lemma 9.1) while the field is a proper class; symmetric back steps; unions at limits, which stay algebraically closed because a polynomial's coefficients occur at an earlier stage. They add these points. Source 10 fixes a common set-sized algebraically closed subfield E pointwise, and notes that a set-sized conjugation-stable $D \subseteq SC^V$ can be enlarged, inside the common algebraically closed class SC^V , to a set-sized conjugation-stable algebraically closed field containing $D \cup \{i\}$, since conjugation carries roots of a polynomial to roots of the conjugate polynomial. Source 11 uses no preselected class transcendence basis and orders elements first by ambient rank. Source 13 extends any prescribed isomorphism of set-sized subfields and shows that every element is eventually included, since the process cannot keep choosing distinct predecessors past the Hartogs ordinal of the element's set of predecessors; its remark stresses that the lemma builds in no order preservation and does not make old and new real closed surreal fields isomorphic. Source 14 assumes that algebraic closures of set-generated subfields are sets and, for a set-indexed family, retains at each stage the set of stage maps for all indices. Source 12 proves that any two proper-class algebraically closed fields of characteristic zero are isomorphic, and records that the theorem fails for “real closed”: the analogous real-closure argument still gives ample transcendence, but a real transcendental has a cut type that cannot be matched arbitrarily, and its whole main construction exploits this difference. No source uses a class Zorn lemma, a class-valued recursion, or an unrestricted class truth predicate.

Proposition 30.2 (Forgetting the involution loses the obstruction; source 12, source 11). *In the setting of theorem 27.4, the pure class field SC^N realizes every complete type in finitely many variables over every*

N-set of parameters, whereas every (SC^N, j_A) of theorem 30.8 omits a complete type over the empty set and is not ω -saturated.

Proof. The first assertion is theorem 11.2 (source 11 reproves it for every proper-class algebraically closed field, as its Proposition 11.4). For the expansion, impose $j_A(X) = X$ and translate p_{r_*} into the fixed field, positive fixed elements being exactly the nonzero squares of fixed elements; the parameter-free type is finitely satisfiable and omitted, and one extends it to a complete type or transports the complete fixed-field cut type through the two-coordinate interpretation of lemma 11.3. $\square$

The obstruction is not the algebraically closed carrier but the real structure selected by the involution; the same class field looks maximally saturated in the pure language while naming one involution exposes specific omitted cuts, and nonconjugacy concerns the added structure, not different isomorphism types of the pure carrier (sources 11, 12).

30.1 Same spectrum, agreement on old constants (source 09)

Theorem 30.3 (Same-spectrum nonconjugate real forms; source 09). *In the setting of theorem 28.5, under Global Choice, there are involutions j_S , $S \in \mathcal{P}(\omega)^M$, of the single field SC^N such that (i) $\text{Fix}(j_S) \cong \mathbf{No}^{M_S}$; (ii) all fixed fields have the same full gap spectrum and none has a set-sized cofinal subset; (iii) j_S and j_T are not conjugate by any class field automorphism when $S \neq T$; (iv) all j_S restrict to ordinary conjugation on the common old complex field $\mathbb{C}^M$. For the Prikry instance the common spectrum is $\{(\omega, \omega)\}$; for the Namba instance it is $\{(\omega, \omega), (\omega_1, \omega_1)\}$.*

Proof. Each SC^{M_S} is algebraically closed and a proper class; extend the identity on the common set-sized subfield $\mathbb{C}^M$ to $\Phi_S : \text{SC}^{M_S} \rightarrow \text{SC}^N$ and put $j_S = \Phi_S c_{M_S} \Phi_S^{-1}$. Then (i), (ii) follow from theorem 28.5(a),(e); a conjugacy would give $\mathbf{No}^{M_S} \cong \mathbf{No}^{M_T}$, contrary to theorem 28.5(b); Φ_S fixes $\mathbb{C}^M$ and c_{M_S} acts there as usual. The spectra are corollaries 28.8 and 28.10. $\square$

A uniform use of the global well-order makes the construction one class relation indexed by the set $\mathcal{P}(\omega)^M$ (source 09).

Proposition 30.4 (Obstruction to agreement on all final constants; source 09). *If $\mathbb{R}^{M_S} \neq \mathbb{R}^N$, no real form isomorphic to $\mathbf{No}^{M_S}$ contains $\mathbb{R}^N$; so no involution whose fixed field is isomorphic to $\mathbf{No}^{M_S}$ agrees with standard conjugation on all of $\mathbb{C}^N$.*

Proof. An inclusion of $\mathbb{R}^N$ into such a real form, followed by an isomorphism to $\mathbf{No}^{M_S}$, is an ordered-field embedding fixing $\mathbb{Q}$, so each real's image realizes that real's rational cut, and theorem 27.3 would put every real of N into $\mathbb{R}^{M_S}$. Agreement with c_N on $\mathbb{C}^N$ fixes $\mathbb{R}^N$ pointwise. $\square$

The common core of theorem 30.3 is $\mathbb{C}^M$, not $\mathbb{C}^N$, and this boundary is necessary for this construction, not an omitted choice of isomorphism: the family answers an infinite-family problem with a prescribed *old* core, not the all-final-constants version that question 19.7 asks for (source 09). The families of theorems 30.5, 30.7, 30.9 and 30.12 do agree on $\mathbb{C}^N$, because their grounds share the reals of N ; they have different spectra.

30.2 Agreement on any set-sized core (source 10)

Theorem 30.5 (Many equally saturated nonconjugate involutions; source 10). *In a (GC) version of the extension of theorem 25.21, let $D \subseteq \text{SC}^V$ be any prescribed set-sized subfield stable under ordinary conjugation. There is a uniform family $(j_S : S \subseteq \chi)$ of involutions of SC^N such that:*

- (i) j_S agrees with c_N on D ; in particular all can agree on every element of $\mathbb{C}^N$;
- (ii) $\text{Fix}(j_S) \cong \mathbf{No}^{M_{K_S}}$, with isolated characters as in theorem 25.21(i), least character κ_* and no set-sized cofinal subset;
- (iii) the j_S are pairwise nonconjugate, and none is conjugate to c_N ;
- (iv) all fixed fields, and all structures (SC^N, j_S) in the ring language with the involution named, have the same saturation spectrum: τ -saturation exactly for infinite $\tau \leq \kappa_*$;
- (v) all (SC^N, j_S) have the same first-order theory, and the maps can be chosen so that they have the same theory over D , each element of D named by a constant.

Proof. Choose a common set-sized conjugation-stable algebraically closed $E \subseteq \text{SC}^V$ containing $D \cup \{i\}$ (remark 30.1), and $\Phi_S : \text{SC}^{M_{K_S}} \rightarrow \text{SC}^N$ with $\Phi_S \upharpoonright E = \text{id}_E$, uniformly in the set parameter S . For $d \in D$, d and $c_N(d)$ lie in E , so $j_S(d) = \Phi_S(c_{M_{K_S}}(d)) = c_N(d)$; no reals were added, so $\mathbb{C}^N = \mathbb{C}^V$ is an allowed core. (ii) and the first part of (iv) transfer along Φ_S . A conjugacy $j_S \sim j_T$ would give $\mathbf{No}^{M_{K_S}} \cong \mathbf{No}^{M_{K_T}}$, contrary to theorem 25.21; c_N has fixed field $\mathbf{No}^N$, which fills every set-sized cut, while every displayed fixed field has a κ_* -gap. For the expansions, $F(i)$ with conjugation and i named is interpreted in F^2 with $(a, b)(u, v) = (au - bv, av + bu)$ and $c(a, b) = (a, -b)$, and conversely the fixed field and its order are definable (lemma 11.3); these finite interpretations preserve saturation at infinite cardinals, a parameter set giving at most twice as many coordinates, and naming or removing i changes nothing (theorem 11.4). For (v), $E_0 = E \cap \mathbf{No}^V$ is real closed with $E = E_0(i)$: real and imaginary parts of elements of E lie in E , the conjugation's fixed field is E_0 , and positive square roots and real roots of odd-degree polynomials over E_0 , which exist in $\mathbf{No}^V$ and are algebraic over E , lie in E_0 . Each transported fixed field contains E_0 with its original embedding, so quantifier elimination makes them elementarily equivalent over E_0 , and the coordinate translation gives equivalence of the conjugation structures over E , hence over D . $\square$

Elementary equivalence over D is distinct from saturation in a language with arbitrarily many named constants; the saturation claim in (iv) is for the fixed finite language (source 10).

Corollary 30.6 (A simultaneous multicharacter answer; source 10). *The multicharacter and old-ground aspects of question 19.7 have positive simultaneous realizations of arbitrary set-indexed size: 2^χ pairwise nonconjugate real forms share one saturation threshold and agree on all ordinary complex numbers, while their higher gap profiles range over every subset of χ designated coordinates.*

The conjugacies and field isomorphisms are pure-field maps: nothing is asserted about the ambient simplicity order, the natural valuation, Hahn summation, an exponential, an omnific integer part or a derivation, and requiring any of these would define a different and potentially far more rigid classification problem (source 10).

30.3 Agreement on $\mathbb{C}^N$ with one saturation spectrum (source 11)

Theorem 30.7 (Many nonconjugate involutions, one field; source 11). *For the family of theorem 25.26, under Global Choice, there is a uniformly indexed family $(j_\xi)_{\xi < \chi}$ of involutive automorphisms of SC^N with $\text{Fix}(j_\xi) \cong \mathbf{No}^{M_\xi}$ as ordered fields, j_ξ ordinary conjugation on $\mathbb{C}^N$, j_ξ and j_η not conjugate for $\xi \neq \eta$, and all (SC^N, j_ξ) elementarily equivalent with the same saturation spectrum.*

Proof. $\mathbb{C}^N$ lies in every SC^{M_ξ} , all reals being shared; take Φ_ξ fixing it. A conjugacy maps fixed fields isomorphically, and positivity is field-definable by $x > 0 \iff x \neq 0 \wedge \exists y \, x = y^2$, so it would identify $\mathbf{No}^{M_\xi}$

with $\mathbf{No}^{M_\eta}$ contrary to theorem 25.26. With i named, every z is uniquely $x + iy$ with $x = (z + j_\xi(z))/2$ and $y = (z - j_\xi(z))/(2i)$ in $\text{Fix}(j_\xi)$, a finite-coordinate bi-interpretation with real closed fields; completeness of RCF gives one theory, and type translation gives saturation at each infinite cardinal; then forget i . $\square$

These are field isomorphisms: the proof does not make them preserve the natural valuation, birthdays, simplicity, Conway normal forms, exponentiation, a derivation or the omnific integer ring, so the involutions are real forms of the pure surcomplex field, not classified symmetries of an enriched structure (source 11).

30.4 Conjugacy and equivariant embeddings in the Boolean family (source 12)

In the setting of theorem 27.4, for each A choose, uniformly using the global choices, $\Phi_A : \mathbf{No}^{M_A}(i) \rightarrow \text{SC}^N$ and put $j_A = \Phi_A \circ \Phi_A^{-1}$; it has order exactly two and $\text{Fix}(j_A) = \Phi_A(\mathbf{No}^{M_A})$ is real closed.

Theorem 30.8 (Boolean patterns of surcomplex involutions; source 12). *(i) Every fixed field has full gap spectrum $\{(\omega, \omega)\}$ with its unique real-closed-field order. (ii) j_A and $j_{A'}$ are conjugate in $\text{Aut}(\text{SC}^N)$ exactly when $A = A'$. (iii) There is a unital field embedding $h : \text{SC}^N \hookrightarrow \text{SC}^N$ with $h \circ j_A = j_{A'} \circ h$ exactly when $A \subseteq A'$. (iv) All (SC^N, j_A) are elementarily equivalent.*

Proof. (i) Isomorphisms of real closed fields preserve order and gap characters. (ii) A conjugacy restricts to an isomorphism of fixed fields, hence $\mathbf{No}^{M_A} \cong \mathbf{No}^{M_{A'}}$ and $A = A'$. (iii) An intertwining embedding restricts to a field embedding of fixed fields, forcing $A \subseteq A'$; conversely, when $A \subseteq A'$ the inclusion $\iota : \mathbf{No}^{M_A}(i) \hookrightarrow \mathbf{No}^{M_{A'}}(i)$ commutes with conjugation and $h = \Phi_{A'} \circ \iota \circ \Phi_A^{-1}$ works. (iv) $u + iv \mapsto (u, v)$ represents $F(i)$ with conjugation in two coordinates, so every sentence translates into one about a real closed field; choosing i or $-i$ changes coordinates but not truth. $\square$

An equivariant embedding need not be onto, which is why a Boolean *embedding* order appears although conjugacy is equality of indices. In the one-Cohen-real case there are continuum many nonconjugate involutions with one fixed-field spectrum; for arbitrary I the number of indexed conjugacy classes is $|\mathcal{P}(I)^V|^N = (2^{|I|})^V$, since the forcing preserves cardinals. All statements concern set-indexed families of class maps (source 12). No preservation of the canonical conjugation, a valuation or real constants is asserted for the maps Φ_A ; the real-field counterexamples of section 27 do not need this transport.

30.5 A continuum of equal-threshold real forms (source 13)

Theorem 30.9 (Simultaneous involutions of the coordinate family; source 13). *In (25.1), under Global Choice in N , there are involutions j_A , $A \subseteq \omega$, of SC^N with $\text{Fix}(j_A) \cong \mathbf{No}^{M_A}$, $j_A \upharpoonright \mathbb{C}^N$ ordinary conjugation, $\text{Char}_N(\text{Fix}(j_A))$ given by theorem 25.15, and $j_A, j_{A'}$ conjugate by an ambient field automorphism if and only if $A = A'$.*

Proof. All intermediate grounds share the reals of N (lemma 25.13), so each SC^{M_A} contains $\mathbb{C}^N = \mathbb{R}^N[i]$; take Φ_A fixing it, uniformly. A conjugacy would equate spectra and proposition 25.17 gives $A = A'$. Lemma 3.3 excludes set-sized cofinal subsets, a property that transfers. $\square$

Corollary 30.10 (A continuum of equal-threshold real forms; source 13). *The subfamily with $0 \notin A$ has cardinality $2^{\aleph_0}$ in N ; its members are pairwise nonconjugate, agree on $\mathbb{C}^N$, and their fixed fields have saturation threshold κ_0 , are ω -saturated and lack set-sized cofinal subsets, while their full gap spectra differ. One may restrict further to infinite omitted sets without reducing the size. In this GCH construction the continuum is $\aleph_1$.*

Proof. Fixing the bit at 0 does not change the number of subsets; distinct parameters give distinct spectra; apply proposition 25.18 and theorem 30.9. Removing the countably many finite sets leaves continuum many. $\square$

For $\kappa_n = \aleph_{n+1}$ these fixed fields are all $\aleph_1$ -saturated and none is $\aleph_2$ -saturated; on the infinite-omitted-set subfamily they all also have an $\aleph_{\omega+1}$ gap, while the intervening characters encode an arbitrary subset of ω with its first bit fixed, which makes precise the information lost by recording only the least omitted type size (source 13). All κ_n are uncountable in this construction; otherwise the common complex-field clause would need separate analysis (source 13). The isomorphisms Φ_A are noncanonical: they preserve field operations and the named ordinary complex field, but no ambient simplicity, monomial map, natural valuation, summability, exponentiation or derivation, and a real form of an algebraically closed field is not an embedding theorem in any richer language (source 13).

30.6 Real forms of one set field, and the class version (source 14)

Theorem 30.11 (Continuum many set-sized real structures, one saturation spectrum; source 14; ZFC). *In the model N of theorem 26.2 there are one algebraically closed field $\Omega = (\mathbf{No}_{<\theta})^N[i]$ of cardinality $\theta = \aleph_{\omega+2}$, containing $\mathbb{C}^N$, and involutions j_A , $0 \notin A \subseteq \omega$, such that: (i) $\text{Fix}(j_A)$ is isomorphic over $\mathbb{R}^N$ to $(\mathbf{No}_{<\theta})^{M_A}$; (ii) all j_A restrict to ordinary conjugation on $\mathbb{C}^N$; (iii) no two are conjugate by an automorphism of Ω ; (iv) all (Ω, j_A) are elementarily equivalent with identical saturation spectra, although their fixed fields have different complete gap-pair spectra. No class choice principle is needed.*

Proof. Each $(\mathbf{No}_{<\theta})^{M_A}[i]$ is algebraically closed of characteristic zero and cardinality θ . The common algebraically closed subfield $\mathbb{C}^N$ has size $\aleph_1 < \theta$, and an algebraic closure of a field generated by $\mathbb{C}^N$ and τ transcendentals has size at most $\max(\aleph_1, \tau)$; so both fields have transcendence degree θ over $\mathbb{C}^N$, and a bijection of transcendence bases extends to $\Phi_A : (\mathbf{No}_{<\theta})^{M_A}[i] \rightarrow \Omega$ fixing $\mathbb{C}^N$. There are set many A and the isomorphisms are sets, so ordinary Choice picks them. A conjugacy would map fixed fields isomorphically, which theorem 26.2 forbids. $(E[i], c)$ is interpretable in E on pairs, E is definable in it by $c(x) = x$ and positivity by nonzero squares, so all expansions have one theory; naming i and translating types preserves the saturation properties, which are then those of corollary 26.3. $\square$

Theorem 30.12 (Full-class real forms with a common threshold; source 14). *Assume (GC) in the product extension of theorem 25.15 with $\kappa_n = \aleph_{n+1}$, and the ground-class predicates of (H). There are $2^{\aleph_0}$ pairwise nonconjugate involutions J_A of SC^N , indexed by $0 \notin A \subseteq \omega$, such that $\text{Fix}(J_A) \cong \mathbf{No}^{M_A}$ with the spectrum of theorem 25.15; J_A agrees with standard conjugation on $\mathbb{C}^N$; all fixed fields are ω_1 -saturated but not ω_2 -saturated, with the same entire saturation spectrum; none has a set-sized cofinal or coinital subset; and the expansions (SC^N, J_A) are elementarily equivalent but pairwise nonisomorphic as class structures.*

Proof. Apply the class back-and-forth to $(\text{SC}^{M_A}, \text{SC}^N)$ over $\mathbb{C}^N$: the fields are proper classes, containing every old ordinal, algebraic closures of set-generated subfields are sets, and real closedness makes each complexification algebraically closed. The properties transfer from theorem 25.15, proposition 25.18, corollary 25.19, and lemma 3.3; conjugacy would give a set-respecting isomorphism of fixed fields, excluded by proposition 25.17; first-order equivalence is the interpretation argument of theorem 30.11. $\square$

The class version needs an actual class construction, not a class-sized transcendence basis selected by ordinary Choice; the set-sized theorem isolates the spectrum result from that foundational issue (source 14). Unstructured does not mean compatible with every expansion: the isomorphisms need not preserve the natural valuation, Conway normal forms, simplicity, the surreal exponential, a derivation or omnific integers; the

involutions are field automorphisms, not strong Hahn automorphisms or automorphisms of a differential or exponential field, and their agreement on $\mathbb{C}^N$ is the one compatibility condition proved (source 14).

30.7 Relation to Part V and to the collection

Theorem 16.6 and corollary 16.7 separated finitely many real forms by distinct *first* thresholds $\delta(M_a, N)$. The families above hold the threshold fixed and separate by higher characters (sources 10, 11, 13, 14), or hold the *whole spectrum* fixed and separate by the real trace (sources 09, 12). The report docs/surcomplex/first-kappa-coefficients already gives, in one universe and without forcing, families of real forms with individually prescribed singleton thresholds (fkc: sb: thm: involutions); the families here are of actual old fields with multicharacter spectra and a common threshold (sources 13, 14). None of these theorems classifies real forms or involutions; they continue, but do not settle, the automorphism report's open real-form classification (remark 16.4), and none addresses question 19.6.

31 Questions of sources 09–14, and which of them the cluster answered

The six sources pose 58 questions: 9 from source 09 (its Questions 11.1–11.9), 12 from source 10 (Questions 1–12), 10 from source 11 (Research questions 14.1–14.10), 9 from source 12 (Questions 12.1–12.9), 8 from source 13 (Questions 10.1–10.8) and 10 from source 14 (Questions 14.1–14.10). Questions asking the same thing are merged below, with every asking source named; those answered by a sibling source are recorded as answered. They are directions suggested by the results, not claims that each has been catalogued as an open problem in the published literature.

Question 31.1 (Equal full spectra, nonisomorphic fields; sources 10, 11, 13, 14; answered for fields). Construct, or rule out under precise hypotheses, two old surreal fields in one universe with identical full set-presented gap spectra that are not isomorphic as ordered fields (source 10 Question 2, source 13 Question 10.5, source 14 Question 14.4, first clause); can this happen with equal gap *and* equal saturation spectra (source 11 Question 14.4); and without a measurable cardinal (source 11 Question 14.3)? Which invariants of the arrangement or multiplicity of cuts detect the difference (sources 10, 13)?

Status: answered for fields. Theorem 27.4 (source 12) gives such fields in ZFC with Cohen forcing only, with equal saturation at every infinite cardinal, and theorem 28.5 (source 09) gives continuum many, without large cardinals under CH (corollary 28.10); theorem 29.2 (source 11) gives a pair with unequal ω -saturation from a measurable. Source 09's Prikry family also answers the equal-saturation clause for distinct proper index sets, none of which is ω - or ω_1 -saturated. The detecting invariants found are the real trace (sources 12, 09) and ω -saturation (source 11); coarse multiplicities cannot help (proposition 26.4). Incidence of set-indexed cuts, their joint realizability in algebraic extensions, algebraic relations among parameters defining gaps and invariants of finitely generated subfields remain candidates (sources 13, 14). The pure-order clause is question 31.4.

Question 31.2 (Equal real traces and equal full spectra; sources 09, 12; open). Suppose $M_0, M_1 \subseteq N$ have the same ordinals, $\mathbb{R}^{M_0} = \mathbb{R}^{M_1}$ and $\text{GapSpec}_N(\mathbf{No}^{M_0}) = \text{GapSpec}_N(\mathbf{No}^{M_1})$. Under which additional hypotheses are the old fields isomorphic? Is there a same-universe counterexample with equal real sets, and one using only set forcing with no large cardinal (source 09 Question 11.1, source 12 Question 12.2)?

All three constructions of section 29's comparison vary the trace and do not decide this. Cuts over a common non-Archimedean countable subfield, where real agreement no longer controls parameterized types, are a promising next step (source 09). Forcing that adds no reals is a natural testing ground, but adding

different subsets of ω_1 is not by itself a proof of field nonisomorphism: an arbitrary field map can move the chosen infinite scale (source 12).

Question 31.3 (Parameterized cut profiles and a trace hierarchy; sources 09, 12; open). (a) For every finitely generated ordered subfield F of an old field $\mathbf{No}^M$, record which cuts of $\text{rc}(F)$ are realized in $\mathbf{No}^M$, compatibly with embeddings between such subfields; with the gap spectrum, does this profile admit a usable classification of old surreal fields (source 09 Question 11.2)? (b) Can residues of canonically specified convex valuations, or realizable cut types over finitely generated subfields, give an intrinsic hierarchy beyond $\text{Tr}_{\mathbb{R}}(F)$; which levels are invariant under all field embeddings, and which depend on naming a noncanonical scale (source 12 Question 12.3)?

The trace supplies only the $F = \mathbb{Q}$ part of the profile; a serious formulation must account for compatibility between parameter systems, not merely collect cardinalities (source 09). An adequate definition should not name ω merely because one presentation has it: the Boolean theorem succeeds because its rational reference scale is fixed by every unital embedding (source 12).

Question 31.4 (The pure-order version; sources 09, 12, 14; open). Can two old surreal fields with equal full gap spectra have nonisomorphic pure order reducts in one universe, and which arrangement or multiplicity invariant of fresh cones would detect it (source 09 Question 11.3, source 14 Question 14.4, second clause)? For the Boolean Cohen family, which fields are isomorphic as pure linear orders; can the whole family collapse to one pure-order type although its field embeddability order is Boolean (source 12 Question 12.1)?

Field arithmetic anchors the rational cuts in every construction above; an arbitrary order isomorphism has no obligation to preserve them, so a solution needs an order-intrinsic obstruction or an order back-and-forth that does not rely on field saturation. This is the most direct unresolved part of question 19.4 (sources 09, 12).

Question 31.5 (Embeddings rather than isomorphisms; sources 12, 14; open). (a) Outside the Cohen family, characterize the pairs with $\mathbb{R}^{M_0} \subseteq \mathbb{R}^{M_1}$ for which $\mathbf{No}^{M_0} \hookrightarrow \mathbf{No}^{M_1}$, and the extra compatibility of omitted cuts that is needed (source 12 Question 12.4). (b) Can every ground set-sized partial order be represented by embeddability of old fields with one common real trace and one common full spectrum; even a two-element antichain (source 12 Question 12.5)? (c) For the coordinate family of theorem 25.15, what is the exact condition for a class ordered-field embedding $\mathbf{No}^{M_A} \rightarrow \mathbf{No}^{M_{A'}}$; does an initial-image condition or a fixed common Hahn core recover inclusion information (source 14 Question 14.5)?

Theorem 27.3 gives necessity, not a converse; a positive theorem would have to extend embeddings through larger old sign domains without meeting an unmatchable cut, and (b) asks for a new mechanism rather than another arrangement of independent Cohen reals (source 12). In (c) the traces coincide, and an embedding need not preserve omitted gaps as omitted gaps, since its image can acquire separators; so the isomorphism argument does not settle the embedding order (source 14).

Question 31.6 (Small-forcing base change beyond a common dense suborder; source 09; open). Can the cardinality hypothesis of theorem 28.3 be replaced by a chain-condition, approximation or Boolean-algebra invariant while keeping both directions, and what is the sharp failure mechanism at $\text{cf } \gamma = |\mathbb{Q}|$ (source 09 Question 11.4)?

Corollary 28.4 already allows the size of a common dense suborder; the canonical-name proof shows where a stronger theorem would need a replacement for the cofinal pigeonhole step.

Question 31.7 (Equal spectra and agreement on $\mathbb{C}^N$; source 09; open). Can a no-new-reals family of intermediate bases give infinitely many nonconjugate real forms with identical entire gap spectra, all agreeing with standard conjugation on $\mathbb{C}^N$ (source 09 Question 11.5)?

Proposition 30.4 rules out rechoosing the isomorphisms in the Cohen family; new invariants with the same real set are needed. The families of sources 10, 11, 13 and 14 agree on $\mathbb{C}^N$ but have different spectra.

Question 31.8 (Family sizes; sources 11, 14 answered; source 09 open). (a) Can one force 2^χ pairwise nonisomorphic old fields in one universe with a fixed uncountable saturation threshold, all containing the ambient reals (source 11 Question 14.2)? (b) For fixed regular κ and set cardinal χ' , when are there χ' pairwise nonisomorphic old real closed fields in one universe with first gap κ , the same real field and explicitly different higher spectra (source 14 Question 14.3)? (c) Can arbitrarily large set-indexed families of pairwise nonisomorphic old fields share one fixed *full* spectrum in one universe while retaining a common prescribed real field (source 09 Question 11.6)?

Status. (a) is answered by theorem 25.21 (source 10): 2^χ fields, threshold κ_* , all containing $\mathbb{R}^V = \mathbb{R}^N$. (b) is answered for every uncountable regular κ and every infinite χ' by theorem 25.26 (source 11), and with 2^χ fields when $\kappa > \chi$ by source 10; the case $\kappa = \omega$ is not covered by these constructions. (c) stays open: a larger Cohen poset changes the range where base change is automatic, and names for reals are no longer coded by one old real, so more coordinates alone do not suffice (source 09).

Question 31.9 (Birthday information; sources 09, 10, 11, 12, 13, 14; open). For each gap character, which birthdays do fresh roots of that cofinality have, which profiles occur when several ground regulars collapse to one external cofinality, and which parts survive an abstract field isomorphism rather than depending on the ambient sign tree (source 09 Question 11.7, source 11 Question 14.5)? Can a pure-field invariant recover the least fresh length or first new birthday, or would naming simplicity or a birthday comparison be needed, and how much (source 10 Question 3, source 13 Question 10.6)? Can two old fields have the same pushed-forward spectrum but different least fresh sign lengths, and is there a useful subclass where lengths are recovered (source 14 Question 14.6)? Which information about fresh-sign *lengths* can be recovered intrinsically (source 12 Question 12.7, second half)?

This continues question 19.5. Proposition 29.4 gives equal spectra with different least new birthdays, but for nonisomorphic fields; a decisive negative answer would need two pairs with different least new birthdays and *isomorphic* old fields, and a positive one needs more than the sign representation, which is not in the field language (source 13). The coincidence of first birthday and κ_* in theorem 25.21 is a feature of that construction, not a reconstruction theorem (source 10). Prikry and Namba give explicit tests (source 09); theorem 23.6 shows a complicated ground spectrum becoming a singleton externally (sources 12, 14).

Question 31.10 (Spectra beyond Easton products; sources 09, 10, 11, 12; open). For a specific forcing with known fresh-function spectrum and changed cofinalities, compute the complete pushforward (22.1), and determine which distinct fresh-function spectra give the same surreal spectrum (source 09 Question 11.8). Which pairs $(\mathfrak{F}(M, N), \lambda \mapsto \text{cf}^N \lambda)$ occur, and can a nontrivial collapse pattern and a multicharacter spectrum be prescribed together while keeping the reals (source 10 Question 5)? Which sets occur as $\text{Char}_N(\mathbf{No}^M)$ for cofinality-preserving set forcing (source 11 Question 14.6)? Given a ground spectrum, which identifications among its members can the pushforward make (source 12 Question 12.7, first half)?

The transfer reduces the surreal question exactly to these realization problems but supplies no general realization theorem; collapse forcings and further Prikry-type forcings are natural candidates, with the generic and the cofinality map specified separately (sources 09, 10, 11).

Question 31.11 (Relative spectra at accumulation points; sources 10, 11, 13, 14; answered for countable sequences). For the fields of theorem 25.21, determine $\text{Char}_N(\mathbf{No}^{M_{\kappa_S}})$ at every regular cardinal: is it $\text{Ecl}^V(\{\kappa_*\} \cup \{\kappa_i : i \in S\})$, or can kept coordinates change the answer (source 10 Question 1)? For infinite omitted sets B in the sparse construction, determine $\text{Char}_N(\mathbf{No}^{M_{A \setminus B}})$ (source 11 Question 14.1). Extend the coordinate calculation to longer Easton products with several regular or singular accumulation points (source 13 Question 10.3), and characterize $\mathfrak{F}(M[G_A], M[G_I])$, $A \subseteq I$, for a general index set I (source 14 Question 14.1).

Status (merge). For index length ω this is answered by theorem 25.15 (sources 13, 14): source 10's family with $\chi = \omega$ is an instance, and its spectra are $\{\kappa_*\} \cup \{\kappa_i : i \in S\} \cup \{\lambda^+ : S \text{ infinite}\}$, which is exactly $\text{Ecl}^V(\{\kappa_*\} \cup \{\kappa_i : i \in S\})$, so the answer to source 10's question is yes for $\chi = \omega$ (example 25.23); likewise infinite omitted sets forming an ω -sequence are covered. Open for uncountable index sets, where heads may be large products, several accumulation points interact, regular-limit and Mahlo phenomena need the chain-condition analysis of the published theory, and the support convention at inaccessible limits must be kept (sources 13, 14). Any proof must treat the quotient as a product of old posets, not a product recomputed in the intermediate model (source 10), and writing $\text{Ecl}(B)$ proves nothing (source 11).

Question 31.12 (Weaker hypotheses than GCH; sources 10, 13, 14; open). Find the weakest cardinal-arithmetic or forcing hypotheses for the simultaneous families (source 10 Question 4); hypotheses under which theorem 25.15 still holds, and its replacement when products collapse cardinals or $\lambda^{\aleph_0} > \lambda^+$ (source 13 Question 10.4); in particular whether local assumptions such as $2^{<\kappa_n} = \kappa_n$ and a true-cofinality hypothesis at λ suffice (source 14 Question 14.2).

The single-coordinate theorem needs no GCH; the obstacles are preservation and localization (small low products, a chain condition for the low part including the test coordinate, a closed high tail), the λ^+ ceiling and the singular-product witness, while transfer and density are unaffected (sources 10, 13, 14). A changed forcing spectrum need not change the gap spectrum, since outer cofinalities can merge its entries (source 13).

Question 31.13 (Approximation beyond initial segments; source 10; open). Absence of fresh functions is an initial-segment approximation condition. Which intrinsic expansions or configurations of surreal cuts detect the stronger approximation properties defined by all small ground subsets, and is there a converse reconstruction (source 10 Question 6)?

Question 31.14 (Accumulation points and large-cardinal structure; source 10; open). Give a direct surreal interpretation of the distinction between non-Mahlo regular accumulation points, Mahlo accumulation points and singular-successor characters in the Easton closure; can the relevant organization of gap families be recognized internally (source 10 Question 7)?

No large cardinal is needed for the Boolean family, whose test coordinates are successors; the question concerns what becomes visible when regular-limit structure is included (source 10).

Question 31.15 (Compatible and enriched structure; sources 10, 11, 12, 13, 14; open). Which of the involutions constructed here can also preserve prescribed valuation data, an omnific integer structure, a surreal exponential or a derivation, perhaps on more than the ordinary complex subfield or on an independently chosen set-sized differential subfield; is there an obstruction already for a small differential or exponential core (source 10 Question 8, source 11 Question 14.8, source 13 Question 10.7, source 14 Question 14.8)? Which parts of the constructions survive naming exponentiation, a derivation, a valuation, a monomial cross-section or an omnific subring (source 13 Question 10.7, source 14 Question 14.7)? What extra assumptions become available for involutions preserving a valuation, integer part or exponential (source 12 Question 12.6, second half)?

This continues questions 19.1 and 19.6. The pure-field back-and-forth uses only algebraic and transcendental one-element extensions and supplies no extension property in these categories; merely transporting the field involution does not transport a preassigned differential or exponential structure. The ordered-field lower bounds remain necessary for saturation of an expansion, but sufficiency used real-closed-field quantifier elimination (sources 10, 11, 13, 14). A tractable first problem is to fix one set-sized expanded substructure and prove the extension lemma it requires (source 13).

Question 31.16 (Class-theoretic strength; sources 10, 11, 12, 13, 14; open). Over a theory weaker than GBC with Global Choice, what is the exact strength of the relative class-field isomorphism principle and of its uniform set-indexed version; can the particular surreal fields here, or the Cohen-coded family, admit the

needed isomorphisms or definable maps without it (source 10 Question 9, source 11 Question 14.7, source 12 Question 12.8, source 13 Question 10.8, first half, source 14 Question 14.9)?

This continues question 19.3. The set-forcing and sign arguments need no Global Choice; the set-valued recursion avoids class truth recursion but not the global choices of enumerations and extensions; a positive answer should give an actual uniform class construction, not infer one from local set-saturation (sources 10, 11, 14). Theorem 30.11 isolates the spectrum result from this issue.

Question 31.17 (Proper-class spectra and class forcing; source 10; open). Under which tameness and preservation assumptions can class forcing produce a prescribed proper-class pattern of gap characters, and can a class-indexed isolated-coordinate argument be carried out while preserving Replacement (source 10 Question 10)?

The forcings here are sets, and theorem 23.2 gives a set bound; a formal union of set-forcing constructions would not establish the class version.

Question 31.18 (Richer inclusion patterns and symmetry; source 10; partly answered). Which partial orders, incidence structures or group actions are realized by inclusions and field automorphisms of such families with one first threshold, and can a refined invariant encode more than isolated membership bits (source 10 Question 11)?

Status. Corollary 27.8 (source 12) realizes every ground set-sized partial order by inclusion, and exactly by field embeddability, among old fields with one full spectrum and threshold ω ; proposition 25.17 (source 13) adds binary intersections for the coordinate family. Group actions, incidence structures and uncountable thresholds remain open, and the inclusion order must still be distinguished from intersections and composita (source 10).

Question 31.19 (Intersections and composita; source 13; open). For $\langle A_m : m < \omega \rangle$, determine $\bigcap_m \mathbf{No}^{M_{A_m}}$ and when it equals $\mathbf{No}^{M_{\bigcap_m A_m}}$, in particular for tails of coordinates (source 13 Question 10.1). Inside $\mathbf{No}^N$, compare the compositum $\mathbf{No}^{M_A} \mathbf{No}^{M_{A'}}$ and its real closure with $\mathbf{No}^{M_{A \cup A'}}$: which elements are algebraic over the compositum, and when is the inclusion an equality (source 13 Question 10.2)?

The binary intersection proof does not iterate through a countable limit, where a reduced product or a common singular-limit quotient may obstruct; a first target is the intersection of the models restricted to sets of ordinals. A finite rational expression in elements of two old fields need not capture every set-theoretic combination of their coordinates; a criterion via names with restricted coordinate dependence is an intermediate goal (source 13).

Question 31.20 (Fixed-field invariants for involutions; source 12; open). Does the pair of the real trace and the full gap spectrum classify any broad, naturally defined family of involutions of SC up to conjugacy (source 12 Question 12.6, first half)?

The involutions here are deliberately unconstrained, and the transport maps preserve no analytic or normal-form data; the questions about enriched automorphisms are separate (question 31.15).

Question 31.21 (Product versus iteration calculus; source 11; open). Develop a relative spectrum calculus distinguishing fixed ground products from iterations whose next factor is recomputed in the intermediate universe (source 11 Question 14.9).

Lemma 29.1 gives one inclusion; a converse needs a criterion for whether intermediate fresh prefixes were already present earlier. Finite towers of this report (corollary 16.7) are iterations; the families of section 25 are products.

Question 31.22 (Formalization; sources 09–14; open). Formalize the fresh-sign correspondence, block codes, transfer and cutoff decomposition first for an abstract initial predicate on a set-sized sign tree or two set-sized

transitive models; then the canonical-name lemma, the filter-base and density bounds, name localization and intersections as abstract forcing lemmas with the truth lemma as an explicit interface; then the Boolean embeddability theorem from real-trace preservation and coordinate detection with its forcing hypotheses kept visibly external; and the class-isomorphism layer as a separate theorem with an explicit Global Choice interface (source 09 Question 11.9, source 10 Question 12, source 11 Question 14.10, source 12 Question 12.9, source 13 Question 10.8, second half, source 14 Question 14.10). Which existing ProveIt and Mathlib interfaces support this?

The sources propose staged routes: ordinal prefix operations, cofinal subsets and a cut-filling axiom for the sign layer; atomic absoluteness for a fixed poset, graph-name restriction, nice-name coding and the truth lemma for the forcing layer (source 09); a sign tree with the three values $- < 0 < +$, then ordinal block combinatorics with the two cofinality computations kept separate, then forcing, then the ground family and class maps, with set-sized models in a larger universe first (source 10); published forcing imports kept separate from the independently checkable sign and cofinality lemmas, and the final spectrum equalities not introduced as axioms (source 11); residues represented by Dedekind cuts, not proper-class equivalence classes, and any change from genuine old fields to an abstract family of real closed fields exposed (source 12); the advanced product facts as named hypotheses rather than unadvertised axioms (sources 13, 14). None of them is a same-universe Lean type of all surreals, no Lean declaration or build is included or claimed, and importing one checked surreal theorem would not certify the forcing above it (sources 10, 13).

32 What sources 09–14 do not claim

Every limitation, disclaimer and priority caveat of the six sources is kept, per source; items written for the merge are marked M. The shipped audit files of each source (section 33) state their boundaries in their own words.

Source 09

- 09.1 AI-assisted manuscript with written proofs; not independently refereed or Lean-verified; priority in the wider literature not certified; open to independent proof review, with the canonical-name lemma and the simultaneous field construction the priority targets.
- 09.2 The fresh-sign correspondence and symmetry are the repository's, reproved and credited, not claimed; the transfer is not claimed as a new forcing-spectrum theorem; FKW Propositions 4.3, 4.10, 5.2, Definitions 4.7, 4.8, Theorems 4.24, 7.21, 7.23 and Corollary 4.25 are imported; the Easton result is an application, not a solution of FKW's realization problems, which remain untouched.
- 09.3 No assertion of first discovery of the common-base-change technique; the lemma keeps the poset fixed and is not a chain-condition assertion; the bound is strict; the countable boundary is not claimed to behave like uncountable cofinalities.
- 09.4 No classification of all old surreal fields, of pure orders or of all forcing spectra; no claim that all gap spectra are Easton closed; no assertion that fields with the same reals and gap spectra must be isomorphic; nonisomorphism is a field claim, not one about pure order reducts.
- 09.5 The Namba instance needs CH and the Prikry instance a measurable; not a consistency proof of ZFC or of a large-cardinal configuration.

- 09.6 The involutions agree on $\mathbb{C}^M$; agreement on $\mathbb{C}^N$ is impossible for most members and is not claimed; no exponential, differential, valuation-preserving or simplicity-preserving isomorphism is claimed, and the pure surcomplex isomorphisms generally preserve none of these.
- 09.7 Global Choice for the simultaneous class isomorphisms, as one indexed class relation; for general pairs the spectrum can be a proper class.
- 09.8 The last row of its comparison table is deliberately not strengthened to equality of ground-regular spectra.
- 09.9 The finite checks test coding only: no generic extension, freshness of an infinite sequence or class recursion is modelled; successful compilation is not a proof check.
- 09.10 Its questions are research directions, not certified novelty statements.

Source 10

- 10.1 Proposed contributions with written proofs; not refereed, not Lean-verified; no priority certification; the content may overlap unlocated consequences of existing literature.
- 10.2 The gap correspondence, class back-and-forth and freshness predate it; the Easton spectrum is an imported published theorem (FKW Definition 4.8, Propositions 4.3, 4.10, Theorem 4.24), and Easton's lemma is used in Cody–Gitman's form.
- 10.3 No proof that equal full spectra imply isomorphism, and no explicit pair with equal full spectra but nonisomorphic fields is asserted (supplied by sources 09, 11, 12; M).
- 10.4 No full-spectrum formula at accumulation points of intermediate quotients; only isolated characters are computed; ground Cohen posets are identified with recomputed ones only at the singleton endpoint where proved.
- 10.5 GCH for the Easton and simultaneous constructions; $V = L$ as a concrete relative setting; no large cardinals; set forcing only; GBC with Global Choice for class maps, as a uniformly indexed relation.
- 10.6 Saturation is claimed for the fixed finite language; elementary equivalence over a core is a separate assertion; the inclusion order does not identify intersections or composita.
- 10.7 No preservation of valuation, simplicity, exponentiation, derivation, Hahn summation or omnific integer parts by the class-field maps; no solution of a general forcing-spectrum conjecture; not a characterization of all cofinality-preserving spectra.
- 10.8 The finite tests check coding and bookkeeping only; no Lean declarations or build.

Source 11

- 11.1 AI-assisted manuscript with conventional proofs; not refereed; no Lean; not a certified priority claim; independent review remains appropriate; novelty not certified by the search.
- 11.2 Its source audit was targeted: the recursive tree, collection and report guides and the *opening* of this report's LaTeX source; not a proof audit of the repository.
- 11.3 Imported: surreal, quantifier-elimination and forcing background; FKW Definition 4.8, Proposition 4.10, Theorem 4.24 and Theorem 7.21; Lévy–Solovay; the Prikry facts.

- 11.4 The finite-complement theorem does not extend to infinite complements without proof; the sparse hypothesis is local terminology.
- 11.5 The Prikry forcing in the tower is computed in the Cohen extension; the tower needs a measurable; nonisomorphism is via ω -saturation, not birthdays, whose field invariance is not claimed.
- 11.6 No claim that equal gaps *and* equal saturation suffice for, or obstruct, isomorphism (answered by source 12; M).
- 11.7 The involutions preserve no valuation, birthdays, simplicity, normal forms, exponential, derivation or omnific ring; they are real forms of the pure field.
- 11.8 No named conjecture is solved; its review priorities (its §§8 and 11) are kept; the finite checks do not verify forcing, freshness, cofinality, saturation or class arguments.

Source 12

- 12.1 A proposed negative answer to the *real-closed-field* direction of question 19.4 only, not to the pure-order question; not refereed or Lean-formalized; priority not certified; not a solution of a named published conjecture.
- 12.2 Credited ingredients: Conway, Gonshor, the repository's Part II, FKW Proposition 2.2, Corollary 2.4, Propositions 5.2–5.3 and 6.1, and the Mitchell and Unger antecedents via FKW; the Archimedean residue is standard.
- 12.3 Forcing statements are relative, not claims that ZFC proves the existence of transitive models or generics; no large cardinal, CH or GCH for the main theorem; Global Choice only for the transports.
- 12.4 “Universality” is only realization of ground set-sized posets, not Borel or effective reducibility; the trace classifies embeddability only within the family, says nothing about equal traces, and is not claimed finitary definable or efficiently computable.
- 12.5 The maps Φ_A preserve no conjugation, valuation or real constants; equivariant embeddings are not surjective.
- 12.6 No repository file was modified and no formalization compiled; the literature check is not exhaustive; the finite checks cover local identities only, and there are no finite fresh signs.

Source 13

- 13.1 Written proofs with an internal audit; not refereed, not Lean-verified; to be reviewed by a set theorist familiar with product forcing and class model theory.
- 13.2 FKW Proposition 4.10 and Corollary 4.21 (Shelah) are imported and not reproved; Proposition 6.1 and Theorems 7.21, 7.23 are used only in the comparison table.
- 13.3 Contributions may be folklore consequences of known results; no first-in-literature breakthrough, no classification of all real forms or all forcings, no solution of the general spectrum-realization problem.
- 13.4 Classification only within the displayed family; binary intersections only, no composita, no arbitrary intersections; the first new birthday is not recovered from the pure field language.
- 13.5 The continuum is $\aleph_1$ in the construction; uncountable κ_n are needed for the complex-core clause; the isomorphisms are noncanonical and preserve no richer structure.

13.6 “The user is not represented as having reviewed or endorsed these proofs.”

13.7 The finite checks do not verify freshness, genericity, cofinality, preservation, the successor-of-singular input, saturation or class arguments; in particular not the λ^+ term.

Source 14

14.1 Proposed contribution with written proofs; not refereed, not Lean-verified; priority of the relative formulations not certified; not to be represented as an independently certified breakthrough.

14.2 The singular-limit witness (FKW Corollary 4.21, Shelah 2000) is imported and not verified; no new pcf theorem; the fresh-sign correspondence and the saturation criterion are credited.

14.3 GBC with Global Choice and ground predicates for the class version; the set version uses ZFC only.

14.4 No preservation of valuation, normal forms, simplicity, exponential, derivation or omnific integers; the involutions are field automorphisms.

14.5 No general equal-spectrum classification, no arbitrary spectrum realization, no elimination of the singular-successor character, no arbitrarily large continuum, no consistency proof.

14.6 Its review order (the localization rectangle, the retention of the singular witness, the fresh-versus-boundary distinction) is kept; the finite checks exclude forcing, pcf, saturation and class recursion; a TeX build is not proof review.

Added by the merge

M.6 Part VIII adds no mathematical claim beyond comparisons stated as merge observations (the instance of theorem 25.15 in example 25.23, and question statuses derived from sibling theorems); proofs printed once from a base are the base’s.

M.7 The corrections of remarks 25.25 and 28.11 concern how two sources describe this report, not their mathematics.

M.8 The source manuscripts were not rebuilt for the merge; their page counts are as delivered and match their build records.

M.9 Nothing in Part VIII is formalized, and placement beside the collection’s Lean development confers no formal status.

This gives 10, 8, 8, 6, 7 and 6 items from sources 09–14 and 4 added by the merge: 49 in all. (The merge items continue the numbering M.1–M.5 of section 20.)

33 Provenance, repository comparison and finite checks of sources 09–14

33.1 Review scope of the sources

All six state the pin `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9` (25 September 2026) and were written on 28 September 2026.

- Source 09 read this report’s `article.tex` and `README` at the pin, and the collection’s reader’s guide `Algebra/SurrealNumbers/docs/README.md`; it cites the labels `univ:thm:gapcorrespondence`,

`univ:cor:gapspectrum`, `univ:thm:classbf`, `univ:q:cutspectrum` and `univ:q:completeness`, all present.

- Source 10 used the report at the fixed commit, especially Part II, Part V and Questions 19.2, 19.4, 19.7 with the batch-32 status note.
- Source 11 inspected the recursive tree, the surreal subproject guide, the collection guide, this report’s guide and the opening of its LaTeX source (targeted, not exhaustive; shipped 11-finite-complements-SOURCE_AUDIT.md).
- Source 12 used the pinned article and Algebra/SurrealNumbers/README.md (shipped 12-real-traces-SOURCES.md).
- Source 13 used this report’s README, its statements on Cohen forcing and its Section 19, and docs/surcomplex/first-kappa-coefficients for comparison (shipped 13-coordinate-spectra-SOURCES.md).
- Source 14 used the article and README, with attention to the fresh-sign correspondence, the single-Cohen example and Questions 19.2, 19.4, 19.7, and the neighbouring report overview.

Each read Fischer–Koelbing–Wohofsky [14] in its author-hosted version; source 11 checked Lévy–Solovay [23], source 14 Shelah [24].

33.2 Repository statements checked against the current collection

This report is unchanged between the pin and the placement commit `0e53d1034`, so no source statement about it is stale in that sense. Checked:

1. The labels and numbers the sources cite exist and mean what they say: “Theorem 1.1” (theorem 1.1), “Theorem 7.1” (theorem 7.1), “Theorems 8.4–8.5” (theorems 8.4 and 8.5), “Proposition 13.2” (proposition 13.2), “Theorem 15.1” (theorem 15.1), Questions 19.2, 19.4, 19.5, 19.7 (questions 19.2, 19.4, 19.5 and 19.7).
2. True at the pin: that proposition 13.2 computed only the first Cohen gap and left larger gaps open (sources 12, 13, 14); that question 19.7 asks for agreement on $\mathbb{C}^N$ (source 09); that its batch-32 note separates single-character forms from multicharacter spectra and families of old fields (source 10); that first-kappa-coefficients already treats singleton thresholds (source 13); and that the report’s own source pin (`4f26455`) differs from the audit pin (source 13). These gaps are now filled by Part VIII; the dated statuses record it.
3. Overstated: source 09’s “implication proposed as a possible classification principle in the repository” (remark 28.11) and source 10’s “suggestion that the least gap size ... could classify the fields” (remark 25.25); the report only asked.
4. Source 11’s shipped audit mentions a “neighboring critical-point-defects guide”; the file is docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms/07-critical-point-defects-PROOF_STATUS.md, a proof-status file rather than a guide, and no claim from it is used.

33.3 Finite checks

Every source shipped a standard-library Python program and a recorded run. There are no finite fresh signs, so none tests freshness, forcing, cofinality, preservation, saturation or class arguments. All six were rerun for

the merge on a copy of this directory with Python 3.14.4, with explicit output paths; each output equals its recorded file apart from line endings.

- `code/09-cohen-amplification-verify_finite.py` (`data/09-cohen-amplification-verification_results.json`): 260,865 canonical-cone cases, 260,865 comparison-antisymmetry cases, 5,704 block round trips, 183,103 block-prefix cases and three symbolic image examples (finite Easton, Prikry, Namba under CH).
- `code/10-isolated-coordinates-check_finite.py` (`data/10-isolated-coordinates-verification_results.json`): 5,461 delimiter-code round trips, 36,409 prefix checks, 3 rejected inputs, 16,065 cone equivalences, 511 profile checks and 87,381 reversed-inclusion pairs.
- `code/11-finite-complements-finite_checks.py` (`data/11-finite-complements_results.json`): 469,107 checks, including 65,025 order comparisons, 65,025 cone equivalences and 330,496 first-disagreement checks.
- `code/12-real-traces-verify_finite.py` (`data/12-real-traces-verification_results.json`): 65,025 dyadic comparisons, 65,025 prefix-separator pairs, 5,698 block round trips, 129,400 prefix dependencies, 2,047 ternary round trips, 11 injectivity checks, and the principal-down-set embedding for all labelled partial orders on at most four points (1, 1, 3, 19, 219 of them; 3,688 pair checks).
- `code/13-coordinate-spectra-verify_finite.py` (`data/13-coordinate-spectra-verification_results.json`): 64,897 separator-cone pairs, 1,365 block round trips, 22,302 prefix-locality checks, 5 rejected encodings, 7,533 condition extensions and 16,384 coordinate-mask pairs.
- `code/14-relative-cutoffs-verify_finite.py` (`data/14-relative-cutoffs-finite_checks.json`): 619,026 assertions, including 260,865 separator-cone checks and the finite and eventually periodic index-set identities behind the coordinate formulas.

33.4 Dependency ledger

Result	Essential inputs	Status
Transfer (theorem 22.7)	Part II; compression; block codes	written proof; related to FKW Props. 5.2–5.3
Filter-base ceiling	truth lemma; cofinal pigeonhole	written proof
Generalized Cohen	closure; universal occurrence; ceiling	written proof; four routes
Square chain condition	bounded maximal antichain	FKW Prop. 2.2, reproved
Easton realization	transfer; FKW Thm. 4.24, Prop. 4.10	imported input
Isolated coordinates, finite complements, cubes	localization; high closure; density; FKW preservation	written relative proofs
Countable coordinate spectra	localization; intersections; density; FKW Cor. 4.21	written proof; named import for λ^+
Cutoff gap pairs	birthday bound; canonical chains; name counting	written proof
Real trace, Boolean family	standard part; Cohen quotients; coordinate detection	written proof; ZFC
Base change, amplification	atomic absoluteness; nice names; trace	written proof
Prikry, Namba spectra	FKW Thms. 7.21, 7.23; transfer	imported inputs
Cohen–Prikry tower	Lévy–Solovay; Prikry spectrum; tower bound; ω criterion	written proof; measurable

Result	Essential inputs	Status
Real-form families	class back-and-forth (Part V); transport; interpretation	Global Choice, except the set version

The ledger records the checks made in assembling the arguments; it is not a proof-assistant log.

A Provenance and repository comparison

A.1 Pinned review scope of the sources

All four sources were written against

VladimirReshetnikov/Surreal, commit 4f2645599121fa872c7104995e47f86f0382351f,

(“Index the newly placed drafts without extending review claims”), a commit that precedes the batch-20 rewrite of the automorphism report. They read the repository through a connector and did not modify or build it.

- Source 01 read docs/README.md, the opening 180 lines of the foundations report, the returned opening text of the automorphism report, and the root README; its scoped code searches used the connector’s default-branch index. It states that this is not a file-by-file audit.
- Source 03 read the main README, the catalogue, the foundations README, and the sections of the automorphism report on the unrestricted field, class back-and-forth and genuinely different real forms.
- Source 04 inspected the recursive file tree, the root description, the catalogue, and the foundations and automorphism sources, not every manuscript in full; a repository-wide code search returned an incomplete result that it did not treat as exhaustive.
- Source 08 read the root README, the documentation inventory, the automorphism report’s README, and selected source sections including the class back-and-forth, the involution conjugacy criterion and the countably cofinal real form.

A.2 Statements corrected against the current collection

1. *Searches for “forcing” and “saturation”.* Source 01 reports that searches for forcing, saturation and distributivity “returned no matches”; source 08 that searches for forcing and saturation “returned no hits”. At the pin, and now, the foundations report contains the phrase “saturation for small cuts” in its discussion of completeness and a subsection “ETR, class forcing, and two invalid moves” (found:sub:etr). Neither contains the results of this report, and both sources presented their searches as limited. Source 04’s statement that the foundations report’s forcing references did not supply the cross-universe spectrum is accurate.
2. *Report counts.* Source 01 describes docs/README.md as listing forty catalogued reports plus four newly placed drafts. That was the state at the pin. The collection now has 44 report directories, including this report and its sibling.
3. *The automorphism report.* It was rewritten after the pin (commit fb182ea, the batch-20 write), and its labels gained the prefix saut:. The sources’ descriptions of what they read in it (“opening

text”, “the newly placed manuscript”, selected sections) refer to the pinned version. This report cites its current numbers and labels: Theorem 9.1 (`saut:thm:acfhomogeneity`), Section 10 (`saut:sec:realforms`), Theorem 10.2 (`saut:thm:inequivalentrealform`), Proposition B.1 (`saut:prop:conjugacycriterion`) and Section 14. Source 04’s remark that the automorphism source contained no matches for `forcing`, `saturat` or `nonconjugate` is true of the pinned text; the current text still has no `saturat` and no `nonconjugate`, and uses “forcing” once only in the unrelated sense of forcing an equation.

4. *A missed prior result.* Sources 03 and 08 correctly credit the collection with class back-and-forth, the fixed-field conjugacy criterion and the countably cofinal real form, but none of the four sources cites the set-cut interpolation criterion (present at the pin as `prop:conjugacycriterion`, now `saut:prop:conjugacycriterion`). Remark 16.4 records what it implies and what remains new.
5. *Formalized degeneracy.* Source 01’s statement that the internal degeneracy of set-indexed convergence is “discussed and formalized” in the repository is accurate: `docs/FORMALIZATION.md` marks the subset, convergent-net and Cauchy-net clauses of `found:thm:discrete` as proved, the last under `HasSmallCutFillers` (remark 14.4).

A.3 Literature boundary

Conway and Gonshor supply the construction and cut theory [1, 2]; van den Dries–Ehrlich with their erratum, and Bournez–Guilmant, the birthday-field background [3, 4, 5]; Marker the quantifier-elimination and saturation machinery [9]; Ehrlich’s absolutely saturated models the class-saturation background, whose terminology concerns all sets of the ambient universe and is not contradicted by the use of a larger universe [6]; van den Dries–Ehrlich’s homogeneous universal H -fields and Ehrlich–Kaplan’s surreal ordered exponential fields the richer structures [7, 8]; Jech, Prikry, Gitik and Benhamou the forcing facts [10, 11, 12, 13]; Fischer–Koelbing–Wohofsky the forcing notion of freshness [14]; Berenbeim the reverse-simplicity question and the suggestion to study surreals under forcing [15]; Holy and coauthors the class-forcing completion theory [16]. Rangel–Mariano and Bertram connect surreal constructions with set-theoretic foundations; their themes and questions are not claimed or resolved here [17, 18]. Each source’s targeted literature search did not locate the exact theorems; none is an exhaustive bibliography.

B Dependency and proof-obligation ledger

Result	Essential inputs	Boundary checked
Old arithmetic and elementarity	Absolute signs, prefix witnesses for old cuts, set-like Conway recursion; RCF/ACF QE	Field elementarity is not universe elementarity.
Ground-set containment	Birthday bounds; common ordinals; old bounded sign sets	The containing set need not be small.
One old side	Ground-set containment; ground cut theorem	Both sides external sets; one literally in M .
Regular first length	Minimal new sequence; old coding maps	Cardinality and cofinality computed in N ; M -regularity by persistence.
Interval tree and decoder	Set-length recursion; explicit splitting; two cuts at limits	Decoder total, with a default value, in M .

Result	Essential inputs	Boundary checked
Absolute interval code	Per-input set recursion; absolute arithmetic and cuts	No set of all children; labels range over Ord.
Block codes	Two-bound cone identity; set-recursive decoding	Ordinal, not natural, sums; successor runs.
Main spectrum	Minimal new sequence; decoder; one-side lemma; RCF QE	Saturation equivalence only for uncountable κ .
First gap	Regularity; nested endpoints; one-side interpolation	Cofinalities in N with set witnesses.
Gap correspondence	Prefix convexity; simplest separators; constant-tail closure	Set-presented gaps only; no Global Choice.
Least gap	Old-object coding; terminal-plus blocks	Sizes computed in N .
First new birthday	Characteristic signs; minimality; ordinal coding in M	β is not a birthday of a given number.
ω -saturation	Countable syntax over finite old parameters; rational cuts	No new reals is not no new countable sequences.
Pure surcomplex saturation	Ground envelopes; algebraic closures of set fields; ACF QE	No class-sized transcendence basis or global isomorphism.
Conjugation spectrum	Fixed field; order by nonzero squares; coordinates	Naming i adds one parameter.
Bounded fragments	Birthday cut bound; sign-cone lemma	Regularity in N separate from equality of short signs.
External Cauchy rigidity	Old positive lower bounds	Fine surreal radii; no cut filling needed.
Closed, nowhere dense	Open cones; old radii below new ones	Needs $M \neq N$ for nowhere-denseness.
Class isomorphism, real forms	Global Choice; set-stage back-and-forth; gap invariance	No valuation, simplicity or exponential preserved.
Set-complete Boolean algebra	Cylinder maps; common levels for set families	Least-level set pairs, not class objects.
No class-complete algebra	Global Choice; proper-class antichain; diagonal subclass	Stronger completeness, extra axiom.

The ledger records the mathematical checks made in assembling the arguments. It is not a proof-assistant log.

C Finite checks

Three sources shipped standard-library Python programs; source 01 shipped none. (The checks of sources 09–14 are described in section 33.) They check finite sign comparisons and interval formulas. There are no finite fresh signs, since a finite sequence of old entries is always old, so none of them tests freshness, forcing, regularity, saturation or any class argument.

- `code/03-fresh-sign-gaps-check_finite_signs.py` checks first-disagreement comparison, convexity of finite prefix cones and the terminal-plus block code with its endpoint ordering, for value lists of length at most four with entries in $\{0, 1, 2, 3\}$ and cones among signs of length at most six. Recorded run (`data/03-fresh-sign-gaps-finite_checks.json`): 127 sign strings, 16,129 ordered-pair comparisons, 127 cone checks, 340 block-code cases and 1,744 nested endpoint pairs, all passing. For the

list $(2, 0, 1)$ the code is $\text{code}^+ = +++-+-++-$, with endpoint lengths $(2, 3), (4, 5), (7, 8)$ and values $2 < \frac{5}{2} < \frac{43}{16} < \frac{173}{64} < \frac{87}{32} < \frac{11}{4} < 3$.

- `code/04-universe-saturation-verify_finite_models.py` checks, in exact rational arithmetic, the successor interval formulas (5.4), the closed formulas of proposition 5.5, separation of children with finite labels, unique finite decoding, and the cut/prefix equivalence of lemma 2.2 for finite strings. Recorded run (`data/04-universe-saturation-finite_check_results.txt`): 1,365 interval nodes, 20,475 child pairs, 1,364 decodings, 65,025 cut/prefix comparisons.
- `code/08-forcing-saturation-verify_sign_code.py` checks sign order against dyadic values, the two-bound identity of lemma 5.6, decoding, nesting and finite cone intersections for $\text{code}(f)$. Recorded run (`data/08-forcing-saturation-verification_results.txt`): 304,000 checks. For $(2, 0, 1)$ the code is $++-+-+ -+ +-+$ with value $\frac{219}{128}$ and intervals $(\frac{3}{2}, 2) \supset (\frac{13}{8}, \frac{7}{4}) \supset (\frac{109}{64}, \frac{55}{32})$.

All three suites were rerun for this merge on a copy of the directory, with Python 3.14; each output agrees with its recorded file except for line endings on Windows. Commands are given in the README.

References

- [1] J. H. Conway, *On Numbers and Games*, 2nd ed., A K Peters, 2001 (first edition, Academic Press, 1976).
- [2] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [3] L. van den Dries and P. Ehrlich, “Fields of surreal numbers and exponentiation,” *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>.
- [4] L. van den Dries and P. Ehrlich, “Erratum to ‘Fields of surreal numbers and exponentiation’,” *Fundamenta Mathematicae* **168** (2001), 295–297. The corrected statements are also restated in [5, Theorems 1.1–1.2 and 3.3].
- [5] O. Bournez and Q. Guilmant, “Surreal fields stable under exponential and logarithmic functions,” arXiv:2201.08199, 2022. <https://arxiv.org/abs/2201.08199>.
- [6] P. Ehrlich, “Absolutely saturated models,” *Fundamenta Mathematicae* **133** (1989), 39–46. <https://doi.org/10.4064/fm-133-1-39-46>.
- [7] L. van den Dries and P. Ehrlich, *Homogeneous universal H-fields*, arXiv:1807.08861, 2018. <https://arxiv.org/abs/1807.08861>.
- [8] P. Ehrlich and E. Kaplan, *Surreal ordered exponential fields*, arXiv:2002.07739, version 3, 2021. <https://arxiv.org/abs/2002.07739>.
- [9] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics 217, Springer, 2002. <https://doi.org/10.1007/b98860>.
- [10] T. Jech, *Set Theory: The Third Millennium Edition, Revised and Expanded*, Springer Monographs in Mathematics, Springer, 2003. <https://doi.org/10.1007/3-540-44761-X>.
- [11] K. L. Prikry, “Changing measurable into accessible cardinals,” *Dissertationes Mathematicae (Rozprawy Matematyczne)* **68** (1970), 55 pp.
- [12] M. Gitik, “Prikry-type forcings,” in M. Foreman and A. Kanamori (eds.), *Handbook of Set Theory*, Springer, 2010, 1351–1447. https://doi.org/10.1007/978-1-4020-5764-9_17.
- [13] T. Benhamou, “Prikry forcing and tree Prikry forcing of various filters,” arXiv:1801.04424, 2018. <https://arxiv.org/abs/1801.04424>.
- [14] V. Fischer, M. Koelbing and W. Wohofsky, “Fresh function spectra,” *Annals of Pure and Applied Logic* **174** (2023), no. 9, 103300. <https://doi.org/10.1016/j.apal.2023.103300>.
- [15] A. Berenbeim, *Review of Surreal Number Construction and Forcing Construction*, research notes dated 8 April 2020, 35 pp. Question 2, p. 32; proposed further directions, p. 34. Listed by the author as incomplete notes. <https://www.berenbeim.com/papers>.
- [16] P. Holy, R. Krapf, P. Lücke, A. Njegomir and P. Schlicht, “Class forcing, the forcing theorem and Boolean completions,” arXiv:1710.10820, 2017. <https://arxiv.org/abs/1710.10820>.
- [17] D. Rocha Rangel and H. L. Mariano, *An algebraic (set) theory of surreal numbers, I*, arXiv:1911.12726, 2019. <https://arxiv.org/abs/1911.12726>.

- [18] W. Bertram, *On Conway's Numbers and Games, the Von Neumann Universe, and Pure Set Theory*, arXiv:2501.04412, version 2, 2025. <https://arxiv.org/abs/2501.04412v2>.
- [19] *Automorphisms of the Surcomplex Numbers*, report in VladimirReshetnikov/Surreal, docs/surcomplex/surcomplex-field-automorphisms/, label prefix saut:. The sources read its pinned version at 4f26455.
- [20] P. Ehrlich, "The absolute arithmetic continuum and the unification of all numbers great and small," *Bulletin of Symbolic Logic* **18** (2012), no. 1, 1–45. <https://doi.org/10.2178/bsl/1327328438>. Cited by sources 09–14 of Part VIII.
- [21] J. D. Hamkins, "Gap forcing," *Israel Journal of Mathematics* **125** (2001), 237–252. Preprint <https://arxiv.org/abs/math/9808011>. Cited by source 09.
- [22] B. Cody and V. Gitman, "Easton's theorem for Ramsey and strongly Ramsey cardinals," *Annals of Pure and Applied Logic* **166** (2015), no. 9, 934–952. <https://arxiv.org/abs/1209.1133>; the form of Easton's lemma used is Lemma 3.8 of arXiv:1209.1133v1. Cited by source 10.
- [23] A. Lévy and R. M. Solovay, "Measurable cardinals and the continuum hypothesis," *Israel Journal of Mathematics* **5** (1967), 234–248. <https://doi.org/10.1007/BF02771612>. Cited by source 11.
- [24] S. Shelah, "Embedding Cohen algebras using pcf theory," *Fundamenta Mathematicae* **166** (2000), no. 1–2, 83–86. <https://doi.org/10.4064/fm-166-1-2-83-86>. Cited by source 14, through [14, Corollary 4.21].